Software Requirements Specification (IEEE 29148 structure)

MathType-style Equation Editor — Web Playground

Complete, testable specification of the embedded editor of https://www.wiris.com/en/mathtype/ and of its output-extraction layer.

Document	SRS-MTE-001
Version / status	1.0 — release candidate for implementation
Date	2026-10-12
Companion document	**FR-DOC**: “MathType Web — Landing-page Playground: functional requirements & UI behaviour reference” (evidence, 4752 screenshots, 1207 scenario steps, 2748 control cases)
Reference system	Wiris MathType Web editor build 7.25.6.804ace, handwriting component 7.11.4.71fa31 (public playground)
Requirement count	172 specified requirements + 700 control requirements (Appendix C) + 12 parity decisions

“SHALL” = mandatory; “SHOULD” = recommended; “MAY” = optional. Verification: T = automated test, I = inspection, D = demonstration. Every behaviour statement was observed and measured on the reference; where the reference shows a defect the decision is recorded in Appendix A (parity mode `exact` by default).

Document control

Version	Date	Author	Change
0.9	2026-10-11	Analysis team	Reference behaviour captured (FR-DOC)
1.0	2026-10-12	Analysis team	Specification completed: requirements, catalogue, data model, services, output extraction, verification

How to use this document

- Chapters 4–7 define **what** the system does, which interfaces it has and which data it produces (each requirement has a unique ID, a reference into FR-DOC and a verification method).
- Appendix C lists **every control** of the reference toolbar (700) with geometry, icon and exact expected MathML for three test states; implementing it exactly gives the reference behaviour.
- Chapter 8 gives the **architecture and technology** choices; chapter 9 the **output extraction** layer (the only functionality added to the reference playground); chapter 10 **verification** and traceability.

Contents

1 Introduction
2 Overall description
3 Requirement conventions
4 Functional requirements
5 Services and non-functional requirements
6 External interface requirements
7 Data requirements
8 Architecture and technology (design constraints)
9 Output extraction layer (addition to the reference)
10 Verification and validation

Appendix A — Parity decisions (reference quirks)
Appendix B — Evidence datasets and reproduction
Appendix C — Control catalogue (700 control requirements)
Appendix D — MathML profile produced by the editor
Appendix E — Recognised words, digit sets and number handling
Appendix F — Colour palette (25 swatches, row-major)
Appendix H — Design tokens
Appendix I — Behaviour differences caused by global modes
Appendix J — Layout measurements of the reference

1 Introduction

1.1 Purpose

This SRS specifies, completely and verifiably, a web-based visual mathematical-equation editor that is functionally and visually equivalent to the editor embedded in the “See MathType in action!” section of the Wiris MathType landing page (“the reference playground”), plus an output-extraction layer that makes the edited equation usable by host applications (MathML, LaTeX, SVG/PNG images, accessible text, clipboard). A team that implements the requirements, the control catalogue and the data model of this document SHALL obtain a component whose behaviour can be proven equal to the reference by the acceptance tests of chapter 10.

1.2 Scope

In scope: the complete editor widget (725 × 415 px): tab strip, ten toolbar tabs with 700 controls, pop-up groups, matrix/table picker, colour palette, special-characters chooser, font and size selectors, clipboard and history controls, formatting toggles, right-to-left and Arabic modes, the editing area with caret/selection/keyboard/mouse/touch behaviour, the Contextual tab (69 distinct controls over 12 structure contexts), elementary mathematics, handwriting mode, accessibility, and the backend services the widget needs. **Additional (the only addition):** the output-extraction API, events and optional output panel (chapter 9).

Out of scope: the marketing page around the widget, cookie banner, third-party scripts; products/tabs that are not part of the reference toolbar configuration (Publishers tab, ChemType, LaTeX/MathML input windows of the Microsoft 365 integration, image-recognition drop zone).

1.3 Definitions

Term	Meaning
Reference playground	The widget at https://www.wiris.com/en/mathtype/, built by `com.wiris.jsEditor.JsEditor.newInstance({language:'en'}).insertInto(#editorContainer)`.
Presentation MathML	W3C MathML presentation markup, the canonical document format of the editor.
Slot	An editable position inside a structure (numerator, radicand, cell …); an empty slot is drawn as a small green box.
Template / symbol	Toolbar controls that insert a structure with slots / a single character.
Contextual control	Control shown in the Contextual tab for the structure at the caret.
Parity mode	Configuration `parity: "exact" | "improved"`; “exact” reproduces the reference including its quirks (Appendix A).
FR-DOC	The companion functional reference containing the evidence.

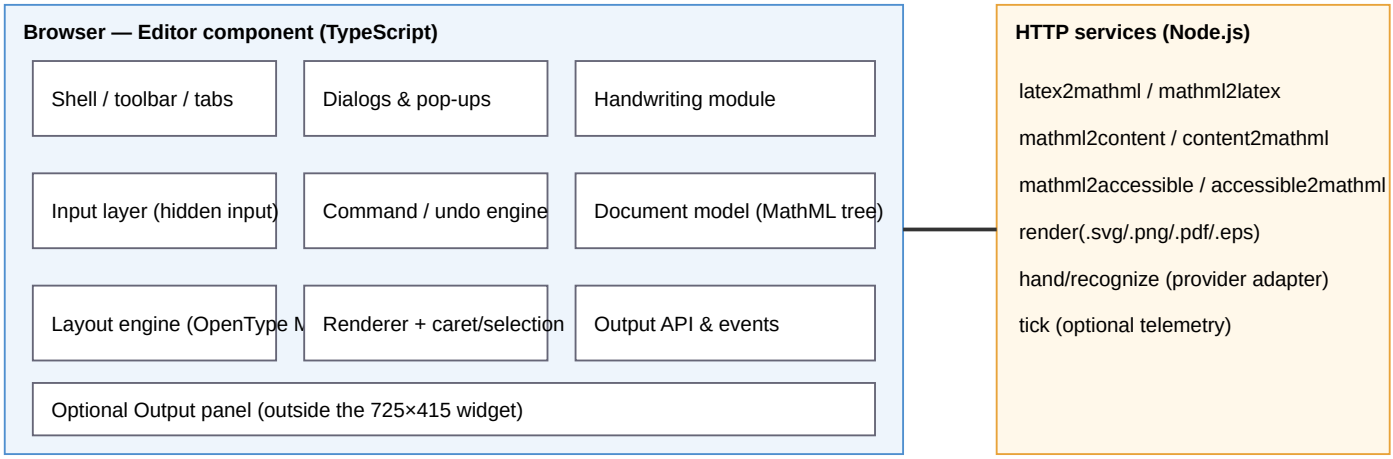
1.4 References

- FR-DOC (companion document, evidence base for every requirement).
- W3C MathML 3.0 / MathML Core; WAI-ARIA 1.2 and ARIA Authoring Practices; WCAG 2.2.
- Vendor documentation (terminology and parameter names): docs.wiris.com — MathType Web interface, Toolbar and icons, Contextual tab, Keyboard shortcuts, Autoformat, Formatting, Insert special characters, Handwritten input, Editor parameters, JavaScript GUI API reference, Conversion / Rendering / Accessibility services references.
- IEEE/ISO/IEC 29148:2018 (requirements engineering); OWASP ASVS 4 (security requirements).

2 Overall description

2.1 Product perspective

The product is a self-contained JavaScript component (no framework dependency) that is embedded into a host page, plus a small set of HTTP services. The component keeps the whole editing state in the browser; services are only used for LaTeX/MathML conversion, spoken-text generation, image rendering and handwriting recognition (FR-DOC §3, §23).



2.2 Product functions (summary)

- Visual construction of equations with 700 toolbar controls in 9 tabs and 24 pop-up groups.
- Keyboard-first editing with structure-aware caret movement, selection, smart delete, line breaks and shortcuts.
- Formatting: bold, italic, automatic italic, regular text, colour (25 swatches + hex), 6 fonts, 16 sizes, RTL, Arabic ligature, digit styles.
- Tables/matrices with size picker and a Contextual tab of 69 controls; elementary-math stacks and long division.
- Handwriting recognition with undo/redo/clear, scribble-erase and in-canvas help.
- Accessibility: ARIA roles, live announcements, spoken formula text, complete keyboard operation.
- **Output extraction**: MathML, LaTeX, SVG, PNG, PDF/EPS (service), accessible text, plain text, clipboard formats, events.

2.3 User classes

Class	Needs
Author (student / teacher / editor)	Build and edit equations quickly by mouse, keyboard, touch or handwriting.
Integrator (developer)	Embed the component, configure parameters, read the result in several formats, store and display it safely.
Assistive-technology user	Operate everything by keyboard, hear the structure and the formula.
Operator	Deploy and monitor the services.

2.4 Operating environment, constraints, assumptions

- Desktop browsers (≥ 1275 px host viewport in the reference page), touch and pen input supported.
- Constraint: the widget geometry, colours and behaviours are fixed by this document (pixel parity, NFR-012).
- Constraint: Wiris artwork, icon sprites and brand names are proprietary; implementers SHALL draw their own icons following the icon sheet (Appendix C) or obtain a licence. Fonts: STIX Two (SIL OFL) are used for the math glyphs.
- Assumption: the host provides HTTPS and a reachable services base URL (or uses the vendor cloud endpoints, SVC-009).

3 Requirement conventions

Requirement IDs have the form `AREA-NNN`. Control requirements have the form `TB-<tab>-[G<group>-]<nnn>`. Each row cites the section of FR-DOC that contains the evidence (screenshots, MathML, network trace). “Ver.” gives the verification method. Priority of every requirement is **Must** unless the text says SHOULD/MAY. Area codes:

Code	Area	Code	Area	Code	Area
UI	Shell and layout	CHR	Toolbar chrome	TAB	Tabs
PUP	Pop-ups	MPK	Matrix picker	EDT	Editing area
INS	Insertion semantics	FMT	Formatting	UNI	Special characters
CLP	Clipboard / history	CTX	Contextual tab	ELM	Elementary math
HND	Handwriting	MOD	Modes	ACC	Accessibility
TCH	Touch	SVC	Services	OUT	Output extraction
NFR	Non-functional	TB	Control catalogue	Q	Parity decisions

4 Functional requirements

Sections 4.1–4.18 contain the area requirements; the per-control requirements are in Appendix C (700 rows).

Area	Title	Requirements
UI	User interface shell and layout	12
CHR	Toolbar chrome, states and tooltips	8
TAB	Tabs and tab navigation	8
PUP	“More” pop-ups (expand groups)	5
MPK	Matrix and table size picker	9
EDT	Editing area: input, caret, selection, deletion	23
INS	Structure insertion semantics (all toolbar templates and symbols)	6
FMT	Text formatting controls	9
UNI	Insert special characters	5
CLP	Clipboard, undo and redo	6
CTX	Contextual tab	9
ELM	Elementary mathematics (stacks and long division)	3
HND	Handwriting recognition mode	12
MOD	Global modes	5
ACC	Accessibility	5
TCH	Touch	2
SVC	Backend services contract	11
OUT	Output extraction (additional to the reference playground)	8
NFR	Non-functional requirements	12
LAY	Layout and typography of the formula	8
TBL	Table / matrix navigation and editing	6
TB	Control catalogue (Appendix C)	700

4.1 User interface shell and layout

ID	Requirement	Reference	Ver.
UI-001	The editor SHALL render inside a host-supplied container element and occupy 725 px (width) × 415 px (height) by default; the container height is taken from the host element (the playground uses a 415 px high container).	FR-DOC §2, §25	T
UI-002	The editor SHALL be composed of four stacked zones: a tab strip (header, 25 px high, y = 1), a toolbar panel (66 px high, x = 3, y = 28, 719 px wide), a formula display (695 × 316 px at x = 0, y = 99) and a handwriting strip (30 px wide, right-aligned, at x = 695, y = 99, 316 px high).	FR-DOC §2	T
UI-003	The formula display SHALL have a 1 px solid #969696 border, a white background, `overflow:auto`, a text cursor, left-to-right direction, and a 5 px inner margin around the formula container.	FR-DOC §2, §17	I
UI-004	The handwriting strip SHALL be a vertical bar filled with #778e9a (the same colour as the toolbar frame) containing a centred white “hand with squiggle” icon; it SHALL show a pointer cursor and open handwriting mode on click, tap or activation of the hidden input titled “Go to handwritten mode”.	FR-DOC §2, §22, §24	T
UI-005	The editor root SHALL use `font-family: sans-serif`, black text, transparent background and 16 px default formula font size; toolbar text SHALL be 11 px bold.	FR-DOC §2	I
UI-006	The toolbar scale parameter `toolbarSize` SHALL accept 1, 1.5 and 2 and scale every toolbar dimension (panel height 66/95/124 px, separator height 60/90/120 px, select height 18/27/36 px, small-button box 22/33/44 px, large-button box 33/50/66 px).	FR-DOC §3	T

UI-007	Selection highlight SHALL be #c3c3c3 on light and #3c3c3c on dark backgrounds, the “edit context” highlight #cddce3, the error highlight #ffaaaa; the caret SHALL be a 1 px #000 vertical line 20 px high at 16 px font size (white on dark backgrounds), blinking: visibility toggles every 500 ms (1 s period).	FR-DOC §15	I
UI-008	Empty placeholder slots SHALL be drawn as small green outlined squares (empty box colour #006400, configurable via `setEmptyBoxColor`); empty script slots SHALL be drawn smaller and lighter.	FR-DOC §6	I
UI-009	The editor SHALL provide a help link (18 × 18 px at x = 705, y = 4) rendered as a dark-blue circled “?”; activation SHALL open the manual URL in a new browsing context (`target=_blank` , tooltip “Manual”).	FR-DOC §2, §3	T
UI-010	The editor SHALL expose no scroll-bars of its own except those of the formula display; the toolbar SHALL never wrap, scroll or resize.	FR-DOC §17, §25	T
UI-011	The editor SHALL not capture global page shortcuts other than the ones listed in section 4.6; focus SHALL be free to leave the widget with Tab / Shift+Tab (no focus trap).	FR-DOC §4, §24	T
UI-012	The host page MAY hide the editor below 1275 px viewport width (reference page behaviour); the editor itself SHALL NOT include a responsive mobile layout: its size is fixed.	FR-DOC §25	I

4.2 Toolbar chrome, states and tooltips

ID	Requirement	Reference	Ver.
CHR-001	Every toolbar control SHALL expose its tooltip text through both the `title` and the `aria-label` attributes; shortcuts SHALL be part of the text (e.g. “Fraction (Ctrl+/)”).	FR-DOC §4, §24	T
CHR-002	A button SHALL have a 2 px transparent border; on hover and keyboard focus the border SHALL become 2 px solid #778e9a with 3 px radius and the face SHALL turn pale yellow; on press the same style SHALL persist.	FR-DOC §4	I
CHR-003	A toggled control (Bold, Italic, Automatic italic, Regular text, Right-to-left, Arabic ligature and contextual toggles) SHALL show the active-button background image (33 px or 22 px variant) with an orange border.	FR-DOC §4, §9	I
CHR-004	A disabled control SHALL be rendered greyed-out, ignore pointer and keyboard activation, carry class `wrs_disabled` and `aria-disabled="true"` and keep a faint focus ring (rgba(119,142,154,0.3)) when focused.	FR-DOC §4, §8	T
CHR-005	Groups of controls SHALL be separated by 1 px vertical lines #bbbdbe, 60 px high in the multi-row panel.	FR-DOC §2	I
CHR-006	The Font family and Font size controls SHALL be drawn as bordered boxes (1 px #b1b3b4, text #b1b3b4, 18 px high, ≤ 73 px wide) with a right-aligned arrow background; their option lists SHALL be those of FMT-006 and FMT-007.	FR-DOC §9	I
CHR-007	The toolbar SHALL contain controls arranged on a 3-row grid (small 22 × 22 px cells, symbols) or a 2-row grid (large 33 × 33 px cells, templates) exactly as listed in the control catalogue (Appendix C) with the recorded x/y positions.	FR-DOC §6	T
CHR-008	Where a control group has additional items, a narrow ▸ expand button (7 × 60 px at the right edge of the group) SHALL be shown (section PUP).	FR-DOC §6, §11	T

4.3 Tabs and tab navigation

ID	Requirement	Reference	Ver.
TAB-001	The editor SHALL provide ten tabs in this order with these accessible names and shortcuts: General (Alt+1), Symbols (Alt+2), Arrows (Alt+3), Greek, letters and numbers (Alt+4), Matrices and elementary (Alt+5), Scripts and layout (Alt+6), Decorations (Alt+7), Big operators (Alt+8), Calculus (Alt+9), Contextual (Alt+0).	FR-DOC §5	T
TAB-002	Tab buttons SHALL be 44 px wide with a 2 px gap; the selected tab SHALL be 25 px high starting at y = 1 with background #f2f3f5 and a 1 px #778e9a top border, non-selected tabs 22 px high at y = 4 with background #556e7b and white icon; the Contextual tab SHALL use the pink background (#dc9e9e) when not selected.	FR-DOC §5	I
TAB-003	Tabs SHALL carry `role="tab"` inside a `role="tablist"` ; arrow keys on a focused tab SHALL select the neighbouring tab immediately.	FR-DOC §5, §24	T
TAB-004	Alt+<digit> (Ctrl+<digit> on macOS) pressed in the editing area SHALL select the tab and move focus to its first symbol (the Contextual tab keeps the focus in the editing area).	FR-DOC §5	T
TAB-005	Alt+→ / Alt+← SHALL move to the next / previous tab and SHALL NOT wrap at the ends.	FR-DOC §5	T
TAB-006	Inside a tab, Tab SHALL focus the first symbol of the grid; arrow keys SHALL move in a 2-D manner between symbols; Enter / Space SHALL insert the symbol and return the focus to the editing area; only the first control of the selected tab is in the page tab order.	FR-DOC §5, §24	T
TAB-007	Selecting a tab SHALL NOT change the document or the selection.	FR-DOC §5	T
TAB-008	The Contextual tab SHALL be empty unless the caret or selection is inside a structure with contextual controls; it SHALL NOT be selected automatically when such a structure is inserted; its content SHALL refresh each time the caret or selection changes while the tab is open.	FR-DOC §19	T

4.4 “More” pop-ups (expand groups)

ID	Requirement	Reference	Ver.
PUP-001	A click on an expand (▸) button SHALL open a pop-up directly below it, mark the button pressed and focus the first item of the pop-up.	FR-DOC §11	T
PUP-002	Inside a pop-up ArrowDown/ArrowUp SHALL move the focus between items; Enter or Space SHALL insert the item at the caret, close the pop-up and return the focus to the editing area; a pointer click on an item SHALL do the same.	FR-DOC §11	T
PUP-003	Escape SHALL close the pop-up without inserting and move focus to the expand button; a click in the editing area, on another tab or on another group's expand button SHALL close it (the other group opens immediately); clicking the same expand button again SHALL keep it open.	FR-DOC §11	T
PUP-004	Pop-up items SHALL be full toolbar buttons (own tooltip, own result) and SHALL behave as specified in the control catalogue (Appendix C).	FR-DOC §6, §11	T
PUP-005	The Greek tab pop-ups SHALL present multi-row scrollable blocks (303 items in 6 groups: double-struck, Fraktur, script, black-letter/special, numerals, chemical elements) and the other tabs single-column lists, with the item counts of Appendix C.	FR-DOC §6, §11	T

4.5 Matrix and table size picker

ID	Requirement	Reference	Ver.
MPK-001	The buttons “Table”, “Matrix with vertical bars”, “Matrix with square brackets” and “Matrix with parenthesis” SHALL open a size picker instead of inserting immediately.	FR-DOC §7	T
MPK-002	The picker SHALL contain a 6 × 6 grid of cells (about 14 px squares, 1 px grey border), a number field “Rows:” and a number field “Columns:” (type number, with spinner arrows).	FR-DOC §7	I
MPK-003	Hovering a grid cell SHALL highlight all cells from the top-left to the hovered cell and SHALL write the corresponding row and column counts into the two fields live.	FR-DOC §7	T
MPK-004	Clicking a cell SHALL insert a table of that size; Enter in either field SHALL insert a table of the typed size; the picker SHALL close afterwards.	FR-DOC §7	T
MPK-005	Typed sizes SHALL be normalised: values < 1, empty and non-numeric → 1; decimals are truncated (2.5 → 2); there SHALL be no upper limit below 100 rows × 50 columns.	FR-DOC §7	T
MPK-006	The inserted structure SHALL be `mtable` (button “Table”) or `mfenced` wrapping an `mtable` with open/close “|”,“[”, or the default parentheses; every cell SHALL be an empty `mtd` .	FR-DOC §7	T
MPK-007	Escape or a click in the editing area SHALL close the picker without inserting; a second click on the same toolbar button SHALL NOT close it.	FR-DOC §7	T
MPK-008	Inserting a matrix over a non-empty selection SHALL replace the selection; with the caret after existing content it SHALL append the matrix after it.	FR-DOC §7	T
MPK-009	(Parity quirk Q-01) After a size is chosen, keyboard focus SHALL be on the document body (typed characters are not received) until the user clicks the editing area; that click places the caret after the matrix. In “improved” mode the focus SHALL return to the editor with the caret in the first cell.	FR-DOC §7, §16	T

4.6 Editing area: input, caret, selection, deletion

ID	Requirement	Reference	Ver.
EDT-001	The editor SHALL receive keyboard input through a visually hidden text input titled “Math input” that is focused whenever the formula display has focus; all printable characters, IME composition and clipboard events SHALL be routed through it.	FR-DOC §24	T
EDT-002	The document model SHALL be an ordered tree of boxes (row, token, fraction, root, scripts, fence, table …) convertible losslessly to Presentation MathML (Appendix D). The empty document SHALL serialise to `<math xmlns="http://www.w3.org/1998/Math/MathML"/>` .	FR-DOC §3, §13	T
EDT-003	Typing a letter SHALL create an identifier token (`mi`), rendered italic when Automatic italic is on; a run of digits SHALL form one number (`mn`); any other printable ASCII symbol SHALL create an operator token (`mo`). Observed examples: “abc”→a b c, “123”→123, “3.14”→3 . 14, “1,000”→1 , 000.	FR-DOC §13	T
EDT-004	With Automatic italic on, the words **sin, cos, tan, log, ln** (case-sensitive, as a run of typed letters) SHALL be merged into one upright identifier; letters following a recognised word remain separate variables (“sinx” → sin x; “abcsin” → a b c sin). The full recognised reserved-word list is Appendix E.	FR-DOC §13, §9	T
EDT-005	Autoformat SHALL be disabled by default (`autoformat=false`): “/”, “^”, “_”, “*”, “>=”, “<=” SHALL remain literal and opening delimiters SHALL NOT be auto-paired. When the parameter is set to true the behaviour of the vendor manual (a/b → fraction, a^b → superscript, a_b → subscript, a*b → ×, >= → ≥, <= → ≤, auto-pairing of ([{) SHALL apply.	FR-DOC §13	T

EDT-006	A typed space SHALL insert a no-break-space operator (`<mo> </mo>`); each press inserts one.	FR-DOC §13, §14	T
EDT-007	Non-ASCII text received from IME or clipboard (accented Latin, Greek, CJK, Hebrew, Arabic, emoji, mathematical symbols) SHALL be accepted; Latin/Greek/CJK/Hebrew letters SHALL create identifiers (adjacent CJK/Hebrew letters inserted in one operation stay in one `mi`), Arabic letters one `mi` per letter (unless Arabic ligature is on), symbols `mo` .	FR-DOC §13	T
EDT-008	The caret SHALL move through the tree in reading order: from the end of “x+1 a/b =2” ArrowLeft visits “2”, “=”, the end of the denominator, the start of the denominator, the end of the numerator, the start of the numerator, “1”, “+”, “x”.	FR-DOC §14, §15	T
EDT-009	ArrowUp / ArrowDown SHALL move between vertically arranged slots (numerator ↔ denominator, over/under scripts, table rows) at the corresponding horizontal position, and between lines created with Enter.	FR-DOC §14, §15, §16	T
EDT-010	Home / End SHALL move to the start / end of the whole line even from inside a structure; Ctrl+Left/Right/Home/End SHALL have no effect.	FR-DOC §14, §15	T
EDT-011	Shift+Arrow SHALL extend or shrink the selection one element at a time (a structure is selected as a unit); Shift+Home / Shift+End SHALL select to the line boundary; Ctrl+A SHALL select the entire formula even when the caret is inside a slot; Shift+click SHALL only move the caret (parity quirk Q-04).	FR-DOC §14, §17	T
EDT-012	Double-click and triple-click SHALL select the whole run of elements on the clicked line; mouse drag SHALL select the covered elements.	FR-DOC §17	T
EDT-013	A click SHALL place the caret at the nearest element boundary; clicks right of or below the content SHALL place it at the end; clicks left of the first element at the start.	FR-DOC §17	T
EDT-014	Typing over a selection SHALL replace the selection with the typed character; Delete SHALL remove the selection.	FR-DOC §14	T
EDT-015	Backspace SHALL delete the previous element; when the previous element is a structure it SHALL first mark the structure (grey highlight, document unchanged) and delete it with its content on the second press (“smart delete”). Backspace in an empty slot SHALL mark the enclosing structure the same way.	FR-DOC §14	T
EDT-016	Forward Delete SHALL remove the following element; at the end of the last cell of a table it SHALL remove the whole table (observed).	FR-DOC §14, §16	T
EDT-017	Enter SHALL insert a line break (`<mspace linebreak="newline"/>`) and start a new line; inside a table cell it SHALL add a new row below; inside an elementary stack it SHALL add a row.	FR-DOC §14, §16, §21	T
EDT-018	Tab SHALL leave the editing area (browser focus moves on); Escape SHALL have no effect in the editing area.	FR-DOC §14	T
EDT-019	Ctrl+/ SHALL insert a fraction, Ctrl+↑ a superscript, Ctrl+↓ a subscript (Ctrl+↑ then Ctrl+↓ builds a sub-and-superscript pair), Ctrl+B / Ctrl+I toggle bold / italic, Ctrl+Shift+Arrow nudge the selected element by 1 px (`mpadded lspace=+1px` / `voffset=-1px`).	FR-DOC §14	T
EDT-020	The following keys SHALL have no effect: Ctrl+0, Ctrl+9, Ctrl+[, Ctrl+{, Ctrl+R, Ctrl+Shift+L, Ctrl+Shift+X, Ctrl+Enter, Alt+Enter, F1, F2, Ctrl+Shift+Z.	FR-DOC §14	T
EDT-021	The formula display SHALL scroll (native scroll-bars) when the formula exceeds 693 × 314 px; it SHALL NOT wrap lines automatically.	FR-DOC §17	T
EDT-022	Nested fractions SHALL wrap inner fractions in `<mstyle displaystyle="true">` as in the reference output (4 nested fractions observed).	FR-DOC §17	T
EDT-023	Right-click SHALL NOT open a custom context menu and SHALL NOT change the document.	FR-DOC §17	T

4.7 Structure insertion semantics (all toolbar templates and symbols)

ID	Requirement	Reference	Ver.
INS-001	Activating a template or symbol with an empty document SHALL insert it at the start; the caret SHALL be placed in the first (leftmost, top) editable slot of a template and after a symbol.	FR-DOC §6	T
INS-002	Activating a **template** (fraction, roots, fences, decorations …) with the caret after existing content SHALL append the template after that content (existing content unchanged); the caret enters the first slot.	FR-DOC §6	T
INS-003	Activating a **template** with a selection SHALL move the selection into the template's first slot (fraction numerator, radicand, fence interior, accent base…).	FR-DOC §6	T
INS-004	Activating a **script** template (superscript, subscript, sub+sup, left scripts) with the caret after a character SHALL make that character the base; with a selection the selection becomes the base; on an empty document an empty base slot is created.	FR-DOC §6	T
INS-005	Activating a **symbol** with a selection SHALL replace the selection; with the caret after content it SHALL append the symbol.	FR-DOC §6	T

INS-006	The behaviour of every individual control (first slot, resulting MathML, differences) SHALL be as listed in the control catalogue (Appendix C).	FR-DOC §6	T

4.8 Text formatting controls

ID	Requirement	Reference	Ver.
FMT-001	Bold (button, Ctrl+B) SHALL toggle bold: on a selection letters get `mathvariant="bold-italic"`, numbers and operators `bold`; with no selection the toggle SHALL apply to following input and the button SHALL show the toggled state; activating again SHALL remove it.	FR-DOC §9	T
FMT-002	Italic (button, Ctrl+I) SHALL toggle italic: numbers become `mathvariant="italic"`; applied to a recognised function name it SHALL split it into italic letters.	FR-DOC §9	T
FMT-003	Automatic italic SHALL be on by default (button toggled). When off, consecutive letters SHALL merge into one upright multi-letter identifier and Greek letters SHALL carry `mathvariant="normal"`.	FR-DOC §9, §12	T
FMT-004	Regular text SHALL start a text run: typed characters go to one `<mtext>` (spaces → no-break spaces, operators literal); applied to a selection it SHALL convert the selection to text; activating again, or inserting a template, SHALL end the run; a symbol inserted during the run joins the text.	FR-DOC §9, §12	T
FMT-005	Colour: the palette SHALL contain 25 swatches in a 5 × 5 grid with the values of Appendix F and a “Color:” hex field pre-filled “#000000”; a swatch SHALL set `mathcolor` on the selection (or on the following input / next inserted structure element); black SHALL remove the attribute; hex input SHALL accept #RRGGBB and #RGB (normalised to upper-case #RRGGBB) and ignore other text; Escape, outside click or a second click on the button SHALL close the palette.	FR-DOC §9	T
FMT-006	Font family SHALL offer: — Font —, Arial, Classic (value “stix”), Courier New, Tahoma, Times New Roman, Verdana; choosing one SHALL set `style="font-family:…"` on the root `math` element for the whole equation regardless of the selection; “— Font —” removes it.	FR-DOC §9	T
FMT-007	Font size SHALL offer: — Size —, 8, 9, 10, 11, 12, 14, 16, 18, 20, 22, 24, 26, 28, 36, 48, 72 px; choosing one SHALL wrap the whole equation in `<mstyle mathsize="Npx">`; “— Size —” removes it.	FR-DOC §9	T
FMT-008	Applying Bold/Italic/colour before inserting a template SHALL style the typed content of every slot; colour before a template SHALL set `mathcolor` on the template element.	FR-DOC §12	T
FMT-009	Buttons “Arabic-Indic numerals” and “Eastern Arabic-Indic numerals” SHALL switch the digit set used for subsequently typed digits (U+0660–0669 / U+06F0–06F9); the mode SHALL persist across document resets until changed (see section 12b of FR-DOC).	FR-DOC §12b	T

4.9 Insert special characters

ID	Requirement	Reference	Ver.
UNI-001	The Ω button SHALL open a panel (225 × 206 px) with a “Code:” text field, a category list and a scrollable 9-column grid of glyph cells (16 px).	FR-DOC §10	I
UNI-002	Categories and glyph counts SHALL be: Symbol 1461, Punctuation 158, Letter 3398, Mark 273, Number 155, Phonetical 294, Other 22 (Unicode categories S, P, L, M, N, F, C of the STIX-covered range).	FR-DOC §10	T
UNI-003	Clicking a glyph SHALL insert it at the caret (replacing a selection) and close the panel.	FR-DOC §10	T
UNI-004	The Code field SHALL filter live by hexadecimal code point (with or without “U+”, 4–6 digits, case-insensitive) and decimal reading; invalid input SHALL show two empty cells; empty input SHALL restore the category; Enter SHALL insert the first result.	FR-DOC §10	T
UNI-005	Escape or an outside click SHALL close the panel; clicking the Ω button again SHALL not close it.	FR-DOC §10	T

4.10 Clipboard, undo and redo

ID	Requirement	Reference	Ver.
CLP-001	Copy and Cut SHALL be disabled when the selection is empty and enabled otherwise; Paste SHALL always be enabled; Undo SHALL be enabled when the history is non-empty; Redo when a step was undone.	FR-DOC §8	T
CLP-002	Copy (button or Ctrl+C) SHALL write the selection as MathML text (`<math xmlns="http://www.w3.org/1998/Math/MathML">…</math>`) to the system clipboard (text/plain); Cut SHALL do the same and delete the selection.	FR-DOC §8	T
CLP-003	Ctrl+V SHALL insert at the caret (replacing a selection): MathML text as structure, LaTeX text via the latex2mathml service (format auto-detected), plain text as typed math characters.	FR-DOC §8, §23	T
CLP-004	(Parity quirk Q-02) The Paste toolbar button SHALL have no visible effect; its tooltip instructs to press Ctrl+V. In improved mode it SHALL read the clipboard through the async Clipboard API when permitted.	FR-DOC §8	T

CLP-005	Undo (button, Ctrl+Z) SHALL revert one input step (a typed character, a structure insertion, a format change); Redo (button, Ctrl+Y) SHALL re-apply; any new edit SHALL clear the redo stack; Ctrl+Shift+Z SHALL have no effect.	FR-DOC §8	T
CLP-006	The history SHALL be cleared when the document is replaced through the API (`setMathML`).	FR-DOC §8, §23	T

4.11 Contextual tab

ID	Requirement	Reference	Ver.
CTX-001	The Contextual tab SHALL show, for the structure containing the caret or selection, exactly the controls listed in Appendix G for that context (matrix 43, piecewise 43, table 37, aligned equations 39, fenced expression 8, operator 2, multi-line 6, elementary stack 17, mspace 3, big operator 2, selected symbol 4, selected number separator 6) and no others.	FR-DOC §19	T
CTX-002	The set of controls SHALL be recomputed on every caret or selection change, whether or not the tab is displayed.	FR-DOC §19	T
CTX-003	Toggle controls (Baseline, Align with axis, Activate solid lines, Align left/center/right, Align rows to baseline ...) SHALL show the state that is currently stored in the structure.	FR-DOC §19	T
CTX-004	Opening bracket, closing bracket and separator selectors SHALL offer: [none], ( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | || (brackets) and , . ; : [space] (separator) and write the `open` , `close` and `separators` attributes of `mfenced` ; ⟨ and ⟩ SHALL be serialised as “<” and “>”, || as “| |”, none as the empty string.	FR-DOC §19	T
CTX-005	Table structure controls SHALL behave as follows relative to the caret cell: Insert row below/above, Insert column at right/left (new empty cells), Remove row, Remove column, Insert/Remove frame (`frame=solid`), Insert/Remove line below and right line (`rowlines` , `columnlines`), Activate solid/dash lines, cell alignment (`columnalign`), row alignment (`rowalign`), Decimal align and decimal separator selector (`wrs:columnalign` , `wrs:decimalseparators`), Relation align, Force equal row heights / column widths (`equalrows` , `equalcolumns`), baseline position (`wrs:valign` , `align`).	FR-DOC §19	T
CTX-006	Length controls SHALL open a small panel (control title + “Value:” input) pre-filled with the current value and focused; Enter SHALL apply it and close the panel; accepted: number+unit (px, em, ex; spaces and case tolerated, “2 px” → “2px”) and a bare number (stored as typed); rejected (document unchanged, panel stays open): letters, negative values, empty.	FR-DOC §20	T
CTX-007	Length controls SHALL map to attributes: column spacing at right → `mtable@columnspacing` (only when the caret column has a column to its right), horizontal / vertical frame spacing → first / second value of `mtable@framespacing` , left / right spacing → `mo@lspace / @rspace` , space width/height/depth → `mspace@width/@height/@depth` ; row spacing below → `rowspacing` (no change observed in the reference for a 2×2 matrix; replicate: no effect).	FR-DOC §19, §20	T
CTX-008	Nudge controls SHALL wrap the selected element in `mpadded` and add 1 px to `lspace` (left/right) or `voffset` (up/down) per activation; they SHALL be identical in effect to Ctrl+Shift+Arrow.	FR-DOC §14, §19	T
CTX-009	The control “Merge/unmerge separators in <mn>” SHALL be offered when a number with a separator is selected (all tokens selected) or the caret is inside a number or next to “,”; with the three tokens `mn mo mn` selected it SHALL merge them into one `mn` (“1.5”, “1,000”); with the caret inside a multi-part `mn` it SHALL split it at its separators into `mn mo mn` .	FR-DOC §20b	T

4.12 Elementary mathematics (stacks and long division)

ID	Requirement	Reference	Ver.
ELM-001	The Matrices tab group “More” SHALL offer: 3 rows stack with line; with line and minus sign; with line and multiplication sign; long division (US/GB style); long division (Spain/Brazil style); long division with remainder (Spain/Brazil); long division with remainder (US/GB); the main tab offers “3 rows stack with line and plus sign” and the long division button.	FR-DOC §21	T
ELM-002	Stacks SHALL serialise as `mstack` (`charalign=center` , `stackalign=right`) with `msrow` , `msline` , `mscarries` ; long division as `mlongdiv` with `msgroup` / `msrow` children (`charalign` , `charspacing` , `stackalign` , optional `longdivstyle`).	FR-DOC §21	T
ELM-003	Inside a stack Enter SHALL add a row and Space SHALL add an indentation (`<none/>`); carry controls SHALL be disabled until “Insert carries in this row” is used; carry location buttons SHALL set `mscarries@location` to nw, n, ne, w, e, sw, s, se; Move to left / right SHALL pad the rows with `none` ; Prepend / Append row and Insert line SHALL add `msrow` / `msline` ; Up diagonal strike SHALL wrap the selection in `menclose notation="updiagonalstrike"` ; “Spacing between digits” SHALL offer 0, 1 ... 10, 12, 15, 20, 25, 30 px and set `charspacing` .	FR-DOC §21	T

4.13 Handwriting recognition mode

ID	Requirement	Reference	Ver.
HND-001	Activating the handwriting strip (click, tap, or keyboard via the “Go to handwritten mode” input) SHALL replace the toolbar and formula display with a handwriting surface of 684 × 415 px (white paper with #d4d4d4 grid lines every 20 px) and a 39 px wide left toolbar (background #e5edf0; keyboard button at its bottom, background #4e5861 with a white keyboard icon); Escape SHALL NOT close it.	FR-DOC §22	T
HND-002	The left toolbar SHALL contain Undo, Redo, Clear (trash), a separator and Help (?); Undo/Redo/Clear SHALL be disabled when not applicable (no strokes / nothing to redo); a keyboard button at the bottom-left SHALL return to the classic editor.	FR-DOC §22	T
HND-003	An empty canvas SHALL show a faint animated hint (squiggle and hand).	FR-DOC §22	T
HND-004	Strokes SHALL be drawn with pointer, touch or pen input in dark blue #284078, about 3 px wide, rounded caps; point coordinates SHALL be recorded per stroke as x/y arrays.	FR-DOC §22	T
HND-005	After the pen is lifted (debounce ≈ 1 s), all strokes SHALL be sent to the recognition service; the returned MathML SHALL be rendered as a small image in the lower-right corner of the canvas (via the render service) and become the document content on switching back.	FR-DOC §22, §23	T
HND-006	Undo SHALL remove the last stroke, Redo restore it, Clear remove all strokes; each of these SHALL trigger a new recognition.	FR-DOC §22	T
HND-007	A dense scribble drawn over existing strokes SHALL turn the covered strokes red and, about 0.6 s later, delete them and the scribble itself.	FR-DOC §22	T
HND-008	Help SHALL play an in-canvas animation that writes “x = y”, scribbles part of it and shows the text “To delete parts of the formula, scribble them out” and a “Powered by wiris”-style logo (the implementer SHALL use own branding); it SHALL not open a new window.	FR-DOC §22	T
HND-009	Switching back to the keyboard editor SHALL transfer the recognised MathML into the formula display; reopening handwriting SHALL regenerate the formula as simulated strokes (Undo disabled for them) and recognise them again.	FR-DOC §22	T
HND-010	While handwriting is open, `getMathML()` SHALL return the classic editor content (not the recognised preview) until the user switches back.	FR-DOC §22	T
HND-011	The stroke-editing controls Edit, Remove point, Insert before, Insert after, Add stroke, Move up, Move down SHALL exist in the DOM as hidden buttons and SHALL NOT be visible (parity with the reference; reserved for future use).	FR-DOC §22, §24	I
HND-012	The recognition service SHALL return JSON `{"result":{"mathml":"…","score":0.0–1.0},"status":"ok"}` and SHALL handle: single-character results, fractions, radicals, scripts, relations, parentheses.	FR-DOC §23	T

4.14 Global modes

ID	Requirement	Reference	Ver.
MOD-001	Right-to-left editing SHALL mirror the entire toolbar (tab order and icons, structure icons, relation symbols), right-align the formula, set `dir="rtl"` on `math`, keep logical typing order and mirror horizontal arrow keys (ArrowLeft moves forward).	FR-DOC §12, §26	T
MOD-002	With RTL on, every control of tabs 1–9 SHALL produce the same MathML as in LTR for the four test states when the selection is made with Shift+ArrowRight (Appendix I lists the differences recorded).	FR-DOC §26	T
MOD-003	Arabic ligature SHALL merge consecutive typed Arabic letters into one `mi`; off → one `mi` per letter.	FR-DOC §12	T
MOD-004	With Automatic italic off the General and Greek tabs SHALL behave as recorded in Appendix I (letters upright, merged).	FR-DOC §26	T
MOD-005	The editor language parameter SHALL select localised tooltips and the RTL/LTR default (languages in `rtlLanguages` default “ar”); the playground uses “en”.	FR-DOC §3	I

4.15 Accessibility

ID	Requirement	Reference	Ver.
ACC-001	The root SHALL have `role="application"`; tabs `role="tablist"/"tab"`; every control a name; disabled state via `aria-disabled`.	FR-DOC §24	T
ACC-002	A polite `aria-live` region SHALL announce the caret location after each move (texts such as “end expression”, “denominator end”, “square root end”, “superscript end”) and a textual description of the formula obtained from the accessibility service.	FR-DOC §24	T
ACC-003	The tab order SHALL be: manual link → selected tab → first control of the selected tab → editing area; roving focus inside grids.	FR-DOC §24	T

ACC-004	All functionality available by pointer SHALL be available by keyboard (tabs, grid, pop-ups, pickers, panels, handwriting entry point); handwriting recognition itself is pointer-only.	FR-DOC §5, §24	T
ACC-005	Colour SHALL NOT be the only means of conveying state: toggle state is also exposed as a border and background change; the minimum contrast of toolbar icons against their background SHALL be ≥ 3:1.	FR-DOC §4	I

4.16 Touch

ID	Requirement	Reference	Ver.
TCH-001	Taps SHALL behave like clicks in the editing area and on controls.	FR-DOC §18	T
TCH-002	When a range is selected on a touch-capable device, two blue (#00aaff) pentagonal selection handles SHALL appear under the start and end of the selection; dragging SHALL adjust the selection; a caret-precision handle SHALL exist for the collapsed caret (hidden in the reference test).	FR-DOC §18	T

4.17 Layout and typography of the formula

ID	Requirement	Reference	Ver.
LAY-001	The default formula font size SHALL be 16 px (line box 18 px high for upright/italic letters, width of “x” 8 px). Script level 1 SHALL be 12 px (75 %), script level 2 10 px; deeper levels SHALL not be smaller than 10 px (measured: x^{2^{3}} → 16/12/10 px).	FR-DOC App. J	T
LAY-002	A fraction SHALL draw a 1 px solid horizontal rule whose width is the wider of numerator and denominator plus about 5 px (a/b: glyph 8.9 px → rule 14 px); numerator and denominator are centred and set at the surrounding size; fractions nested inside a numerator or denominator SHALL be set at 75 % (12 px) while the outer denominator keeps 16 px (measured).	FR-DOC App. J	T
LAY-003	Large operators (∑, ∏, ∫, ⋃, ⋂ …) SHALL be drawn at 150 % (∑ at 24 px) with under/over limits at 12 px (script size), horizontally centred.	FR-DOC App. J	T
LAY-004	Square roots SHALL draw the radical sign plus a 1 px overbar spanning the radicand; the index of an n-th root SHALL be set at script size to the upper left.	FR-DOC App. J	T
LAY-005	Round brackets SHALL be taken from pre-sized glyph sets for 18, 22, 26, 30 and 34 px content height and composed from top/middle/bottom pieces above that; square/curly/angle/vertical-bar fences stretch analogously; fence glyph at base size is 4.3 px wide.	FR-DOC App. J, CSS font list	T
LAY-006	Superscripts SHALL be raised and subscripts lowered relative to the baseline so that the measured offsets of Appendix J are reproduced within 1 px; sub/superscript pairs share the base and stack with the superscript first.	FR-DOC App. J	T
LAY-007	Tables SHALL be laid out with column spacing 0.8 em default, row spacing 0.5 ex default, cell alignment as per `columnalign`; delimiters of matrices stretch to the full table height.	FR-DOC §7, §19	T
LAY-008	Glyph fonts SHALL be STIX-compatible: letters/digits (regular + italic), symbols, script, Fraktur, double-struck, brackets, horizontal braces/arrows and RTL-mirrored symbols; Arabic letters from an Arabic maths font.	FR-DOC App. F fonts	I

4.18 Table / matrix navigation and editing

ID	Requirement	Reference	Ver.
TBL-001	In a table the caret SHALL traverse cells in reading order; ArrowLeft from the end of a 2×2 parenthesised matrix visits: end-inside-fence, cell (2,2), (2,1), (1,2), (1,1), start-inside-fence, before the fence (observed ×1… ×7).	FR-DOC §16	T
TBL-002	ArrowRight from the last cell SHALL leave the table but stay inside the fence; ArrowUp moves to the cell above (same column); ArrowDown in the last row stays in the cell; Home/End jump to the start/end of the line (before/after the fence).	FR-DOC §16	T
TBL-003	Enter in a cell SHALL insert a row below the current row and put the caret in the new row (2×2 → 3×2).	FR-DOC §16	T
TBL-004	Backspace in a cell deletes the previous element; at the start of the matrix content the first Backspace marks the structure (document unchanged) and the second removes the whole matrix.	FR-DOC §16	T
TBL-005	(Q-07) Forward Delete at the end of the last cell SHALL remove the whole table (exact parity).	FR-DOC §16	T
TBL-006	Tab SHALL leave the editor from inside a cell (no cell-to-cell Tab navigation).	FR-DOC §16	T

5 Services and non-functional requirements

5.1 Backend services contract

ID	Requirement	Reference	Ver.
SVC-001	All services SHALL be exposed under a configurable base URL (default `/editor-services/`) and accept GET and POST (`application/x-www-form-urlencoded`); POST SHALL be used by the client for MathML/LaTeX payloads.	FR-DOC §3, §23	T
SVC-002	**mathml2accessible**: parameters `mml`, `lang`; response `text/plain; charset=UTF-8` with an English description in the style "x plus 1 fraction numerator a greater than over denominator blank end fraction"; empty fractions/slots SHALL be spoken as "blank"; an empty formula SHALL return "blank". Recommended implementation: Speech Rule Engine (MathSpeak or ClearSpeak rules) with a post-processor producing the same phrases.	FR-DOC §23, §24	T
SVC-003	**latex2mathml**: parameters `latex`, `pasteformat=auto` (optional); response Presentation MathML as text; \frac{a}{b} → mfrac, \sqrt{x^2+1} → msqrt(msup, +, 1).	FR-DOC §8, §23	T
SVC-004	**mathml2latex**, **mathml2content**, **content2mathml**: parameter `mml`; responses as text.	FR-DOC §3	T
SVC-005	**accessible2mathml**: parameter `accessible` (+ `lang`); inverse of SVC-002 for the supported phrase grammar.	FR-DOC §3	T
SVC-006	**render / render.svg / render.png / render.pdf / render.eps**: parameters `mml` or `latex`, `format`, `zoom`, `dpi` (default 96), `backgroundColor`, `centerbaseline`, `lang`; response the image; PNG response SHALL carry the baseline offset (header `X-Baseline` or metadata) so that clients can align the image with text.	FR-DOC §22, §23	T
SVC-007	**hand/recognize**: POST JSON `{"x":[[…]],"y":[[…]]}` (one array per stroke) plus optional `t` (timestamps) and context; response as HND-012; latency target ≤ 800 ms p95 for ≤ 20 strokes.	FR-DOC §22, §23	T
SVC-008	Every service SHALL return HTTP 200 with the error embedded unless `httpstatus=true` is supplied, in which case failures SHALL return HTTP 5xx (compatible with the reference services).	FR-DOC §23	T
SVC-009	The client SHALL work with the vendor-hosted endpoints `https://www.wiris.net/demo/editor/<service>` or any self-hosted implementation of this contract by changing a single configuration value.	FR-DOC §3	I
SVC-010	Telemetry (`POST …/tick` with `wrs-license`, `wrs-deployment`, `wrs-cause`) is OPTIONAL and SHALL be disabled by default in new implementations.	FR-DOC §23	I
SVC-011	The client SHALL issue a `mathml2accessible` request only when the formula text changed since the last description and the editor lost keyboard focus to a toolbar control or the contextual tab was opened (observed triggers); it SHALL debounce ≥ 250 ms and cancel superseded requests.	FR-DOC §23	T

5.2 Non-functional requirements

ID	Requirement	Reference	Ver.
NFR-001	Initial load: JavaScript ≤ 400 KB gzip for the core, fonts ≤ 1.5 MB total, time-to-interactive ≤ 2.5 s on a 4G profile; fonts SHALL be lazily loaded (STIX symbols/letters first).	FR-DOC perf	T
NFR-002	Input latency: keystroke to repaint ≤ 50 ms p95 for documents ≤ 500 elements; structure insertion ≤ 100 ms; handwriting stroke rendering ≥ 60 fps.	FR-DOC perf	T
NFR-003	Supported browsers: last two stable versions of Chrome, Edge, Firefox, Safari (macOS/iOS) and Chrome Android; no IE.	FR-DOC compat	T
NFR-004	The component SHALL not leak global CSS/JS: all styles scoped under `.wrs_editor` (or Shadow DOM), a single global namespace, no dependency on host frameworks.	FR-DOC quality	I
NFR-005	Accessibility: WCAG 2.2 AA for the shell; keyboard operability per section ACC; reduced-motion media query SHALL disable the handwriting hint animation.	FR-DOC a11y	T
NFR-006	Security: Content-Security-Policy compatible (no inline script/eval), all services reachable only over HTTPS, request size limit 1 MB, rate-limit 60 req/min/IP, input validation of MathML/LaTeX, no execution of MathML; handwriting strokes SHALL not be stored and logs SHALL not contain user content.	FR-DOC sec	T
NFR-007	Privacy: no cookies; optional telemetry only with explicit opt-in; user content leaves the browser only for the services listed in SVC.	FR-DOC sec	I
NFR-008	Internationalisation: all UI strings (tooltips, labels, panels) in message catalogues; RTL languages mirror the toolbar; numerals per locale parameters (arabicIndicLanguages, easternArabicIndicLanguages, europeanLanguages).	FR-DOC i18n	T
NFR-009	Testability: every requirement of this document SHALL have an automated test (Playwright for UI, unit tests for model/serialiser, contract tests for services); the reference evidence of FR-DOC SHALL be the golden data.	FR-DOC qa	T

NFR-010	Reliability: service failures SHALL never block editing; failed services show an inline retry and the editor continues offline.	FR-DOC rel	T
NFR-011	Maintainability: TypeScript strict mode, ≥ 85 % unit-test coverage of the model and serialiser, semantic versioning, public API documented with TypeDoc.	FR-DOC quality	I
NFR-012	Pixel parity: screenshots of the toolbar tabs, pop-ups and dialogs SHALL differ from the reference by ≤ 1 % of pixels at 100 % zoom and device scale factors 1 and 2 (anti-aliasing tolerance).	FR-DOC qa	T

6 External interface requirements

6.1 User interface — geometry and design tokens

The values below were measured on the reference (device scale 1, widget at 725 × 415 px) with coordinates relative to the widget's top-left corner. They are normative for pixel parity (UI-001…UI-012, NFR-012). The complete token list is Appendix H.

Element	x, y, w, h (px)	Style
Editor root	0, 0, 725, 415	`bg rgba(0, 0, 0, 0) · border-top 0px none rgb(0, 0, 0) · radius 0px · font 16px/400`
Toolbar block	0, 0, 725, 97	`bg rgba(0, 0, 0, 0) · border-top 0px none rgb(0, 0, 0) · radius 0px · font 11px/700`
Tab strip (header)	0, 1, 725, 25	`bg rgba(0, 0, 0, 0) · border-top 0px none rgb(0, 0, 0) · radius 0px · font 11px/700`
Selected tab	0, 1, 44, 25	`bg rgb(242, 243, 245) · border-top 1px solid rgb(119, 142, 154) · radius 5px 5px 0px 0px · font 11px/400`
Non-selected tab (example #2)	46, 4, 44, 22	`bg rgb(85, 110, 123) · border-top 0px none rgb(255, 255, 255) · radius 5px 5px 0px 0px · font 11px/400`
Contextual tab (not selected)	414, 4, 44, 22	`bg rgb(220, 158, 158) · border-top 0px none rgb(255, 255, 255) · radius 5px 5px 0px 0px · font 11px/400`
Help link	705, 4, 18, 18	`bg rgba(0, 0, 0, 0) · border-top 0px none rgb(24, 33, 47) · radius 0px · font 11px/700`
Toolbar panel	3, 28, 719, 66	`bg rgba(0, 0, 0, 0) · border-top 0px none rgb(0, 0, 0) · radius 0px · font 11px/700`
Formula display	0, 99, 695, 316	`bg rgb(255, 255, 255) · border-top 1px solid rgb(150, 150, 150) · radius 0px · font 16px/400`
First toolbar button	3, 28, 33, 33	`bg rgba(0, 0, 0, 0) · border-top 2px solid rgba(0, 0, 0, 0) · radius 0px · font 11px/700`

6.2 Software interfaces

Host API: chapter 9. Service API: section 5.1. Browser APIs used: DOM, Pointer Events, Clipboard API (async + legacy `execCommand` fallback), Canvas 2D (handwriting), `beforeinput`/composition events, `matchMedia`, `ResizeObserver`, FontFace API.

6.3 Communications interfaces

HTTPS, JSON and form-encoded bodies, CORS enabled for configured origins, `Content-Type: text/plain; charset=UTF-8` for text results and `image/svg+xml`, `image/png` for rendering.

6.4 Fonts and assets

- Conv_a (letters/digits, stix_regular_letters_numbers)
- Conv_math1 (symbols, stix_symbols), Conv_math1RTL (mirrored symbols for RTL)
- Conv_horizontal (horizontal braces/arrows), Conv_brackets / Conv_brack_sm (large brackets)
- Conv_script, Conv_fraktur, Conv_doubleStruck (alphabet variants)
- Conv_roundBrackets1854…3454 (parenthesis sizes), moe-arabic (Arabic letters)

Math glyphs: STIX Two Math / STIX fonts (SIL Open Font License). UI icons: to be drawn by the implementer following Appendix C icon sheet (each icon is a 22–33 px raster/vector glyph; "icons.png" of the reference is a proprietary sprite).

7 Data requirements

7.1 Document format

The document is a single Presentation MathML `<math>` tree. The reference editor emitted 6606 different documents during testing; the element and attribute inventory of Appendix D is the closed set that the component SHALL produce and accept (other markup is preserved on a best-effort basis when pasted).

7.2 Serialisation rules (normative)

1. Root: `<math xmlns="http://www.w3.org/1998/Math/MathML">` ; the empty document is `<math xmlns="…"/>` (self-closing). Editor extension namespace `xmlns:wrs="http://www.wiris.com/xml/mathml-extension"` is declared on the root only when a `wrs:` attribute is present.
2. No whitespace between elements; empty elements are self-closed (`<mrow/>` , `<mtd/>` , `<msqrt/>`).
3. Non-ASCII characters are written as numeric character references with upper-case hexadecimal (`×` , `π`); the five XML special characters as `< > & "` (apostrophe is literal).
4. Letters: `<mi>` one per letter (merged only for recognised function names, Automatic-italic-off runs, Arabic ligature runs and inserted CJK/Hebrew strings); digit runs: one `<mn>` ; everything else `<mo>` ; text mode: `<mtext>` .
5. Empty slots serialise as `<mrow/>` ; the base slot of a script created on an empty document is `<maction actiontype="argument"><mrow/></maction>` .
6. Whole-equation style is stored on the root (`style="font-family:…"` , `dir="rtl"`) or as a single wrapping `<mstyle mathsize="…">` .

7.3 Canonical outputs (selection)

Action	MathML
Fraction on empty editor	`<math><mfrac><mrow/><mrow/></mfrac></math>`
Square root with x selected	`<math><msqrt><mi>x</mi></msqrt></math>`
Superscript after x	`<math><msup><mi>x</mi><mrow/></msup></math>`
Parentheses	`<math><mfenced><mrow/></mfenced></math>`
Square brackets	`<math><mfenced open="[" close="]"><mrow/></mfenced></math>`
2×2 matrix (parentheses)	`<math><mfenced><mtable><mtr><mtd/><mtd/></mtr><mtr><mtd/><mtd/></mtr></mtable></mfenced></math>`
Bold x	`<math><mi mathvariant="bold-italic">x</mi></math>`
Red x	`<math><mi mathcolor="#FF0000">x</mi></math>`
Regular text	`<math><mtext>hello world</mtext></math>`
Line break	`<math><mi>a</mi><mspace linebreak="newline"/><mi>b</mi></math>`
RTL	`<math dir="rtl"><mi>a</mi></math>`
Font size 24px	`<math><mstyle mathsize="24px"><mi>x</mi></mstyle></math>`

Full per-control expectations: Appendix C.

8 Architecture and technology (design constraints)

8.1 Architectural principles

1. **Model–command–view:** an immutable document tree (MathML-shaped) is changed only through commands (insert template, insert symbol, delete, format …); a command stack provides undo/redo; every command returns a new tree and the new caret/selection.
2. **Layout independent of rendering:** a layout engine computes boxes (width, height, baseline) from OpenType MATH constants of STIX Two Math; the renderer positions DOM nodes (or SVG) from those boxes and paints caret, selection, highlights and empty-slot boxes as overlays.
3. **Input isolation:** one hidden input element receives keyboard, IME and clipboard events; pointer events are handled on the display; handwriting uses its own canvas module.
4. **Declarative toolbar:** tabs, groups, controls, pop-ups and contextual controls are data (JSON) generated from Appendix C/G; the same data drives rendering, keyboard navigation, tooltips and tests.
5. **Services behind adapters:** all network access is through a typed `Services` interface with default HTTP implementation, so vendors or self-hosted engines can be swapped (SVC-009).
6. **Parity switch:** quirks of Appendix A are isolated behind `parity`.

8.2 Recommended technology stack

Concern	Choice	Rationale
Language / build	TypeScript 5 (strict), Vite 6 + Rollup library mode, ESM + UMD bundle, no runtime framework dependency	Type safety, small bundle, embeddable via script tag or npm.
UI shell	Custom Elements (`<math-editor>`) with Shadow DOM or scoped CSS (`.wrs_editor`); Lit 3 allowed for dialogs	Host isolation (NFR-004).
Layout / fonts	OpenType MATH table via `opentype.js` (or HarfBuzz WASM) with STIX Two Math (OFL), subset per script	Typographically correct fractions/scripts/radicals; metric parity.
Rendering	Absolutely positioned DOM spans for glyph runs + SVG for rules/radicals/stretchy delimiters; Canvas 2D for handwriting	Matches reference DOM model; crisp at all zoom levels.
State / undo	Immutable tree (structural sharing), command objects, inverse patches	Cheap history, deterministic tests.
Input	`beforeinput` , `compositionstart/end` , hidden `<input title="Math input">`	IME and mobile keyboards.
Accessibility	ARIA APG patterns (tablist, toolbar with roving tabindex), `aria-live="polite"` region	ACC-001…005.
Services runtime	Node.js 22 LTS, Fastify 5, zod validation, pino logging, Docker, stateless horizontal scaling	Small, fast, easily self-hosted.
LaTeX ↔ MathML	MathJax 3 TeX input → MathML (`mathjax-full`); `mathml-to-latex` (or a custom serializer for the editor's closed MathML profile)	The profile of Appendix D is small, so a deterministic serializer is feasible and testable.
Spoken text	Speech Rule Engine (SRE 4) with a phrase post-processor matching SVC-002	Industry standard for MathML speech.
Images	MathJax SVG output (or own layout engine server-side) → `resvg-js` for PNG, `svg-to-pdfkit` for PDF	One layout truth for SVG/PNG/PDF.
Handwriting recognition	Adapter interface `Recognizer.recognize(strokes): {mathml, score}` with a self-hosted model (sequence-to-sequence OCR for online ink) or a commercial recognition API behind the adapter	Recognition is the only ML component; isolate it.
Quality	Vitest (unit), Playwright (UI/e2e, screenshot parity), pact-style contract tests for services, axe-core (a11y), Lighthouse CI (perf)	NFR-009, NFR-012.
CI/CD, ops	GitHub Actions, semantic-release, container registry, OpenTelemetry traces, structured logs without content	NFR-006/007/011.

8.3 Security design

- Treat MathML/LaTeX as untrusted: parse with a strict parser, enforce the closed element/attribute set (Appendix D), cap depth (≤ 64) and node count (≤ 20 000), reject DOCTYPE/entities.
- Services: HTTPS only, CORS allow-list, 1 MB body limit, per-IP rate limit, no persistence of payloads, timeouts (3 s conversion, 5 s render, 8 s recognition), SSRF-safe (no remote fetch), dependency scanning.
- Output sanitisation before DOM insertion (OUT-006) and CSP-compatible output (no inline scripts, SVG without script/foreignObject).

9 Output extraction layer (addition to the reference)

This chapter is the only functional addition to the reference playground. It does **not** alter the widget (725 × 415 px, behaviour of chapters 4–5). Requirements OUT-001…008 (section 9.5) define it normatively; the rest of the chapter gives the design and the formats researched from the vendor documentation.

9.1 Output formats

Format	How obtained	Use	Notes / best practice
Presentation MathML	Local: `getMathML()`	Canonical storage, rendering in browsers (MathML Core), exchange	Store this as the source of truth; always keep the namespace; round-trip safe (OUT-008).
MathML + semantics/annotation	Local: `getMathML({semantics:true})`	Carry LaTeX/strokes with the equation	`<semantics><mrow/><annotation encoding="application/x-tex">`.
LaTeX	Service `mathml2latex` (or local serializer)	Authoring systems, LMS, publishing	Meaning-preserving; some editor-specific layout (nudges, spacing) is not representable.
Content MathML	Service `mathml2content`	Evaluation, CAS	Only for semantically unambiguous input.
SVG	Service `render.svg` / local	Vector image in HTML/PDF, best quality	**Recommended image format**; embed fonts or paths; set `role="img"` and `aria-label` from the accessible text.
PNG	Service `render.png` (dpi, zoom)	Email, legacy systems	Use dpi 192+ for retina; keep the baseline offset to align with text.
PDF / EPS	Service `render.pdf` / `render.eps`	Print workflows	Vector; fonts embedded.
Accessible text	Service `mathml2accessible`	`alt` / `aria-label`, screen-reader fallback	Pass the user language (`lang`); regenerate when the formula changes.
Plain text	Local linearisation	Search indexing, previews	Unicode math, e.g. “x+1=2”, “√(x)”, “a/b”.
Clipboard	Editor Copy / Output panel Copy	Paste into other editors	MathML text + HTML + plain (OUT-005).

9.2 JavaScript API (normative signatures)

```
interface EditorOptions {
  language?: string;                // default 'en'
  autoformat?: boolean;             // default false (EDT-005)
  toolbarSize?: 1 | 1.5 | 2;        // default 1 (UI-006)
  detectHand?: boolean;             // default true (HND-*)
  fontStyle?: 'initial'|'normal'|'italic';   // default 'initial'
  reservedWords?: string[];         // default Appendix E
  globalStyles?: boolean;           // default true
  rtlLanguages?: string[]; arabicIndicLanguages?: string[]; easternArabicIndicLanguages?: string[];
  parity?: 'exact'|'improved';      // default 'exact' (Appendix A)
  services?: { baseUrl: string; headers?: Record<string,string>; telemetry?: boolean };
  mml?: string;                     // initial content
}
interface MathEditor {
  insertInto(el: HTMLElement): void;  destroy(): void;
  getMathML(o?: { semantics?: boolean; annotations?: boolean }): string;
  setMathML(mml: string, o?: { focus?: boolean }): Promise<void>;   // clears history
  isEmpty(): boolean;  isReady(): boolean;  focus(): void;  blur(): void;
  undo(): void;  redo(): void;  reset(): void;
  getLaTeX(): Promise<string>;  getAccessibleText(lang?: string): Promise<string>;
  getSVG(o?: RenderOptions): Promise<string>;  getPNG(o?: RenderOptions): Promise<Blob>;
  getPlainText(): string;  getImageDataURL(format: 'svg'|'png', o?: RenderOptions): Promise<string>;
  setParams(p: Partial<EditorOptions>): void;  getVersion(): string;
  on(ev: 'ready'|'change'|'focus'|'blur'|'tabchange'|'handstart'|'handend', cb: (e: any) => void): () => void;
}
interface RenderOptions { zoom?: number; dpi?: number; background?: string; color?: string; fontSize?: string }
```

Compatibility: `getMathML`, `setMathML`, `isFormulaEmpty` (alias of `isEmpty`), `focus`, `blur`, `getVersion`, `setParams`, `insertInto` keep the vendor names (see vendor JsEditor/EditorInterface reference) so existing integrations work.

9.3 Output panel (optional UI, outside the widget)

A strip of 725 px width and 180 px height directly below the widget: a segmented control (MathML · LaTeX · SVG · PNG · Accessible text · Plain text), a read-only content area (monospace for code, image preview for SVG/PNG), a **Copy** button, a **Download** button for images, a status line ("Updated 12:01:03 · 1.2 KB") and an inline error with **Retry**. It updates on `change` (debounce 150 ms), shows a skeleton while a service request is in flight and never blocks the editor. Keyboard: tabs by arrow keys, Copy by Enter.

9.4 Integration sequence

1. Host loads the bundle and calls `MathEditor.create(options)`, `insertInto(container)`.
2. Editor emits `ready`; host optionally calls `setMathML(stored)`.
3. User edits → `change` events (debounced).
4. On save: `getMathML()` (sync) and, in parallel, `getAccessibleText(lang)` + `getSVG()` for display fallbacks.
5. Host stores MathML (source of truth) and derived artefacts; renders with MathML + SVG fallback.

9.5 Output extraction (additional to the reference playground)

ID	Requirement	Reference	Ver.
OUT-001	The component SHALL expose `getMathML(options?)` returning Presentation MathML as a string with the namespace on the root; options: `semantics` (wrap in `<semantics>` with a LaTeX `<annotation encoding="application/x-tex">`), `annotations` (include handwriting strokes annotation when available).	FR-DOC API	T
OUT-002	The component SHALL expose asynchronous `getLaTeX()`, `getAccessibleText(lang?)`, `getSVG(options?)`, `getPNG(options?)` (options: zoom, dpi, background, color, fontSize), `getPlainText()` (linearised Unicode math, e.g. "x+1=2") and `getImageDataURL(format)`.	FR-DOC API	T
OUT-003	The component SHALL emit events `ready`, `change` (debounced 100 ms, payload MathML + isEmpty), `focus`, `blur`, `tabchange`, `handstart`, `handend` and SHALL expose `setMathML`, `isEmpty`, `focus`, `blur`, `undo`, `redo`, `reset` (clear history), `destroy`.	FR-DOC API	T
OUT-004	Optionally the integrator SHALL be able to show an **Output panel** directly below the editor (outside the 725 × 415 px widget so the widget stays pixel-identical): tabs MathML | LaTeX | SVG | PNG | Accessible text | Plain text; each tab SHALL show the live value, a Copy button (clipboard) and, for images, a Download button; updates SHALL be debounced 150 ms; errors SHALL be shown inline with a retry action.	FR-DOC UI	T
OUT-005	MathML copied from the panel or from the editor Copy action SHALL be written to the clipboard simultaneously as `text/plain` (MathML source), `text/html` (MathML wrapped in a `<span>` with an `<img>` fallback) and `application/mathml+xml` where supported.	FR-DOC UI	T
OUT-006	When MathML is inserted into the host DOM by the integrator, it SHALL be sanitised with an allow-list of the elements and attributes of Appendix D (no scripts, no event handlers, no `href` / `xlink:href`, no `style` beyond `font-family`).	FR-DOC SEC	T
OUT-007	Recommended storage format: Presentation MathML (canonical) plus LaTeX and SVG as derived caches; the accessible text SHALL be stored as `alt` / `aria-label` of the rendered image.	FR-DOC BP	I
OUT-008	Round-trip requirement: for every document produced by the editor, `setMathML(getMathML())` followed by `getMathML()` SHALL return an identical string.	FR-DOC API	T

10 Verification and validation

10.1 Strategy

- **Golden-master testing:** the evidence files of FR-DOC (MathML per control and state, scenario steps with screenshots, network traces) are the oracle; the implementation SHALL reproduce them byte-for-byte for MathML and within NFR-012 for pixels.
- **Layers:** unit tests (model, commands, serialiser, layout metrics), component tests (toolbar data, dialogs), end-to-end tests (Playwright, the same scripts used for the reference), service contract tests, accessibility (axe, keyboard-only script) and performance (NFR-001/002).
- **Differential run:** the same Playwright scripts SHALL be run against the reference URL and the implementation; the report lists any MathML/screenshot difference. Zero differences in “exact” parity mode is the exit criterion.

10.2 Acceptance test suites

Suite	Content	Evidence / script	Requirements	Pass criterion
TS-01	Toolbar control crawl (4 states × 687 controls + 13 special)	`behav.js → out/tab0…8.json`	TB-*, PUP-004	For every control: MathML in states A (empty), A2 (typed 7), B (after x), C (x selected) equals the golden value.
TS-02	Chrome, states, tab navigation	`s01_layout, s02_tabs_kbd`	CHR-*, TAB-*	Hover/pressed/disabled/focus; Alt+digit, Alt+arrows, grid keys.
TS-03	Matrix picker	`s09_matrix_picker, s20_matrix_nav`	MPK-*	Hover preview, sizes 1…100, normalisation, close behaviour, focus quirk, keyboard navigation inside matrix.
TS-04	Clipboard and history	`s07_clipboard_undo`	CLP-*	Button states, clipboard payloads, paste of MathML/LaTeX/text, undo/redo.
TS-05	Formatting and colour	`s05_format, s22_swatches, s12_modes`	FMT-*	Bold/italic/auto-italic/text/colour (25 swatches, hex), fonts, sizes, mode interplay.
TS-06	Special characters	`s06_unicode`	UNI-*	Categories and counts, code search, insertion, closing.
TS-07	Typing and recognition	`s03_typing, s26_reserved_words, s25_digit_modes`	EDT-003…007, FMT-009	Token creation, function names, autoformat off, Unicode, digit modes.
TS-08	Keyboard, caret, mouse, scrolling	`s08_keyboard_edit, s21_caret_probe, s13_mouse_scroll`	EDT-008…023	Caret traversal table, selection, smart delete, Enter/Space/Tab/Esc, shortcuts.
TS-09	Contextual tab	`s10b_ctx_*, s23_length_fields, s16_elementary`	CTX-*, ELM-*	Control sets per context; every control; length values; carries; stacks.
TS-10	Handwriting	`s11_handwriting, s24_hand_edit`	HND-*	Open/close, strokes, recognition requests, undo/redo/clear, scribble erase, help.
TS-11	Pop-ups	`s19_more_popup`	PUP-*	Open, keyboard, escape, switch, toggle.
TS-12	Modes	`out_rtl/*, out_noitalic/*`	MOD-*	Whole-toolbar re-crawl in RTL and automatic-italic-off.
TS-13	Accessibility	`s14_a11y_net`	ACC-*	ARIA tree, live region texts, tab order, spoken text.
TS-14	Services and network	`s17_network_ui_only`	SVC-*	Which user action triggers which request, payloads, responses.
TS-15	Touch and responsive	`s18_touch, s15_responsive`	TCH-*, UI-012	Selection handles, hidden below 1275 px host width.

10.3 Exit criteria

1. All suites pass in Chrome, Firefox, Safari and Edge (latest) at device scale 1 and 2.

2. Differential run: 0 MathML differences over the 2748 control cases and 1207 scenario steps; ≤ 1 % pixel difference on toolbar/pop-up/dialog screenshots.
3. Accessibility: 0 critical axe violations; full keyboard script passes.
4. Security: dependency audit clean, OWASP ASVS L2 checklist for services.

10.4 Traceability matrix

Requirement area	FR-DOC sections	Evidence datasets	Test suite
UI, CHR	§2, §4, §25	`s01_layout, s15_responsive, style probe`	TS-02, TS-15
TAB	§5	`s02_tabs_kbd`	TS-02
PUP	§6, §11	`out/*_popup, s19_more_popup`	TS-01, TS-11
MPK	§7, §16	`s09_matrix_picker, s20_matrix_nav`	TS-03
EDT, INS	§6, §13–§17	`s03_typing, s08, s13, s21, s26`	TS-01, TS-07, TS-08
FMT, UNI	§9, §10, §12b	`s05, s06, s12, s22, s25`	TS-05, TS-06, TS-07
CLP	§8	`s07_clipboard_undo`	TS-04
CTX, ELM	§19–§21	`s10b_ctx_*, s16, s23`	TS-09
HND	§22	`s11, s24`	TS-10
MOD	§12, §26	`s12, out_rtl, out_noitalic`	TS-12
ACC, TCH	§18, §24	`s14, s18`	TS-13, TS-15
SVC, OUT	§3, §23	`s17_network_ui_only`	TS-14
TB-*	§6	`out/tab*.json`	TS-01

Appendix A — Parity decisions (reference quirks)

Parity mode `exact` (default) reproduces the reference; `improved` applies the listed fix. Both modes SHALL be covered by tests (suite TS-xx with `parity` parameter).

ID	Observed in the reference	exact	improved	Evidence
Q-01	After choosing a size in the matrix/table picker, keyboard focus is left on ; typed characters are lost until the user clicks the editing area, and that click puts the caret after the matrix.	exact: replicate	improved: focus returns to the editor, caret in first cell	FR-DOC §7, §16
Q-02	The Paste toolbar button does nothing visible (tooltip says “Press Ctrl+V”).	exact: inert	improved: reads clipboard via async Clipboard API when permission is granted, otherwise shows the same hint as a toast	FR-DOC §8
Q-03	Shift+click only moves the caret; it does not extend the selection.	exact: replicate	improved: Shift+click extends the selection	FR-DOC §17
Q-04	Ctrl+A inside a fraction slot selects the whole formula, not just the slot.	exact: replicate (this is also conventional)	same	FR-DOC §14
Q-05	An invalid value in a contextual length field leaves the small panel open with the rejected text and no message.	exact: replicate	improved: panel marks the field invalid (aria-invalid, red border) and shows a hint	FR-DOC §20
Q-06	“Row spacing below” had no effect on the 2×2 matrix.	exact: replicate	improved: sets mtable@rowspacing	FR-DOC §19
Q-07	Forward Delete at the end of the last table cell removes the whole table.	exact: replicate	improved: Delete at the end of a cell is a no-op	FR-DOC §16
Q-08	The digit style set by “Arabic-Indic numerals” / “Eastern Arabic-Indic numerals” persists even when the document is reset programmatically.	exact: replicate	improved: mode is reset by reset()/setMathML()	FR-DOC §12b
Q-09	The Undo button is enabled on a pristine document (but disabled after the history is cleared by the API).	exact: enabled at load	improved: disabled until the first change	FR-DOC §8
Q-10	After a font-family or font-size change the MathML occasionally read as empty for a short time in one automated run (not reproducible in isolation).	exact: n/a (race in reference)	required: the model SHALL be consistent synchronously after the change	FR-DOC §9
Q-11	Handwriting “Help” plays a demo that writes “x = y” and leaves the previous strokes untouched; there is no close button (it ends by itself).	exact: replicate	same	FR-DOC §22
Q-12	Typed superscript/subscript on an empty document creates an empty base slot (`maction actiontype="argument"`) instead of a placeholder-only script.	exact: replicate (serialisation includes the maction wrapper)	same	FR-DOC §6

Appendix B — Evidence datasets and reproduction

All behaviour statements are backed by machine-readable evidence produced by the harness (Node.js + Playwright, Chrome). To reproduce or to run the differential comparison, execute the scripts below against any URL that hosts the editor.

Dataset / script	Content
`out/tab0…8.json (behav.js)`	Every toolbar control and pop-up item: states A, A2, B, C (MathML + screenshots + icon crop + coordinates)
`out_rtl/*, out_noitalic/* (behav_mode.js)`	Same crawl with RTL / automatic italic off
`sc/s01…s26 (*.js)`	Scenarios: chrome, tabs, typing, formatting, clipboard, special characters, matrix picker and navigation, keyboard, caret probes, mouse, touch, contextual controls, length fields, elementary math, handwriting, network, accessibility, responsive, digit modes, reserved words
`sc/s17_calls.json`	UI-only network trace with payloads and responses
`mathml_profile.json`	Element/attribute inventory of all produced MathML
`style_probe.json, template.css`	Computed geometry and style of the reference
`docs/*, docs2/*`	Vendor documentation excerpts used for terminology

Appendix C — Control catalogue (700 control requirements)

Each row is a requirement: **“Activating the control with the tooltip … SHALL yield the MathML shown for states A, B and C and place the caret as stated.”** Coordinates are relative to the widget top-left (x, y of the control box and its size); for pop-up items the position inside the pop-up follows the order of the table (single column for the Symbols/Arrows/Matrices/Big-operator/Calculus pop-ups; multi-row blocks for the Greek tab). State A = empty editor; B = after typing “x”, caret at end; C = “x” typed and selected.

C.1 Tab 1 — General (Alt+1)

Main panel (38 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-1-001		Fraction (Ctrl+/)	3, 28, 33, 33	**A** `<math><mfrac><mrow/><mrow/></mfrac></math>` **B** `<math><mi>x</mi><mfrac><mrow/><mrow/></mfrac></math>` **C** `<math><mfrac><mi>x</mi><mrow/></mfrac></math>` Empty editor → inserts frac(□ ⁄ □). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → frac(7 ⁄ □)).
TB-1-002		Square root	36, 28, 33, 33	**A** `<math><msqrt/></math>` **B** `<math><mi>x</mi><msqrt/></math>` **C** `<math><msqrt><mi>x</mi></msqrt></math>` Empty editor → inserts √(□). The caret is placed inside the square root; the next typed character goes there (typed “7” → √(7)).
TB-1-003		Bevelled fraction	3, 61, 33, 33	**A** `<math><mfrac bevelled="true"><mrow/><mrow/></mfrac></math>` **B** `<math><mi>x</mi><mfrac bevelled="true"><mrow/><mrow/></mfrac></math>` **C** `<math><mfrac bevelled="true"><mi>x</mi><mrow/></mfrac></math>` Empty editor → inserts bevelledFrac(□ ⁄ □). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → bevelled Frac(7 ⁄ □)).
TB-1-004		Root	36, 61, 33, 33	**A** `<math><mroot><mrow/><mrow/></mroot></math>` **B** `<math><mi>x</mi><mroot><mrow/><mrow/></mroot></math>` **C** `<math><mroot><mi>x</mi><mrow/></mroot></math>` Empty editor → inserts root(radicand □, index □). The caret is placed in the radicand of the root; the next typed character goes there (typed “7” → root(radicand 7, index □)).
TB-1-005		Superscript (Ctrl+Up)	72, 28, 33, 33	**A** `<math><msup><maction actiontype="argument"><mrow/></maction><mrow/></msup></math>` **B** `<math><msup><mi>x</mi><mrow/></msup></math>` **C** `<math><msup><mi>x</mi><mrow/></msup></math>` Empty editor → inserts □^{□}. The caret is placed in the base of the superscript; the next typed character goes there (typed “7” → 7^{□}).
TB-1-006		Subscript (Ctrl+Down)	72, 61, 33, 33	**A** `<math><msub><maction actiontype="argument"><mrow/></maction><mrow/></msub></math>` **B** `<math><msub><mi>x</mi><mrow/></msub></math>` **C** `<math><msub><mi>x</mi><mrow/></msub></math>` Empty editor → inserts □_{□}. The caret is placed in the base of the subscript; the next typed character goes there (typed “7” → 7_{□}).
TB-1-007		Parentheses	108, 28, 33, 33	**A** `<math><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mfenced><mrow/></mfenced></math>` **C** `<math><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced (7)).
TB-1-008		Square brackets	141, 28, 33, 33	**A** `<math><mfenced open="[" close="]"><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mfenced open="[" close="]"><mrow/></mfenced></math>` **C** `<math><mfenced open="[" close="]"><mi>x</mi></mfenced></math>` Empty editor → inserts fenced [□]. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced [7]).
TB-1-009		Vertical bars	108, 61, 33, 33	**A** `<math><mfenced open="|" close="|"><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mfenced open="|" close="|"><mrow/></mfenced></math>` **C** `<math><mfenced open="|" close="|"><mi>x</mi></mfenced></math>` Empty editor → inserts fenced | □ |. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced | 7 |).
TB-1-010		Curly brackets	141, 61, 33, 33	**A** `<math><mfenced open="{" close="}"><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mfenced open="{" close="}"><mrow/></mfenced></math>` **C** `<math><mfenced open="{" close="}"><mi>x</mi></mfenced></math>` Empty editor → inserts fenced { □ }. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced { 7 }).
TB-1-011		Plus sign	177, 28, 22, 22	**A** `<math><mo>+</mo></math>` **B** `<math><mi>x</mi><mo>+</mo></math>` **C** `<math><mo>+</mo></math>` Empty editor → inserts +. The caret ends after the inserted element; typed characters follow it (typed “7” → + 7).

TB-1-012		Forward slash	199, 28, 22, 22	**A** <math><mo>/</mo></math> **B** <math><mi>x</mi><mo>/</mo></math> **C** <math><mo>/</mo></math> Empty editor → inserts /. The caret ends after the inserted element; typed characters follow it (typed “7” → / 7).
TB-1-013		Multiplication sign	177, 50, 22, 22	**A** <math><mo>×</mo></math> **B** <math><mi>x</mi><mo>×</mo></math> **C** <math><mo>×</mo></math> Empty editor → inserts ×. The caret ends after the inserted element; typed characters follow it (typed “7” → × 7).
TB-1-014		Plus-minus sign	199, 50, 22, 22	**A** <math><mo>±</mo></math> **B** <math><mi>x</mi><mo>±</mo></math> **C** <math><mo>±</mo></math> Empty editor → inserts ±. The caret ends after the inserted element; typed characters follow it (typed “7” → ± 7).
TB-1-015		Minus sign	177, 72, 22, 22	**A** <math><mo>-</mo></math> **B** <math><mi>x</mi><mo>-</mo></math> **C** <math><mo>-</mo></math> Empty editor → inserts -. The caret ends after the inserted element; typed characters follow it (typed “7” → - 7).
TB-1-016		Division sign	199, 72, 22, 22	**A** <math><mo>÷</mo></math> **B** <math><mi>x</mi><mo>÷</mo></math> **C** <math><mo>÷</mo></math> Empty editor → inserts ÷. The caret ends after the inserted element; typed characters follow it (typed “7” → ÷ 7).
TB-1-017		Greater-than or equal to	224, 28, 22, 22	**A** <math><mo>≥</mo></math> **B** <math><mi>x</mi><mo>≥</mo></math> **C** <math><mo>≥</mo></math> Empty editor → inserts ≥. The caret ends after the inserted element; typed characters follow it (typed “7” → ≥ 7).
TB-1-018		Less-than or equal to	246, 28, 22, 22	**A** <math><mo>≤</mo></math> **B** <math><mi>x</mi><mo>≤</mo></math> **C** <math><mo>≤</mo></math> Empty editor → inserts ≤. The caret ends after the inserted element; typed characters follow it (typed “7” → ≤ 7).
TB-1-019		Element of	224, 50, 22, 22	**A** <math><mo>∈</mo></math> **B** <math><mi>x</mi><mo>∈</mo></math> **C** <math><mo>∈</mo></math> Empty editor → inserts ∈. The caret ends after the inserted element; typed characters follow it (typed “7” → ∈ 7).
TB-1-020		Subset of	246, 50, 22, 22	**A** <math><mo>⊂</mo></math> **B** <math><mi>x</mi><mo>⊂</mo></math> **C** <math><mo>⊂</mo></math> Empty editor → inserts ⊂. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊂ 7).
TB-1-021		Union	224, 72, 22, 22	**A** <math><mo>∪</mo></math> **B** <math><mi>x</mi><mo>∪</mo></math> **C** <math><mo>∪</mo></math> Empty editor → inserts ∪. The caret ends after the inserted element; typed characters follow it (typed “7” → ∪ 7).
TB-1-022		Intersection	246, 72, 22, 22	**A** <math><mo>∩</mo></math> **B** <math><mi>x</mi><mo>∩</mo></math> **C** <math><mo>∩</mo></math> Empty editor → inserts ∩. The caret ends after the inserted element; typed characters follow it (typed “7” → ∩ 7).
TB-1-023		Empty set	271, 28, 22, 22	**A** <math><mo>∅</mo></math> **B** <math><mi>x</mi><mo>∅</mo></math> **C** <math><mo>∅</mo></math> Empty editor → inserts ∅. The caret ends after the inserted element; typed characters follow it (typed “7” → ∅ 7).
TB-1-024		Infinity	271, 50, 22, 22	**A** <math><mo>∞</mo></math> **B** <math><mi>x</mi><mo>∞</mo></math> **C** <math><mo>∞</mo></math> Empty editor → inserts ∞. The caret ends after the inserted element; typed characters follow it (typed “7” → ∞ 7).
TB-1-025		Number pi	271, 72, 22, 22	**A** <math><mi mathvariant="normal">π</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">π</mi></math> **C** <math><mi mathvariant="normal">π</mi></math> Empty editor → inserts π. The caret ends after the inserted element; typed characters follow it (typed “7” → π 7).
TB-1-026		Press Ctrl+C to copy the selection	290.5, 22.5, 33, 33	Dedicated requirements: see CLP-001..004
TB-1-027		Undo	312.5, 22.5, 33, 33	Dedicated requirements: see CLP-005
TB-1-028		Press Ctrl+X to cut the selection	290.5, 44.5, 33, 33	Dedicated requirements: see CLP-001..004

TB-1-029		Redo	312.5, 44.5, 33, 33	Dedicated requirements: see CLP-005
TB-1-030		Press Ctrl+V to paste the content of the clipboard	290.5, 66.5, 33, 33	Dedicated requirements: see CLP-001..004
TB-1-031		Bold (Ctrl+B)	337.5, 22.5, 33, 33	Dedicated requirements: see FMT-001..004
TB-1-032		Color	359.5, 22.5, 33, 33	Dedicated requirements: see FMT-005
TB-1-033		Italic (Ctrl+I)	337.5, 44.5, 33, 33	Dedicated requirements: see FMT-001..004
TB-1-034		Insert special characters	359.5, 44.5, 33, 33	Dedicated requirements: see UNI-001..005
TB-1-035		Automatic italic	337.5, 66.5, 33, 33	Dedicated requirements: see FMT-001..004
TB-1-036		Regular text	359.5, 66.5, 33, 33	Dedicated requirements: see FMT-001..004
TB-1-037		Right to left editing	384.5, 22.5, 33, 33	Dedicated requirements: see MOD-001..003
TB-1-038		Arabic ligature	384.5, 44.5, 33, 33	Dedicated requirements: see MOD-001..003

C.2 Tab 2 — Symbols (Alt+2)

Main panel (52 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-001		Negate operator	3, 28, 22, 22	**A** <math><menclose notation="updiagonalstrike"/></math> **B** <math><mi>x</mi><menclose notation="updiagonalstrike"/></math> **C** <math><menclose notation="updiagonalstrike"><mi>x</mi></menclose></math> Empty editor → inserts enclose[updiagonalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[updiagonalstrike](7)).
TB-2-002		Plus sign	28, 28, 22, 22	**A** <math><mo>+</mo></math> **B** <math><mi>x</mi><mo>+</mo></math> **C** <math><mo>+</mo></math> Empty editor → inserts +. The caret ends after the inserted element; typed characters follow it (typed “7” → + 7).
TB-2-003		Multiplication sign	50, 28, 22, 22	**A** <math><mo>×</mo></math> **B** <math><mi>x</mi><mo>×</mo></math> **C** <math><mo>×</mo></math> Empty editor → inserts ×. The caret ends after the inserted element; typed characters follow it (typed “7” → × 7).
TB-2-004		Middle dot	72, 28, 22, 22	**A** <math><mo>·</mo></math> **B** <math><mi>x</mi><mo>·</mo></math> **C** <math><mo>·</mo></math> Empty editor → inserts ·. The caret ends after the inserted element; typed characters follow it (typed “7” → · 7).
TB-2-005		Minus sign	28, 50, 22, 22	**A** <math><mo>-</mo></math> **B** <math><mi>x</mi><mo>-</mo></math> **C** <math><mo>-</mo></math> Empty editor → inserts -. The caret ends after the inserted element; typed characters follow it (typed “7” → - 7).
TB-2-006		Division sign	50, 50, 22, 22	**A** <math><mo>÷</mo></math> **B** <math><mi>x</mi><mo>÷</mo></math> **C** <math><mo>÷</mo></math> Empty editor → inserts ÷. The caret ends after the inserted element; typed characters follow it (typed “7” → ÷ 7).

TB-2-007		Forward slash	72, 50, 22, 22	**A** <math><mo>/</mo></math> **B** <math><mi>x</mi><mo>/</mo></math> **C** <math><mo>/</mo></math> Empty editor → inserts /. The caret ends after the inserted element; typed characters follow it (typed “7” → / 7).
TB-2-008		Plus-minus sign	28, 72, 22, 22	**A** <math><mo>±</mo></math> **B** <math><mi>x</mi><mo>±</mo></math> **C** <math><mo>±</mo></math> Empty editor → inserts ±. The caret ends after the inserted element; typed characters follow it (typed “7” → ± 7).
TB-2-009		Asterisk	50, 72, 22, 22	**A** <math><mo>*</mo></math> **B** <math><mi>x</mi><mo>*</mo></math> **C** <math><mo>*</mo></math> Empty editor → inserts *. The caret ends after the inserted element; typed characters follow it (typed “7” → * 7).
TB-2-010		Ring operator	72, 72, 22, 22	**A** <math><mo>∘</mo></math> **B** <math><mi>x</mi><mo>∘</mo></math> **C** <math><mo>∘</mo></math> Empty editor → inserts ∘. The caret ends after the inserted element; typed characters follow it (typed “7” → ∘ 7).
TB-2-011		Number pi	105, 28, 22, 22	**A** <math><mi mathvariant="normal">π</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">π</mi></math> **C** <math><mi mathvariant="normal">π</mi></math> Empty editor → inserts π. The caret ends after the inserted element; typed characters follow it (typed “7” → π 7).
TB-2-012		Partial differential	127, 28, 22, 22	**A** <math><mo>∂</mo></math> **B** <math><mi>x</mi><mo>∂</mo></math> **C** <math><mo>∂</mo></math> Empty editor → inserts ∂. The caret ends after the inserted element; typed characters follow it (typed “7” → ∂ 7).
TB-2-013		Degree sign	149, 28, 22, 22	**A** <math><mo>°</mo></math> **B** <math><mi>x</mi><mo>°</mo></math> **C** <math><mo>°</mo></math> Empty editor → inserts °. The caret ends after the inserted element; typed characters follow it (typed “7” → ° 7).
TB-2-014		Infinity	105, 50, 22, 22	**A** <math><mo>∞</mo></math> **B** <math><mi>x</mi><mo>∞</mo></math> **C** <math><mo>∞</mo></math> Empty editor → inserts ∞. The caret ends after the inserted element; typed characters follow it (typed “7” → ∞ 7).
TB-2-015		Increment	127, 50, 22, 22	**A** <math><mo>∆</mo></math> **B** <math><mi>x</mi><mo>∆</mo></math> **C** <math><mo>∆</mo></math> Empty editor → inserts Δ. The caret ends after the inserted element; typed characters follow it (typed “7” → Δ 7).
TB-2-016		Apostrophe	149, 50, 22, 22	**A** <math><mo>'</mo></math> **B** <math><mi>x</mi><mo>'</mo></math> **C** <math><mo>'</mo></math> Empty editor → inserts '. The caret ends after the inserted element; typed characters follow it (typed “7” → ' 7).
TB-2-017		Empty set	105, 72, 22, 22	**A** <math><mo>∅</mo></math> **B** <math><mi>x</mi><mo>∅</mo></math> **C** <math><mo>∅</mo></math> Empty editor → inserts ∅. The caret ends after the inserted element; typed characters follow it (typed “7” → ∅ 7).
TB-2-018		Nabla	127, 72, 22, 22	**A** <math><mo>∇</mo></math> **B** <math><mi>x</mi><mo>∇</mo></math> **C** <math><mo>∇</mo></math> Empty editor → inserts ∇. The caret ends after the inserted element; typed characters follow it (typed “7” → ∇ 7).
TB-2-019		Double apostrophe	149, 72, 22, 22	**A** <math><mo>''</mo></math> **B** <math><mi>x</mi><mo>''</mo></math> **C** <math><mo>'</mo><mi>x</mi><mo>'</mo></math> Empty editor → inserts ''. The caret ends after the inserted element; typed characters follow it (typed “7” → ' 7 ').
TB-2-020		Equals sign	182, 28, 22, 22	**A** <math><mo>=</mo></math> **B** <math><mi>x</mi><mo>=</mo></math> **C** <math><mo>=</mo></math> Empty editor → inserts =. The caret ends after the inserted element; typed characters follow it (typed “7” → = 7).
TB-2-021		Identical to	204, 28, 22, 22	**A** <math><mo>≡</mo></math> **B** <math><mi>x</mi><mo>≡</mo></math> **C** <math><mo>≡</mo></math> Empty editor → inserts ≡. The caret ends after the inserted element; typed characters follow it (typed “7” → ≡ 7).
TB-2-022		Tilde operator	182, 50, 22, 22	**A** <math><mo>~</mo></math> **B** <math><mi>x</mi><mo>~</mo></math> **C** <math><mo>~</mo></math> Empty editor → inserts ~. The caret ends after the inserted element; typed characters follow it (typed “7” → ~ 7).

TB-2-023		Almost equal to	204, 50, 22, 22	**A** <math><mo>≈</mo></math> **B** <math><mi>x</mi><mo>≈</mo></math> **C** <math><mo>≈</mo></math> Empty editor → inserts ≈. The caret ends after the inserted element; typed characters follow it (typed “7” → ≈ 7).
TB-2-024		Asymptotically equal to	182, 72, 22, 22	**A** <math><mo>≃</mo></math> **B** <math><mi>x</mi><mo>≃</mo></math> **C** <math><mo>≃</mo></math> Empty editor → inserts ≃. The caret ends after the inserted element; typed characters follow it (typed “7” → ≃ 7).
TB-2-025		Approximately equal to	204, 72, 22, 22	**A** <math><mo>≅</mo></math> **B** <math><mi>x</mi><mo>≅</mo></math> **C** <math><mo>≅</mo></math> Empty editor → inserts ≅. The caret ends after the inserted element; typed characters follow it (typed “7” → ≅ 7).
TB-2-026		Greater-than sign	237, 28, 22, 22	**A** <math><mo>></mo></math> **B** <math><mi>x</mi><mo>></mo></math> **C** <math><mo>></mo></math> Empty editor → inserts >. The caret ends after the inserted element; typed characters follow it (typed “7” → > 7).
TB-2-027		Less-than sign	259, 28, 22, 22	**A** <math><mo><</mo></math> **B** <math><mi>x</mi><mo><</mo></math> **C** <math><mo><</mo></math> Empty editor → inserts <. The caret ends after the inserted element; typed characters follow it (typed “7” → < 7).
TB-2-028		Greater-than or equal to	237, 50, 22, 22	**A** <math><mo>≥</mo></math> **B** <math><mi>x</mi><mo>≥</mo></math> **C** <math><mo>≥</mo></math> Empty editor → inserts ≥. The caret ends after the inserted element; typed characters follow it (typed “7” → ≥ 7).
TB-2-029		Less-than or equal to	259, 50, 22, 22	**A** <math><mo>≤</mo></math> **B** <math><mi>x</mi><mo>≤</mo></math> **C** <math><mo>≤</mo></math> Empty editor → inserts ≤. The caret ends after the inserted element; typed characters follow it (typed “7” → ≤ 7).
TB-2-030		Greater-than or slanted equal to	237, 72, 22, 22	**A** <math><mo>⩾</mo></math> **B** <math><mi>x</mi><mo>⩾</mo></math> **C** <math><mo>⩾</mo></math> Empty editor → inserts ⩾. The caret ends after the inserted element; typed characters follow it (typed “7” → ⩾ 7).
TB-2-031		Less-than or slanted equal to	259, 72, 22, 22	**A** <math><mo>⩽</mo></math> **B** <math><mi>x</mi><mo>⩽</mo></math> **C** <math><mo>⩽</mo></math> Empty editor → inserts ⩽. The caret ends after the inserted element; typed characters follow it (typed “7” → ⩽ 7).
TB-2-032		Element of	292, 28, 22, 22	**A** <math><mo>∈</mo></math> **B** <math><mi>x</mi><mo>∈</mo></math> **C** <math><mo>∈</mo></math> Empty editor → inserts ∈. The caret ends after the inserted element; typed characters follow it (typed “7” → ∈ 7).
TB-2-033		Contains as member	314, 28, 22, 22	**A** <math><mo>∋</mo></math> **B** <math><mi>x</mi><mo>∋</mo></math> **C** <math><mo>∋</mo></math> Empty editor → inserts ∋. The caret ends after the inserted element; typed characters follow it (typed “7” → ∋ 7).
TB-2-034		Union	292, 50, 22, 22	**A** <math><mo>∪</mo></math> **B** <math><mi>x</mi><mo>∪</mo></math> **C** <math><mo>∪</mo></math> Empty editor → inserts ∪. The caret ends after the inserted element; typed characters follow it (typed “7” → ∪ 7).
TB-2-035		Intersection	314, 50, 22, 22	**A** <math><mo>∩</mo></math> **B** <math><mi>x</mi><mo>∩</mo></math> **C** <math><mo>∩</mo></math> Empty editor → inserts ∩. The caret ends after the inserted element; typed characters follow it (typed “7” → ∩ 7).
TB-2-036		Subset of	292, 72, 22, 22	**A** <math><mo>⊂</mo></math> **B** <math><mi>x</mi><mo>⊂</mo></math> **C** <math><mo>⊂</mo></math> Empty editor → inserts ⊂. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊂ 7).
TB-2-037		Superset of	314, 72, 22, 22	**A** <math><mo>⊃</mo></math> **B** <math><mi>x</mi><mo>⊃</mo></math> **C** <math><mo>⊃</mo></math> Empty editor → inserts ⊃. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊃ 7).
TB-2-038		Logical and	347, 28, 22, 22	**A** <math><mo>∧</mo></math> **B** <math><mi>x</mi><mo>∧</mo></math> **C** <math><mo>∧</mo></math> Empty editor → inserts ∧. The caret ends after the inserted element; typed characters follow it (typed “7” → ∧ 7).

TB-2-039		Logical or	369, 28, 22, 22	**A** <math><mo>∨</mo></math> **B** <math><mi>x</mi><mo>∨</mo></math> **C** <math><mo>∨</mo></math> Empty editor → inserts ∨. The caret ends after the inserted element; typed characters follow it (typed “7” → ∨ 7).
TB-2-040		Not sign	347, 50, 22, 22	**A** <math><mo>¬</mo></math> **B** <math><mi>x</mi><mo>¬</mo></math> **C** <math><mo>¬</mo></math> Empty editor → inserts ¬. The caret ends after the inserted element; typed characters follow it (typed “7” → ¬ 7).
TB-2-041		For all	369, 50, 22, 22	**A** <math><mo>∀</mo></math> **B** <math><mi>x</mi><mo>∀</mo></math> **C** <math><mo>∀</mo></math> Empty editor → inserts ∀. The caret ends after the inserted element; typed characters follow it (typed “7” → ∀ 7).
TB-2-042		There exists	347, 72, 22, 22	**A** <math><mo>∃</mo></math> **B** <math><mi>x</mi><mo>∃</mo></math> **C** <math><mo>∃</mo></math> Empty editor → inserts ∃. The caret ends after the inserted element; typed characters follow it (typed “7” → ∃ 7).
TB-2-043		There does not exist	369, 72, 22, 22	**A** <math><mo>∄</mo></math> **B** <math><mi>x</mi><mo>∄</mo></math> **C** <math><mo>∄</mo></math> Empty editor → inserts ∄. The caret ends after the inserted element; typed characters follow it (typed “7” → ∄ 7).
TB-2-044		Angle	402, 28, 22, 22	**A** <math><mo>∠</mo></math> **B** <math><mi>x</mi><mo>∠</mo></math> **C** <math><mo>∠</mo></math> Empty editor → inserts ∠. The caret ends after the inserted element; typed characters follow it (typed “7” → ∠ 7).
TB-2-045		Parallel to	402, 50, 22, 22	**A** <math><mo>∥</mo></math> **B** <math><mi>x</mi><mo>∥</mo></math> **C** <math><mo>∥</mo></math> Empty editor → inserts ∥. The caret ends after the inserted element; typed characters follow it (typed “7” → ∥ 7).
TB-2-046		Perpendicular	402, 72, 22, 22	**A** <math><mo>⊥</mo></math> **B** <math><mi>x</mi><mo>⊥</mo></math> **C** <math><mo>⊥</mo></math> Empty editor → inserts ⊥. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊥ 7).
TB-2-047		Square	435, 28, 22, 22	**A** <math><mo>□</mo></math> **B** <math><mi>x</mi><mo>□</mo></math> **C** <math><mo>□</mo></math> Empty editor → inserts □. The caret ends after the inserted element; typed characters follow it (typed “7” → □ 7).
TB-2-048		Triangle	435, 50, 22, 22	**A** <math><mo>△</mo></math> **B** <math><mi>x</mi><mo>△</mo></math> **C** <math><mo>△</mo></math> Empty editor → inserts △. The caret ends after the inserted element; typed characters follow it (typed “7” → △ 7).
TB-2-049		Circle	435, 72, 22, 22	**A** <math><mo>○</mo></math> **B** <math><mi>x</mi><mo>○</mo></math> **C** <math><mo>○</mo></math> Empty editor → inserts ○. The caret ends after the inserted element; typed characters follow it (typed “7” → ○ 7).
TB-2-050		Circled plus	468, 28, 22, 22	**A** <math><mo>⊕</mo></math> **B** <math><mi>x</mi><mo>⊕</mo></math> **C** <math><mo>⊕</mo></math> Empty editor → inserts ⊕. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊕ 7).
TB-2-051		Circled times	468, 50, 22, 22	**A** <math><mo>⊗</mo></math> **B** <math><mi>x</mi><mo>⊗</mo></math> **C** <math><mo>⊗</mo></math> Empty editor → inserts ⊗. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊗ 7).
TB-2-052		Circled dot	468, 72, 22, 22	**A** <math><mo>⊙</mo></math> **B** <math><mi>x</mi><mo>⊙</mo></math> **C** <math><mo>⊙</mo></math> Empty editor → inserts ⊙. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊙ 7).

“More” group 1 (3 controls) — expand button at 95.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected

TB-2-G1-001		Reverse solidus	98, 94, 22, 22 (open pop-up)	**A** <math><mo>\</mo></math> **B** <math><mi>x</mi><mo>\</mo></math> **C** <math><mo>\</mo></math> Empty editor → inserts \. The caret ends after the inserted element; typed characters follow it (typed “7” → \ 7).
TB-2-G1-002		Set minus	98, 116, 22, 22 (open pop-up)	**A** <math><mo>∖</mo></math> **B** <math><mi>x</mi><mo>∖</mo></math> **C** <math><mo>∖</mo></math> Empty editor → inserts \. The caret ends after the inserted element; typed characters follow it (typed “7” → \ 7).
TB-2-G1-003		Minus-or-plus sign	98, 138, 22, 22 (open pop-up)	**A** <math><mo>∓</mo></math> **B** <math><mi>x</mi><mo>∓</mo></math> **C** <math><mo>∓</mo></math> Empty editor → inserts ∓. The caret ends after the inserted element; typed characters follow it (typed “7” → ∓ 7).

“More” group 2 (3 controls) — expand button at 172.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G2-001		Triple apostrophe	175, 94, 22, 22 (open pop-up)	**A** <math><mo>'''</mo></math> **B** <math><mi>x</mi><mo>'''</mo></math> **C** <math><mo>'</mo><mi>x</mi><mo>''</mo></math> Empty editor → inserts '''. The caret ends after the inserted element; typed characters follow it (typed “7” → ' 7 '').
TB-2-G2-002		Quadruple apostrophe	175, 116, 22, 22 (open pop-up)	**A** <math><mo>''''</mo></math> **B** <math><mi>x</mi><mo>''''</mo></math> **C** <math><mo>''</mo><mi>x</mi><mo>''</mo></math> Empty editor → inserts ''''. The caret ends after the inserted element; typed characters follow it (typed “7” → '' 7 '').
TB-2-G2-003		Reversed prime	175, 138, 22, 22 (open pop-up)	**A** <math><mo>‵</mo></math> **B** <math><mi>x</mi><mo>‵</mo></math> **C** <math><mo>‵</mo></math> Empty editor → inserts `. The caret ends after the inserted element; typed characters follow it (typed “7” → ` 7).

“More” group 3 (4 controls) — expand button at 227.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G3-001		Not equal to	230, 94, 22, 22 (open pop-up)	**A** <math><mo>≠</mo></math> **B** <math><mi>x</mi><mo>≠</mo></math> **C** <math><mo>≠</mo></math> Empty editor → inserts ≠. The caret ends after the inserted element; typed characters follow it (typed “7” → ≠ 7).
TB-2-G3-002		Not almost equal to	252, 94, 22, 22 (open pop-up)	**A** <math><mo>≉</mo></math> **B** <math><mi>x</mi><mo>≉</mo></math> **C** <math><mo>≉</mo></math> Empty editor → inserts ≠. The caret ends after the inserted element; typed characters follow it (typed “7” → ≠ 7).
TB-2-G3-003		Not identical to	230, 116, 22, 22 (open pop-up)	**A** <math><mo>≢</mo></math> **B** <math><mi>x</mi><mo>≢</mo></math> **C** <math><mo>≢</mo></math> Empty editor → inserts ≠. The caret ends after the inserted element; typed characters follow it (typed “7” → ≠ 7).
TB-2-G3-004		Not tilde	230, 138, 22, 22 (open pop-up)	**A** <math><mo>≁</mo></math> **B** <math><mi>x</mi><mo>≁</mo></math> **C** <math><mo>≁</mo></math> Empty editor → inserts ≁. The caret ends after the inserted element; typed characters follow it (typed “7” → ≁ 7).

“More” group 4 (9 controls) — expand button at 282.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G4-001		Less-than and single-line not equal to	285, 94, 22, 22 (open pop-up)	**A** <math><mo>⪇</mo></math> **B** <math><mi>x</mi><mo>⪇</mo></math> **C** <math><mo>⪇</mo></math> Empty editor → inserts ⪇. The caret ends after the inserted element; typed characters follow it (typed “7” → ⪇ 7).

TB-2-G4-002		Much greater-than	307, 94, 22, 22 (open pop-up)	**A** <math><mo>≫</mo></math> **B** <math><mi>x</mi><mo>≫</mo></math> **C** <math><mo>≫</mo></math> Empty editor → inserts ≫. The caret ends after the inserted element; typed characters follow it (typed “7” → ≫ 7).
TB-2-G4-003		Succeeds	329, 94, 22, 22 (open pop-up)	**A** <math><mo>≻</mo></math> **B** <math><mi>x</mi><mo>≻</mo></math> **C** <math><mo>≻</mo></math> Empty editor → inserts ≻. The caret ends after the inserted element; typed characters follow it (typed “7” → ≻ 7).
TB-2-G4-004		Greater-than and single-line not equal to	285, 116, 22, 22 (open pop-up)	**A** <math><mo>⪈</mo></math> **B** <math><mi>x</mi><mo>⪈</mo></math> **C** <math><mo>⪈</mo></math> Empty editor → inserts ⪈. The caret ends after the inserted element; typed characters follow it (typed “7” → ⪈ 7).
TB-2-G4-005		Proportional to	307, 116, 22, 22 (open pop-up)	**A** <math><mo>∝</mo></math> **B** <math><mi>x</mi><mo>∝</mo></math> **C** <math><mo>∝</mo></math> Empty editor → inserts ∝. The caret ends after the inserted element; typed characters follow it (typed “7” → ∝ 7).
TB-2-G4-006		Normal subgroup of	329, 116, 22, 22 (open pop-up)	**A** <math><mo>⊲</mo></math> **B** <math><mi>x</mi><mo>⊲</mo></math> **C** <math><mo>⊲</mo></math> Empty editor → inserts ⊲. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊲ 7).
TB-2-G4-007		Much less-than	285, 138, 22, 22 (open pop-up)	**A** <math><mo>≪</mo></math> **B** <math><mi>x</mi><mo>≪</mo></math> **C** <math><mo>≪</mo></math> Empty editor → inserts ≪. The caret ends after the inserted element; typed characters follow it (typed “7” → ≪ 7).
TB-2-G4-008		Precedes	307, 138, 22, 22 (open pop-up)	**A** <math><mo>≺</mo></math> **B** <math><mi>x</mi><mo>≺</mo></math> **C** <math><mo>≺</mo></math> Empty editor → inserts ≺. The caret ends after the inserted element; typed characters follow it (typed “7” → ≺ 7).
TB-2-G4-009		Contains as normal subgroup	329, 138, 22, 22 (open pop-up)	**A** <math><mo>⊳</mo></math> **B** <math><mi>x</mi><mo>⊳</mo></math> **C** <math><mo>⊳</mo></math> Empty editor → inserts ⊳. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊳ 7).

“More” group 5 (10 controls) — expand button at 337.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G5-001		Not an element of	340, 94, 22, 22 (open pop-up)	**A** <math><mo>∉</mo></math> **B** <math><mi>x</mi><mo>∉</mo></math> **C** <math><mo>∉</mo></math> Empty editor → inserts ∉. The caret ends after the inserted element; typed characters follow it (typed “7” → ∉ 7).
TB-2-G5-002		Does not contain as member	362, 94, 22, 22 (open pop-up)	**A** <math><mo>∌</mo></math> **B** <math><mi>x</mi><mo>∌</mo></math> **C** <math><mo>∌</mo></math> Empty editor → inserts ∌. The caret ends after the inserted element; typed characters follow it (typed “7” → ∌ 7).
TB-2-G5-003		Subset of or equal to	384, 94, 22, 22 (open pop-up)	**A** <math><mo>⊆</mo></math> **B** <math><mi>x</mi><mo>⊆</mo></math> **C** <math><mo>⊆</mo></math> Empty editor → inserts ⊆. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊆ 7).
TB-2-G5-004		Superset of or equal to	406, 94, 22, 22 (open pop-up)	**A** <math><mo>⊇</mo></math> **B** <math><mi>x</mi><mo>⊇</mo></math> **C** <math><mo>⊇</mo></math> Empty editor → inserts ⊇. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊇ 7).
TB-2-G5-005		Square subset of	340, 116, 22, 22 (open pop-up)	**A** <math><mo>⊏</mo></math> **B** <math><mi>x</mi><mo>⊏</mo></math> **C** <math><mo>⊏</mo></math> Empty editor → inserts ⊏. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊏ 7).
TB-2-G5-006		Square superset of	362, 116, 22, 22 (open pop-up)	**A** <math><mo>⊐</mo></math> **B** <math><mi>x</mi><mo>⊐</mo></math> **C** <math><mo>⊐</mo></math> Empty editor → inserts ⊐. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊐ 7).

TB-2-G5-007	⊑	Square subset of or equal to	384, 116, 22, 22 (open pop-up)	**A** <math><mo>⊑</mo></math> **B** <math><mi>x</mi><mo>⊑</mo></math> **C** <math><mo>⊑</mo></math> Empty editor → inserts ⊑. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊑ 7).
TB-2-G5-008	⊒	Square superset of or equal to	406, 116, 22, 22 (open pop-up)	**A** <math><mo>⊒</mo></math> **B** <math><mi>x</mi><mo>⊒</mo></math> **C** <math><mo>⊒</mo></math> Empty editor → inserts ⊒. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊒ 7).
TB-2-G5-009	⊓	Square cap	340, 138, 22, 22 (open pop-up)	**A** <math><mo>⊓</mo></math> **B** <math><mi>x</mi><mo>⊓</mo></math> **C** <math><mo>⊓</mo></math> Empty editor → inserts ⊓. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊓ 7).
TB-2-G5-010	⊔	Square cup	362, 138, 22, 22 (open pop-up)	**A** <math><mo>⊔</mo></math> **B** <math><mi>x</mi><mo>⊔</mo></math> **C** <math><mo>⊔</mo></math> Empty editor → inserts ⊔. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊔ 7).

“More” group 6 (2 controls) — expand button at 392.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G6-001	∴	Therefore	395, 94, 22, 22 (open pop-up)	**A** <math><mo>∴</mo></math> **B** <math><mi>x</mi><mo>∴</mo></math> **C** <math><mo>∴</mo></math> Empty editor → inserts ∴. The caret ends after the inserted element; typed characters follow it (typed “7” → ∴ 7).
TB-2-G6-002	∵	Because	395, 116, 22, 22 (open pop-up)	**A** <math><mo>∵</mo></math> **B** <math><mi>x</mi><mo>∵</mo></math> **C** <math><mo>∵</mo></math> Empty editor → inserts ∵. The caret ends after the inserted element; typed characters follow it (typed “7” → ∵ 7).

“More” group 7 (4 controls) — expand button at 425.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G7-001	∦	Not parallel to	428, 94, 22, 22 (open pop-up)	**A** <math><mo>∦</mo></math> **B** <math><mi>x</mi><mo>∦</mo></math> **C** <math><mo>∦</mo></math> Empty editor → inserts ∦. The caret ends after the inserted element; typed characters follow it (typed “7” → ∦ 7).
TB-2-G7-002	⋄	Diamond	450, 94, 22, 22 (open pop-up)	**A** <math><mo>⋄</mo></math> **B** <math><mi>x</mi><mo>⋄</mo></math> **C** <math><mo>⋄</mo></math> Empty editor → inserts ⋄. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋄ 7).
TB-2-G7-003	∡	Measured angle	428, 116, 22, 22 (open pop-up)	**A** <math><mo>∡</mo></math> **B** <math><mi>x</mi><mo>∡</mo></math> **C** <math><mo>∡</mo></math> Empty editor → inserts ∡. The caret ends after the inserted element; typed characters follow it (typed “7” → ∡ 7).
TB-2-G7-004	∢	Spherical angle	428, 138, 22, 22 (open pop-up)	**A** <math><mo>∢</mo></math> **B** <math><mi>x</mi><mo>∢</mo></math> **C** <math><mo>∢</mo></math> Empty editor → inserts ∢. The caret ends after the inserted element; typed characters follow it (typed “7” → ∢ 7).

“More” group 8 (2 controls) — expand button at 458.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G8-001	▭	Rectangle	461, 94, 22, 22 (open pop-up)	**A** <math><mo>▭</mo></math> **B** <math><mi>x</mi><mo>▭</mo></math> **C** <math><mo>▭</mo></math> Empty editor → inserts ▭. The caret ends after the inserted element; typed characters follow it (typed “7” → ▭ 7).

TB-2-G8-002		Parallelogram	461, 116, 22, 22 (open pop-up)	**A** <math><mo>▱</mo></math> **B** <math><mi>x</mi><mo>▱</mo></math> **C** <math><mo>▱</mo></math> Empty editor → inserts ▱. The caret ends after the inserted element; typed characters follow it (typed “7” → ▱ 7).

“More” group 9 (4 controls) — expand button at 491.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-2-G9-001		Circled dash	494, 94, 22, 22 (open pop-up)	**A** <math><mo>⊝</mo></math> **B** <math><mi>x</mi><mo>⊝</mo></math> **C** <math><mo>⊝</mo></math> Empty editor → inserts ⊝. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊝ 7).
TB-2-G9-002		Bullet	516, 94, 22, 22 (open pop-up)	**A** <math><mo>∙</mo></math> **B** <math><mi>x</mi><mo>∙</mo></math> **C** <math><mo>∙</mo></math> Empty editor → inserts ∙. The caret ends after the inserted element; typed characters follow it (typed “7” → ∙ 7).
TB-2-G9-003		Circled asterisk	494, 116, 22, 22 (open pop-up)	**A** <math><mo>⊛</mo></math> **B** <math><mi>x</mi><mo>⊛</mo></math> **C** <math><mo>⊛</mo></math> Empty editor → inserts ⊛. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊛ 7).
TB-2-G9-004		Circled division	494, 138, 22, 22 (open pop-up)	**A** <math><mo>⨸</mo></math> **B** <math><mi>x</mi><mo>⨸</mo></math> **C** <math><mo>⨸</mo></math> Empty editor → inserts ⨸. The caret ends after the inserted element; typed characters follow it (typed “7” → ⨸ 7).

C.3 Tab 3 — Arrows (Alt+3)

Main panel (26 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-3-001		Leftwards arrow	3, 28, 22, 22	**A** <math><mo>←</mo></math> **B** <math><mi>x</mi><mo>←</mo></math> **C** <math><mo>←</mo></math> Empty editor → inserts ←. The caret ends after the inserted element; typed characters follow it (typed “7” → ← 7).
TB-3-002		Rightwards arrow	25, 28, 22, 22	**A** <math><mo>→</mo></math> **B** <math><mi>x</mi><mo>→</mo></math> **C** <math><mo>→</mo></math> Empty editor → inserts →. The caret ends after the inserted element; typed characters follow it (typed “7” → → 7).
TB-3-003		Left right arrow	47, 28, 22, 22	**A** <math><mo>↔</mo></math> **B** <math><mi>x</mi><mo>↔</mo></math> **C** <math><mo>↔</mo></math> Empty editor → inserts ↔. The caret ends after the inserted element; typed characters follow it (typed “7” → ↔ 7).
TB-3-004		Leftwards double arrow	3, 50, 22, 22	**A** <math><mo>⇐</mo></math> **B** <math><mi>x</mi><mo>⇐</mo></math> **C** <math><mo>⇐</mo></math> Empty editor → inserts ⇐. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇐ 7).
TB-3-005		Rightwards double arrow	25, 50, 22, 22	**A** <math><mo>⇒</mo></math> **B** <math><mi>x</mi><mo>⇒</mo></math> **C** <math><mo>⇒</mo></math> Empty editor → inserts ⇒. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇒ 7).
TB-3-006		Left right double arrow	47, 50, 22, 22	**A** <math><mo>⇔</mo></math> **B** <math><mi>x</mi><mo>⇔</mo></math> **C** <math><mo>⇔</mo></math> Empty editor → inserts ⇔. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇔ 7).
TB-3-007		Leftwards arrow from bar	3, 72, 22, 22	**A** <math><mo>↤</mo></math> **B** <math><mi>x</mi><mo>↤</mo></math> **C** <math><mo>↤</mo></math> Empty editor → inserts ↤. The caret ends after the inserted element; typed characters follow it (typed “7” → ↤ 7).

TB-3-008		Rightwards arrow from bar	25, 72, 22, 22	**A** <math><mo>↦</mo></math> **B** <math><mi>x</mi><mo>↦</mo></math> **C** <math><mo>↦</mo></math> Empty editor → inserts ↦. The caret ends after the inserted element; typed characters follow it (typed “7” → ↦ 7).
TB-3-009		Vertical ellipsis	80, 28, 22, 22	**A** <math><mo>⋮</mo></math> **B** <math><mi>x</mi><mo>⋮</mo></math> **C** <math><mo>⋮</mo></math> Empty editor → inserts ⋮. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋮ 7).
TB-3-010		Up right diagonal ellipsis	102, 28, 22, 22	**A** <math><mo>⋰</mo></math> **B** <math><mi>x</mi><mo>⋰</mo></math> **C** <math><mo>⋰</mo></math> Empty editor → inserts ⋰. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋰ 7).
TB-3-011		Horizontal ellipsis	80, 50, 22, 22	**A** <math><mo>…</mo></math> **B** <math><mi>x</mi><mo>…</mo></math> **C** <math><mo>…</mo></math> Empty editor → inserts ….. The caret ends after the inserted element; typed characters follow it (typed “7” → … 7).
TB-3-012		Down right diagonal ellipsis	102, 50, 22, 22	**A** <math><mo>⋱</mo></math> **B** <math><mi>x</mi><mo>⋱</mo></math> **C** <math><mo>⋱</mo></math> Empty editor → inserts ⋱. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋱ 7).
TB-3-013		Midline horizontal ellipsis	80, 72, 22, 22	**A** <math><mo>⋯</mo></math> **B** <math><mi>x</mi><mo>⋯</mo></math> **C** <math><mo>⋯</mo></math> Empty editor → inserts ⋯. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋯ 7).
TB-3-014		Short dash	127, 28, 22, 22	**A** <math><mo>‐</mo></math> **B** <math><mi>x</mi><mo>‐</mo></math> **C** <math><mo>‐</mo></math> Empty editor → inserts -. The caret ends after the inserted element; typed characters follow it (typed “7” → - 7).
TB-3-015		Dash	127, 50, 22, 22	**A** <math><mo>-</mo></math> **B** <math><mi>x</mi><mo>-</mo></math> **C** <math><mo>-</mo></math> Empty editor → inserts -. The caret ends after the inserted element; typed characters follow it (typed “7” → - 7).
TB-3-016		Long dash	127, 72, 22, 22	**A** <math><mo>—</mo></math> **B** <math><mi>x</mi><mo>—</mo></math> **C** <math><mo>—</mo></math> Empty editor → inserts —. The caret ends after the inserted element; typed characters follow it (typed “7” → — 7).
TB-3-017		Right arrow with over script	152, 28, 33, 33	**A** <math><mover><mo>→</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>→</mo><mrow/></mover></math> **C** <math><mover><mo>→</mo><mi>x</mi></mover></math> Empty editor → inserts over(→; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(→; 7)).
TB-3-018		Right arrow with under script	185, 28, 33, 33	**A** <math><munder><mo>→</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>→</mo><mrow/></munder></math> **C** <math><munder><mo>→</mo><mi>x</mi></munder></math> Empty editor → inserts under(→; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(→; 7)).
TB-3-019		Right arrow with under and over script	218, 28, 33, 33	**A** <math><munderover><mo>→</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>→</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>→</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(→; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(→; □; 7)).
TB-3-020		Left arrow with over script	152, 61, 33, 33	**A** <math><mover><mo>←</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>←</mo><mrow/></mover></math> **C** <math><mover><mo>←</mo><mi>x</mi></mover></math> Empty editor → inserts over(←; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(←; 7)).
TB-3-021		Left arrow with under script	185, 61, 33, 33	**A** <math><munder><mo>←</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>←</mo><mrow/></munder></math> **C** <math><munder><mo>←</mo><mi>x</mi></munder></math> Empty editor → inserts under(←; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(←; 7)).
TB-3-022		Left arrow with under and over script	218, 61, 33, 33	**A** <math><munderover><mo>←</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>←</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>←</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(←; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(←; □; 7)).

TB-3-023		Vector accent	262, 28, 33, 33	**A** <math><mover><mrow/><mo>⇀</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>⇀</mo></mover></math> **C** <math><mover><mi>x</mi><mo>⇀</mo></mover></math> Empty editor → inserts over(□; ⇀). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ⇀)).
TB-3-024		Left right arrow accent	295, 28, 33, 33	**A** <math><mover><mrow/><mo>↔</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>↔</mo></mover></math> **C** <math><mover><mi>x</mi><mo>↔</mo></mover></math> Empty editor → inserts over(□; ↔). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ↔)).
TB-3-025		Arrow accent	262, 61, 33, 33	**A** <math><mover><mrow/><mo>→</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>→</mo></mover></math> **C** <math><mover><mi>x</mi><mo>→</mo></mover></math> Empty editor → inserts over(□; →). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; →)).
TB-3-026		Bar accent	295, 61, 33, 33	**A** <math><mover><mrow/><mo>¯</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>¯</mo></mover></math> **C** <math><mover><mi>x</mi><mo>¯</mo></mover></math> Empty editor → inserts over(□; ¯). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ¯)).

“More” group 1 (31 controls) — expand button at 70.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-3-G1-001		North east arrow	73, 94, 22, 22 (open pop-up)	**A** <math><mo>↗</mo></math> **B** <math><mi>x</mi><mo>↗</mo></math> **C** <math><mo>↗</mo></math> Empty editor → inserts ↗. The caret ends after the inserted element; typed characters follow it (typed “7” → ↗ 7).
TB-3-G1-002		South east arrow	95, 94, 22, 22 (open pop-up)	**A** <math><mo>↘</mo></math> **B** <math><mi>x</mi><mo>↘</mo></math> **C** <math><mo>↘</mo></math> Empty editor → inserts ↘. The caret ends after the inserted element; typed characters follow it (typed “7” → ↘ 7).
TB-3-G1-003		North west arrow	117, 94, 22, 22 (open pop-up)	**A** <math><mo>↖</mo></math> **B** <math><mi>x</mi><mo>↖</mo></math> **C** <math><mo>↖</mo></math> Empty editor → inserts ↖. The caret ends after the inserted element; typed characters follow it (typed “7” → ↖ 7).
TB-3-G1-004		South west arrow	139, 94, 22, 22 (open pop-up)	**A** <math><mo>↙</mo></math> **B** <math><mi>x</mi><mo>↙</mo></math> **C** <math><mo>↙</mo></math> Empty editor → inserts ↙. The caret ends after the inserted element; typed characters follow it (typed “7” → ↙ 7).
TB-3-G1-005		North west and south east arrow	161, 94, 22, 22 (open pop-up)	**A** <math><mo>⤡</mo></math> **B** <math><mi>x</mi><mo>⤡</mo></math> **C** <math><mo>⤡</mo></math> Empty editor → inserts ⤡. The caret ends after the inserted element; typed characters follow it (typed “7” → ⤡ 7).
TB-3-G1-006		North east and south west arrow	183, 94, 22, 22 (open pop-up)	**A** <math><mo>⤢</mo></math> **B** <math><mi>x</mi><mo>⤢</mo></math> **C** <math><mo>⤢</mo></math> Empty editor → inserts ⤢. The caret ends after the inserted element; typed characters follow it (typed “7” → ⤢ 7).
TB-3-G1-007		Leftwards arrow with hook	205, 94, 22, 22 (open pop-up)	**A** <math><mo>↩</mo></math> **B** <math><mi>x</mi><mo>↩</mo></math> **C** <math><mo>↩</mo></math> Empty editor → inserts ↩. The caret ends after the inserted element; typed characters follow it (typed “7” → ↩ 7).
TB-3-G1-008		Rightwards arrow with hook	227, 94, 22, 22 (open pop-up)	**A** <math><mo>↪</mo></math> **B** <math><mi>x</mi><mo>↪</mo></math> **C** <math><mo>↪</mo></math> Empty editor → inserts ↪. The caret ends after the inserted element; typed characters follow it (typed “7” → ↪ 7).
TB-3-G1-009		Leftwards harpoon with barb upwards	249, 94, 22, 22 (open pop-up)	**A** <math><mo>↼</mo></math> **B** <math><mi>x</mi><mo>↼</mo></math> **C** <math><mo>↼</mo></math> Empty editor → inserts ↼. The caret ends after the inserted element; typed characters follow it (typed “7” → ↼ 7).
TB-3-G1-010		Rightwards harpoon with barb upwards	271, 94, 22, 22 (open pop-up)	**A** <math><mo>⇀</mo></math> **B** <math><mi>x</mi><mo>⇀</mo></math> **C** <math><mo>⇀</mo></math> Empty editor → inserts ⇀. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇀ 7).

TB-3-G1-011		Upwards arrow	293, 94, 22, 22 (open pop-up)	**A** <math><mo>↑</mo></math> **B** <math><mi>x</mi><mo>↑</mo></math> **C** <math><mo>↑</mo></math> Empty editor → inserts ↑. The caret ends after the inserted element; typed characters follow it (typed “7” → ↑ 7).
TB-3-G1-012		Downwards arrow	73, 116, 22, 22 (open pop-up)	**A** <math><mo>↓</mo></math> **B** <math><mi>x</mi><mo>↓</mo></math> **C** <math><mo>↓</mo></math> Empty editor → inserts ↓. The caret ends after the inserted element; typed characters follow it (typed “7” → ↓ 7).
TB-3-G1-013		Upwards double arrow	95, 116, 22, 22 (open pop-up)	**A** <math><mo>⇑</mo></math> **B** <math><mi>x</mi><mo>⇑</mo></math> **C** <math><mo>⇑</mo></math> Empty editor → inserts ⇑. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇑ 7).
TB-3-G1-014		Downwards double arrow	117, 116, 22, 22 (open pop-up)	**A** <math><mo>⇓</mo></math> **B** <math><mi>x</mi><mo>⇓</mo></math> **C** <math><mo>⇓</mo></math> Empty editor → inserts ⇓. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇓ 7).
TB-3-G1-015		Leftwards harpoon with barb up above long dash	139, 116, 22, 22 (open pop-up)	**A** <math><mo>⥪</mo></math> **B** <math><mi>x</mi><mo>⥪</mo></math> **C** <math><mo>⥪</mo></math> Empty editor → inserts ⥪. The caret ends after the inserted element; typed characters follow it (typed “7” → ⥪ 7).
TB-3-G1-016		Rightwards harpoon with barb down below long dash	161, 116, 22, 22 (open pop-up)	**A** <math><mo>⥭</mo></math> **B** <math><mi>x</mi><mo>⥭</mo></math> **C** <math><mo>⥭</mo></math> Empty editor → inserts ⥭. The caret ends after the inserted element; typed characters follow it (typed “7” → ⥭ 7).
TB-3-G1-017		Leftwards harpoon over rightwards harpoon	183, 116, 22, 22 (open pop-up)	**A** <math><mo>⇋</mo></math> **B** <math><mi>x</mi><mo>⇋</mo></math> **C** <math><mo>⇋</mo></math> Empty editor → inserts ⇋. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇋ 7).
TB-3-G1-018		Rightwards harpoon over leftwards harpoon	205, 116, 22, 22 (open pop-up)	**A** <math><mo>⇌</mo></math> **B** <math><mi>x</mi><mo>⇌</mo></math> **C** <math><mo>⇌</mo></math> Empty editor → inserts ⇌. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇌ 7).
TB-3-G1-019		Leftwards harpoon with barb downwards	227, 116, 22, 22 (open pop-up)	**A** <math><mo>↽</mo></math> **B** <math><mi>x</mi><mo>↽</mo></math> **C** <math><mo>↽</mo></math> Empty editor → inserts ↽. The caret ends after the inserted element; typed characters follow it (typed “7” → ↽ 7).
TB-3-G1-020		Rightwards harpoon with barb downwards	249, 116, 22, 22 (open pop-up)	**A** <math><mo>⇁</mo></math> **B** <math><mi>x</mi><mo>⇁</mo></math> **C** <math><mo>⇁</mo></math> Empty editor → inserts ⇁. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇁ 7).
TB-3-G1-021		Leftwards arrow over rightwards arrow	271, 116, 22, 22 (open pop-up)	**A** <math><mo>⇆</mo></math> **B** <math><mi>x</mi><mo>⇆</mo></math> **C** <math><mo>⇆</mo></math> Empty editor → inserts ⇆. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇆ 7).
TB-3-G1-022		Rightwards arrow over leftwards arrow	293, 116, 22, 22 (open pop-up)	**A** <math><mo>⇄</mo></math> **B** <math><mi>x</mi><mo>⇄</mo></math> **C** <math><mo>⇄</mo></math> Empty editor → inserts ⇄. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇄ 7).
TB-3-G1-023		Upwards arrow leftwards of downwards arrow	73, 138, 22, 22 (open pop-up)	**A** <math><mo>⇅</mo></math> **B** <math><mi>x</mi><mo>⇅</mo></math> **C** <math><mo>⇅</mo></math> Empty editor → inserts ⇅. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇅ 7).
TB-3-G1-024		Downwards arrow leftwards of upwards arrow	95, 138, 22, 22 (open pop-up)	**A** <math><mo>⇵</mo></math> **B** <math><mi>x</mi><mo>⇵</mo></math> **C** <math><mo>⇵</mo></math> Empty editor → inserts ⇵. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇵ 7).
TB-3-G1-025		Upwards harpoon with barb left beside downwards harpoon with barb right	117, 138, 22, 22 (open pop-up)	**A** <math><mo>⥮</mo></math> **B** <math><mi>x</mi><mo>⥮</mo></math> **C** <math><mo>⥮</mo></math> Empty editor → inserts ⥮. The caret ends after the inserted element; typed characters follow it (typed “7” → ⥮ 7).

TB-3-G1-026		Downwards harpoon with barb left beside upwards harpoon with barb right	139, 138, 22, 22 (open pop-up)	**A** <math><mo>⥯</mo></math> **B** <math><mi>x</mi><mo>⥯</mo></math> **C** <math><mo>⥯</mo></math> Empty editor → inserts ⥯. The caret ends after the inserted element; typed characters follow it (typed “7” → ⥯ 7).
TB-3-G1-027		Rightwards arrow over short leftwards arrow	161, 138, 22, 22 (open pop-up)	**A** <math><mo>⥂</mo></math> **B** <math><mi>x</mi><mo>⥂</mo></math> **C** <math><mo>⥂</mo></math> Empty editor → inserts ⥂. The caret ends after the inserted element; typed characters follow it (typed “7” → ⥂ 7).
TB-3-G1-028		Short rightwards arrow over leftwards arrow	183, 138, 22, 22 (open pop-up)	**A** <math><mo>⥄</mo></math> **B** <math><mi>x</mi><mo>⥄</mo></math> **C** <math><mo>⥄</mo></math> Empty editor → inserts ⥄. The caret ends after the inserted element; typed characters follow it (typed “7” → ⥄ 7).
TB-3-G1-029		Up down arrow	205, 138, 22, 22 (open pop-up)	**A** <math><mo>↕</mo></math> **B** <math><mi>x</mi><mo>↕</mo></math> **C** <math><mo>↕</mo></math> Empty editor → inserts ↕. The caret ends after the inserted element; typed characters follow it (typed “7” → ↕ 7).
TB-3-G1-030		Up down double arrow	227, 138, 22, 22 (open pop-up)	**A** <math><mo>⇕</mo></math> **B** <math><mi>x</mi><mo>⇕</mo></math> **C** <math><mo>⇕</mo></math> Empty editor → inserts ⇕. The caret ends after the inserted element; typed characters follow it (typed “7” → ⇕ 7).
TB-3-G1-031		Downwards arrow with corner leftwards	249, 138, 22, 22 (open pop-up)	**A** <math><mo>↵</mo></math> **B** <math><mi>x</mi><mo>↵</mo></math> **C** <math><mo>↵</mo></math> Empty editor → inserts ↵. The caret ends after the inserted element; typed characters follow it (typed “7” → ↵ 7).

“More” group 2 (21 controls) — expand button at 252.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-3-G2-001		Right left arrow with over script	255, 94, 33, 33 (open pop-up)	**A** <math><mover><mo>↔</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>↔</mo><mrow/></mover></math> **C** <math><mover><mo>↔</mo><mi>x</mi></mover></math> Empty editor → inserts over(↔; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(↔; 7)).
TB-3-G2-002		Right left arrow with under script	288, 94, 33, 33 (open pop-up)	**A** <math><munder><mo>↔</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>↔</mo><mrow/></munder></math> **C** <math><munder><mo>↔</mo><mi>x</mi></munder></math> Empty editor → inserts under(↔; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(↔; 7)).
TB-3-G2-003		Right left arrow with under and over script	321, 94, 33, 33 (open pop-up)	**A** <math><munderover><mo>↔</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>↔</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>↔</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(↔; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(↔; □; 7)).
TB-3-G2-004		Left arrow with over right arrow with over script	354, 94, 33, 33 (open pop-up)	**A** <math><mover><mo>⇆</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>⇆</mo><mrow/></mover></math> **C** <math><mover><mo>⇆</mo><mi>x</mi></mover></math> Empty editor → inserts over(⇆; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(⇆; 7)).
TB-3-G2-005		Left arrow over right arrow with under script	387, 94, 33, 33 (open pop-up)	**A** <math><munder><mo>⇆</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>⇆</mo><mrow/></munder></math> **C** <math><munder><mo>⇆</mo><mi>x</mi></munder></math> Empty editor → inserts under(⇆; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(⇆; 7)).
TB-3-G2-006		Left arrow over right arrow with under and over script	420, 94, 33, 33 (open pop-up)	**A** <math><munderover><mo>⇆</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>⇆</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>⇆</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(⇆; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(⇆; □; 7)).
TB-3-G2-007		Right arrow over left arrow with over script	453, 94, 33, 33 (open pop-up)	**A** <math><mover><mo>⇄</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>⇄</mo><mrow/></mover></math> **C** <math><mover><mo>⇄</mo><mi>x</mi></mover></math> Empty editor → inserts over(⇄; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(⇄; 7)).

TB-3-G2-008		Right arrow over left arrow with under script	486, 94, 33, 33 (open pop-up)	**A** <math><munder><mo>⇄</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>⇄</mo><mrow/></munder></math> **C** <math><munder><mo>⇄</mo><mi>x</mi></munder></math> Empty editor → inserts under(⇄; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(⇄; 7)).
TB-3-G2-009		Right arrow over left arrow with under and over script	519, 94, 33, 33 (open pop-up)	**A** <math><munderover><mo>⇄</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>⇄</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>⇄</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(⇄; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(⇄; □; 7)).
TB-3-G2-010		Left harpoon over right harpoon with over script	552, 94, 33, 33 (open pop-up)	**A** <math><mover><mo>⇋</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>⇋</mo><mrow/></mover></math> **C** <math><mover><mo>⇋</mo><mi>x</mi></mover></math> Empty editor → inserts over(⇋; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(⇋; 7)).
TB-3-G2-011		Left harpoon over right harpoon with under script	585, 94, 33, 33 (open pop-up)	**A** <math><munder><mo>⇋</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>⇋</mo><mrow/></munder></math> **C** <math><munder><mo>⇋</mo><mi>x</mi></munder></math> Empty editor → inserts under(⇋; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(⇋; 7)).
TB-3-G2-012		Left harpoon over right harpoon with under and over script	255, 127, 33, 33 (open pop-up)	**A** <math><munderover><mo>⇋</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>⇋</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>⇋</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(⇋; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(⇋; □; 7)).
TB-3-G2-013		Right harpoon over left harpoon with over script	288, 127, 33, 33 (open pop-up)	**A** <math><mover><mo>⇌</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>⇌</mo><mrow/></mover></math> **C** <math><mover><mo>⇌</mo><mi>x</mi></mover></math> Empty editor → inserts over(⇌; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(⇌; 7)).
TB-3-G2-014		Right harpoon over left harpoon with under script	321, 127, 33, 33 (open pop-up)	**A** <math><munder><mo>⇌</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>⇌</mo><mrow/></munder></math> **C** <math><munder><mo>⇌</mo><mi>x</mi></munder></math> Empty editor → inserts under(⇌; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(⇌; 7)).
TB-3-G2-015		Right harpoon over left harpoon with under and over script	354, 127, 33, 33 (open pop-up)	**A** <math><munderover><mo>⇌</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>⇌</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>⇌</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(⇌; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(⇌; □; 7)).
TB-3-G2-016		Rightwards arrow over short leftwards arrow with over script	387, 127, 33, 33 (open pop-up)	**A** <math><mover><mo>⥂</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>⥂</mo><mrow/></mover></math> **C** <math><mover><mo>⥂</mo><mi>x</mi></mover></math> Empty editor → inserts over(⥂; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(⥂; 7)).
TB-3-G2-017		Rightwards arrow over short leftwards arrow with under script	420, 127, 33, 33 (open pop-up)	**A** <math><munder><mo>⥂</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>⥂</mo><mrow/></munder></math> **C** <math><munder><mo>⥂</mo><mi>x</mi></munder></math> Empty editor → inserts under(⥂; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(⥂; 7)).
TB-3-G2-018		Rightwards arrow over short leftwards arrow with under and over script	453, 127, 33, 33 (open pop-up)	**A** <math><munderover><mo>⥂</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>⥂</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>⥂</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(⥂; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(⥂; □; 7)).
TB-3-G2-019		Short rightwards arrow over leftwards arrow with over script	486, 127, 33, 33 (open pop-up)	**A** <math><mover><mo>⥄</mo><mrow/></mover></math> **B** <math><mi>x</mi><mover><mo>⥄</mo><mrow/></mover></math> **C** <math><mover><mo>⥄</mo><mi>x</mi></mover></math> Empty editor → inserts over(⥄; □). The caret is placed in the over-script of the over-script; the next typed character goes there (typed “7” → over(⥄; 7)).
TB-3-G2-020		Short rightwards arrow over leftwards arrow with under script	519, 127, 33, 33 (open pop-up)	**A** <math><munder><mo>⥄</mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo>⥄</mo><mrow/></munder></math> **C** <math><munder><mo>⥄</mo><mi>x</mi></munder></math> Empty editor → inserts under(⥄; □). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → under(⥄; 7)).

TB-3-G2-021		Short rightwards arrow over leftwards arrow with under and over script	552, 127, 33, 33 (open pop-up)	**A** <math><munderover><mo>⥄</mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo>⥄</mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mo>⥄</mo><mrow/><mi>x</mi></munderover></math> Empty editor → inserts underover(⥤; □; □). The caret is placed in the over-script of the under+over scripts; the next typed character goes there (typed “7” → underover(⥤; □; 7)).

C.4 Tab 4 — Greek, letters and numbers (Alt+4)

Main panel (51 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-001		Alpha	3, 28, 22, 22	**A** <math><mi>α</mi></math> **B** <math><mi>x</mi><mi>α</mi></math> **C** <math><mi>α</mi></math> Empty editor → inserts α. The caret ends after the inserted element; typed characters follow it (typed “7” → α 7).
TB-4-002		Beta	25, 28, 22, 22	**A** <math><mi>β</mi></math> **B** <math><mi>x</mi><mi>β</mi></math> **C** <math><mi>β</mi></math> Empty editor → inserts β. The caret ends after the inserted element; typed characters follow it (typed “7” → β 7).
TB-4-003		Gamma	47, 28, 22, 22	**A** <math><mi>γ</mi></math> **B** <math><mi>x</mi><mi>γ</mi></math> **C** <math><mi>γ</mi></math> Empty editor → inserts γ. The caret ends after the inserted element; typed characters follow it (typed “7” → γ 7).
TB-4-004		Delta	69, 28, 22, 22	**A** <math><mi>δ</mi></math> **B** <math><mi>x</mi><mi>δ</mi></math> **C** <math><mi>δ</mi></math> Empty editor → inserts δ. The caret ends after the inserted element; typed characters follow it (typed “7” → δ 7).
TB-4-005		Epsilon	91, 28, 22, 22	**A** <math><mi>ε</mi></math> **B** <math><mi>x</mi><mi>ε</mi></math> **C** <math><mi>ε</mi></math> Empty editor → inserts ε. The caret ends after the inserted element; typed characters follow it (typed “7” → ε 7).
TB-4-006		Zeta	113, 28, 22, 22	**A** <math><mi>ζ</mi></math> **B** <math><mi>x</mi><mi>ζ</mi></math> **C** <math><mi>ζ</mi></math> Empty editor → inserts ζ. The caret ends after the inserted element; typed characters follow it (typed “7” → ζ 7).
TB-4-007		Eta	135, 28, 22, 22	**A** <math><mi>η</mi></math> **B** <math><mi>x</mi><mi>η</mi></math> **C** <math><mi>η</mi></math> Empty editor → inserts η. The caret ends after the inserted element; typed characters follow it (typed “7” → η 7).
TB-4-008		Theta	157, 28, 22, 22	**A** <math><mi>θ</mi></math> **B** <math><mi>x</mi><mi>θ</mi></math> **C** <math><mi>θ</mi></math> Empty editor → inserts θ. The caret ends after the inserted element; typed characters follow it (typed “7” → θ 7).
TB-4-009		Theta	179, 28, 22, 22	**A** <math><mi>ϑ</mi></math> **B** <math><mi>x</mi><mi>ϑ</mi></math> **C** <math><mi>ϑ</mi></math> Empty editor → inserts ϑ. The caret ends after the inserted element; typed characters follow it (typed “7” → ϑ 7).
TB-4-010		Iota	201, 28, 22, 22	**A** <math><mi>ι</mi></math> **B** <math><mi>x</mi><mi>ι</mi></math> **C** <math><mi>ι</mi></math> Empty editor → inserts ι. The caret ends after the inserted element; typed characters follow it (typed “7” → ι 7).
TB-4-011		Kappa	3, 50, 22, 22	**A** <math><mi>κ</mi></math> **B** <math><mi>x</mi><mi>κ</mi></math> **C** <math><mi>κ</mi></math> Empty editor → inserts κ. The caret ends after the inserted element; typed characters follow it (typed “7” → κ 7).
TB-4-012		Lambda	25, 50, 22, 22	**A** <math><mi>λ</mi></math> **B** <math><mi>x</mi><mi>λ</mi></math> **C** <math><mi>λ</mi></math> Empty editor → inserts λ. The caret ends after the inserted element; typed characters follow it (typed “7” → λ 7).

TB-4-013		Mu	47, 50, 22, 22	**A** <math><mi>μ</mi></math> **B** <math><mi>x</mi><mi>μ</mi></math> **C** <math><mi>μ</mi></math> Empty editor → inserts μ. The caret ends after the inserted element; typed characters follow it (typed “7” → μ 7).
TB-4-014		Nu	69, 50, 22, 22	**A** <math><mi>ν</mi></math> **B** <math><mi>x</mi><mi>ν</mi></math> **C** <math><mi>ν</mi></math> Empty editor → inserts ν. The caret ends after the inserted element; typed characters follow it (typed “7” → ν 7).
TB-4-015		Xi	91, 50, 22, 22	**A** <math><mi>ξ</mi></math> **B** <math><mi>x</mi><mi>ξ</mi></math> **C** <math><mi>ξ</mi></math> Empty editor → inserts ξ. The caret ends after the inserted element; typed characters follow it (typed “7” → ξ 7).
TB-4-016		Omicron	113, 50, 22, 22	**A** <math><mi>ο</mi></math> **B** <math><mi>x</mi><mi>ο</mi></math> **C** <math><mi>ο</mi></math> Empty editor → inserts ο. The caret ends after the inserted element; typed characters follow it (typed “7” → ο 7).
TB-4-017		Pi	135, 50, 22, 22	**A** <math><mi>π</mi></math> **B** <math><mi>x</mi><mi>π</mi></math> **C** <math><mi>π</mi></math> Empty editor → inserts π. The caret ends after the inserted element; typed characters follow it (typed “7” → π 7).
TB-4-018		Pi	157, 50, 22, 22	**A** <math><mi>ϖ</mi></math> **B** <math><mi>x</mi><mi>ϖ</mi></math> **C** <math><mi>ϖ</mi></math> Empty editor → inserts ϖ. The caret ends after the inserted element; typed characters follow it (typed “7” → ϖ 7).
TB-4-019		Rho	179, 50, 22, 22	**A** <math><mi>ρ</mi></math> **B** <math><mi>x</mi><mi>ρ</mi></math> **C** <math><mi>ρ</mi></math> Empty editor → inserts ρ. The caret ends after the inserted element; typed characters follow it (typed “7” → ρ 7).
TB-4-020		Final sigma	201, 50, 22, 22	**A** <math><mi>ς</mi></math> **B** <math><mi>x</mi><mi>ς</mi></math> **C** <math><mi>ς</mi></math> Empty editor → inserts ς. The caret ends after the inserted element; typed characters follow it (typed “7” → ς 7).
TB-4-021		Sigma	3, 72, 22, 22	**A** <math><mi>σ</mi></math> **B** <math><mi>x</mi><mi>σ</mi></math> **C** <math><mi>σ</mi></math> Empty editor → inserts σ. The caret ends after the inserted element; typed characters follow it (typed “7” → σ 7).
TB-4-022		Tau	25, 72, 22, 22	**A** <math><mi>τ</mi></math> **B** <math><mi>x</mi><mi>τ</mi></math> **C** <math><mi>τ</mi></math> Empty editor → inserts τ. The caret ends after the inserted element; typed characters follow it (typed “7” → τ 7).
TB-4-023		Upsilon	47, 72, 22, 22	**A** <math><mi>υ</mi></math> **B** <math><mi>x</mi><mi>υ</mi></math> **C** <math><mi>υ</mi></math> Empty editor → inserts υ. The caret ends after the inserted element; typed characters follow it (typed “7” → υ 7).
TB-4-024		Phi	69, 72, 22, 22	**A** <math><mi>φ</mi></math> **B** <math><mi>x</mi><mi>φ</mi></math> **C** <math><mi>φ</mi></math> Empty editor → inserts φ. The caret ends after the inserted element; typed characters follow it (typed “7” → φ 7).
TB-4-025		Phi	91, 72, 22, 22	**A** <math><mi>ϕ</mi></math> **B** <math><mi>x</mi><mi>ϕ</mi></math> **C** <math><mi>ϕ</mi></math> Empty editor → inserts ϕ. The caret ends after the inserted element; typed characters follow it (typed “7” → ϕ 7).
TB-4-026		Chi	113, 72, 22, 22	**A** <math><mi>χ</mi></math> **B** <math><mi>x</mi><mi>χ</mi></math> **C** <math><mi>χ</mi></math> Empty editor → inserts χ. The caret ends after the inserted element; typed characters follow it (typed “7” → χ 7).
TB-4-027		Psi	135, 72, 22, 22	**A** <math><mi>ψ</mi></math> **B** <math><mi>x</mi><mi>ψ</mi></math> **C** <math><mi>ψ</mi></math> Empty editor → inserts ψ. The caret ends after the inserted element; typed characters follow it (typed “7” → ψ 7).
TB-4-028		Omega	157, 72, 22, 22	**A** <math><mi>ω</mi></math> **B** <math><mi>x</mi><mi>ω</mi></math> **C** <math><mi>ω</mi></math> Empty editor → inserts ω. The caret ends after the inserted element; typed characters follow it (typed “7” → ω 7).

TB-4-029		Natural numbers (double-struck capital n)	234, 28, 22, 22	**A** <math><mi mathvariant="normal">ℕ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℕ</mi></math> **C** <math><mi mathvariant="normal">ℕ</mi></math> Empty editor → inserts ℕ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℕ 7).
TB-4-030		Integer numbers (double-struck capital z)	256, 28, 22, 22	**A** <math><mi mathvariant="normal">ℤ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℤ</mi></math> **C** <math><mi mathvariant="normal">ℤ</mi></math> Empty editor → inserts ℤ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℤ 7).
TB-4-031		Rational numbers (double-struck capital q)	234, 50, 22, 22	**A** <math><mi mathvariant="normal">ℚ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℚ</mi></math> **C** <math><mi mathvariant="normal">ℚ</mi></math> Empty editor → inserts ℚ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℚ 7).
TB-4-032		Complex numbers (double-struck capital c)	256, 50, 22, 22	**A** <math><mi mathvariant="normal">ℂ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℂ</mi></math> **C** <math><mi mathvariant="normal">ℂ</mi></math> Empty editor → inserts ℂ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℂ 7).
TB-4-033		Real numbers (double-struck capital r)	234, 72, 22, 22	**A** <math><mi mathvariant="normal">ℝ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℝ</mi></math> **C** <math><mi mathvariant="normal">ℝ</mi></math> Empty editor → inserts ℝ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℝ 7).
TB-4-034		Prime numbers (double-struck capital p)	256, 72, 22, 22	**A** <math><mi mathvariant="normal">ℙ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℙ</mi></math> **C** <math><mi mathvariant="normal">ℙ</mi></math> Empty editor → inserts ℙ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℙ 7).
TB-4-035		Fraktur capital a	289, 28, 22, 22	**A** <math><mi mathvariant="fraktur">A</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">A</mi></math> **C** <math><mi mathvariant="fraktur">A</mi></math> Empty editor → inserts A. The caret ends after the inserted element; typed characters follow it (typed “7” → A 7).
TB-4-036		Fraktur capital b	289, 50, 22, 22	**A** <math><mi mathvariant="fraktur">B</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">B</mi></math> **C** <math><mi mathvariant="fraktur">B</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B 7).
TB-4-037		Fraktur capital c	289, 72, 22, 22	**A** <math><mi mathvariant="fraktur">C</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">C</mi></math> **C** <math><mi mathvariant="fraktur">C</mi></math> Empty editor → inserts C. The caret ends after the inserted element; typed characters follow it (typed “7” → C 7).
TB-4-038		Script capital a	322, 28, 22, 22	**A** <math><mi mathvariant="script">A</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">A</mi></math> **C** <math><mi mathvariant="script">A</mi></math> Empty editor → inserts A. The caret ends after the inserted element; typed characters follow it (typed “7” → A 7).
TB-4-039		Script capital b	322, 50, 22, 22	**A** <math><mi mathvariant="script">B</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">B</mi></math> **C** <math><mi mathvariant="script">B</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B 7).
TB-4-040		Script capital c	322, 72, 22, 22	**A** <math><mi mathvariant="script">C</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">C</mi></math> **C** <math><mi mathvariant="script">C</mi></math> Empty editor → inserts C. The caret ends after the inserted element; typed characters follow it (typed “7” → C 7).
TB-4-041		Imaginary part (black-letter capital i)	355, 28, 22, 22	**A** <math><mi mathvariant="fraktur">I</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">I</mi></math> **C** <math><mi mathvariant="fraktur">I</mi></math> Empty editor → inserts I. The caret ends after the inserted element; typed characters follow it (typed “7” → I 7).
TB-4-042		Real part (black-letter capital r)	355, 50, 22, 22	**A** <math><mi mathvariant="fraktur">R</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">R</mi></math> **C** <math><mi mathvariant="fraktur">R</mi></math> Empty editor → inserts R. The caret ends after the inserted element; typed characters follow it (typed “7” → R 7).
TB-4-043		Ell	355, 72, 22, 22	**A** <math><mi mathvariant="script">l</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">l</mi></math> **C** <math><mi mathvariant="script">l</mi></math> Empty editor → inserts l. The caret ends after the inserted element; typed characters follow it (typed “7” → l 7).
TB-4-044		Arabic-Indic numerals	388, 28, 22, 22	**A** <math/> **B** <math><mi>x</mi></math> **C** <math><mi>x</mi></math> On an empty editor the click inserts nothing visible, but typed text afterwards appears: ∨. With “x” typed before the caret: x.

TB-4-045	٧٧	Eastern Arabic-Indic numerals	388, 50, 22, 22	**A** <math/> **B** <math><mi>x</mi></math> **C** <math><mi>x</mi></math> On an empty editor the click inserts nothing visible, but typed text afterwards appears: ٧. With “x” typed before the caret: x.
TB-4-046	H	Hydrogen	413, 28, 22, 22	**A** <math><mi mathvariant="normal">H</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">H</mi></math> **C** <math><mi mathvariant="normal">H</mi></math> Empty editor → inserts H. The caret ends after the inserted element; typed characters follow it (typed “7” → H ٧).
TB-4-047	C	Carbon	435, 28, 22, 22	**A** <math><mi mathvariant="normal">C</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">C</mi></math> **C** <math><mi mathvariant="normal">C</mi></math> Empty editor → inserts C. The caret ends after the inserted element; typed characters follow it (typed “7” → C ٧).
TB-4-048	N	Nitrogen	413, 50, 22, 22	**A** <math><mi mathvariant="normal">N</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">N</mi></math> **C** <math><mi mathvariant="normal">N</mi></math> Empty editor → inserts N. The caret ends after the inserted element; typed characters follow it (typed “7” → N ٧).
TB-4-049	O	Oxygen	435, 50, 22, 22	**A** <math><mi mathvariant="normal">O</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">O</mi></math> **C** <math><mi mathvariant="normal">O</mi></math> Empty editor → inserts O. The caret ends after the inserted element; typed characters follow it (typed “7” → O ٧).
TB-4-050	F	Fluorine	413, 72, 22, 22	**A** <math><mi mathvariant="normal">F</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">F</mi></math> **C** <math><mi mathvariant="normal">F</mi></math> Empty editor → inserts F. The caret ends after the inserted element; typed characters follow it (typed “7” → F ٧).
TB-4-051	S	Sulfur	435, 72, 22, 22	**A** <math><mi mathvariant="normal">S</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">S</mi></math> **C** <math><mi mathvariant="normal">S</mi></math> Empty editor → inserts S. The caret ends after the inserted element; typed characters follow it (typed “7” → S ٧).

“More” group 1 (24 controls) — expand button at 224.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-G1-001	A	Capital alpha	227, 94, 22, 22 (open pop-up)	**A** <math><mi>Α</mi></math> **B** <math><mi>x</mi><mi>Α</mi></math> **C** <math><mi>Α</mi></math> Empty editor → inserts A. The caret ends after the inserted element; typed characters follow it (typed “7” → A ٧).
TB-4-G1-002	B	Capital beta	249, 94, 22, 22 (open pop-up)	**A** <math><mi>Β</mi></math> **B** <math><mi>x</mi><mi>Β</mi></math> **C** <math><mi>Β</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B ٧).
TB-4-G1-003	Γ	Capital gamma	271, 94, 22, 22 (open pop-up)	**A** <math><mi>Γ</mi></math> **B** <math><mi>x</mi><mi>Γ</mi></math> **C** <math><mi>Γ</mi></math> Empty editor → inserts Γ. The caret ends after the inserted element; typed characters follow it (typed “7” → Γ ٧).
TB-4-G1-004	Δ	Capital delta	293, 94, 22, 22 (open pop-up)	**A** <math><mi>Δ</mi></math> **B** <math><mi>x</mi><mi>Δ</mi></math> **C** <math><mi>Δ</mi></math> Empty editor → inserts Δ. The caret ends after the inserted element; typed characters follow it (typed “7” → Δ ٧).
TB-4-G1-005	E	Capital epsilon	315, 94, 22, 22 (open pop-up)	**A** <math><mi>Ε</mi></math> **B** <math><mi>x</mi><mi>Ε</mi></math> **C** <math><mi>Ε</mi></math> Empty editor → inserts E. The caret ends after the inserted element; typed characters follow it (typed “7” → E ٧).
TB-4-G1-006	Z	Capital zeta	337, 94, 22, 22 (open pop-up)	**A** <math><mi>Ζ</mi></math> **B** <math><mi>x</mi><mi>Ζ</mi></math> **C** <math><mi>Ζ</mi></math> Empty editor → inserts Z. The caret ends after the inserted element; typed characters follow it (typed “7” → Z ٧).
TB-4-G1-007	H	Capital eta	359, 94, 22, 22 (open pop-up)	**A** <math><mi>Η</mi></math> **B** <math><mi>x</mi><mi>Η</mi></math> **C** <math><mi>Η</mi></math> Empty editor → inserts H. The caret ends after the inserted element; typed characters follow it (typed “7” → H ٧).

TB-4-G1-008		Capital theta	381, 94, 22, 22 (open pop-up)	**A** <math><mi>Θ</mi></math> **B** <math><mi>x</mi><mi>Θ</mi></math> **C** <math><mi>Θ</mi></math> Empty editor → inserts Θ. The caret ends after the inserted element; typed characters follow it (typed “7” → Θ ∀).
TB-4-G1-009		Capital iota	227, 116, 22, 22 (open pop-up)	**A** <math><mi>Ι</mi></math> **B** <math><mi>x</mi><mi>Ι</mi></math> **C** <math><mi>Ι</mi></math> Empty editor → inserts Ι. The caret ends after the inserted element; typed characters follow it (typed “7” → Ι ∀).
TB-4-G1-010		Capital kappa	249, 116, 22, 22 (open pop-up)	**A** <math><mi>Κ</mi></math> **B** <math><mi>x</mi><mi>Κ</mi></math> **C** <math><mi>Κ</mi></math> Empty editor → inserts Κ. The caret ends after the inserted element; typed characters follow it (typed “7” → Κ ∀).
TB-4-G1-011		Capital lambda	271, 116, 22, 22 (open pop-up)	**A** <math><mi>Λ</mi></math> **B** <math><mi>x</mi><mi>Λ</mi></math> **C** <math><mi>Λ</mi></math> Empty editor → inserts Λ. The caret ends after the inserted element; typed characters follow it (typed “7” → Λ ∀).
TB-4-G1-012		Capital mu	293, 116, 22, 22 (open pop-up)	**A** <math><mi>Μ</mi></math> **B** <math><mi>x</mi><mi>Μ</mi></math> **C** <math><mi>Μ</mi></math> Empty editor → inserts Μ. The caret ends after the inserted element; typed characters follow it (typed “7” → Μ ∀).
TB-4-G1-013		Capital nu	315, 116, 22, 22 (open pop-up)	**A** <math><mi>Ν</mi></math> **B** <math><mi>x</mi><mi>Ν</mi></math> **C** <math><mi>Ν</mi></math> Empty editor → inserts Ν. The caret ends after the inserted element; typed characters follow it (typed “7” → Ν ∀).
TB-4-G1-014		Capital xi	337, 116, 22, 22 (open pop-up)	**A** <math><mi>Ξ</mi></math> **B** <math><mi>x</mi><mi>Ξ</mi></math> **C** <math><mi>Ξ</mi></math> Empty editor → inserts Ξ. The caret ends after the inserted element; typed characters follow it (typed “7” → Ξ ∀).
TB-4-G1-015		Capital omicron	359, 116, 22, 22 (open pop-up)	**A** <math><mi>Ο</mi></math> **B** <math><mi>x</mi><mi>Ο</mi></math> **C** <math><mi>Ο</mi></math> Empty editor → inserts Ο. The caret ends after the inserted element; typed characters follow it (typed “7” → Ο ∀).
TB-4-G1-016		Capital pi	381, 116, 22, 22 (open pop-up)	**A** <math><mi>Π</mi></math> **B** <math><mi>x</mi><mi>Π</mi></math> **C** <math><mi>Π</mi></math> Empty editor → inserts Π. The caret ends after the inserted element; typed characters follow it (typed “7” → Π ∀).
TB-4-G1-017		Capital rho	227, 138, 22, 22 (open pop-up)	**A** <math><mi>Ρ</mi></math> **B** <math><mi>x</mi><mi>Ρ</mi></math> **C** <math><mi>Ρ</mi></math> Empty editor → inserts Ρ. The caret ends after the inserted element; typed characters follow it (typed “7” → Ρ ∀).
TB-4-G1-018		Capital sigma	249, 138, 22, 22 (open pop-up)	**A** <math><mi>Σ</mi></math> **B** <math><mi>x</mi><mi>Σ</mi></math> **C** <math><mi>Σ</mi></math> Empty editor → inserts Σ. The caret ends after the inserted element; typed characters follow it (typed “7” → Σ ∀).
TB-4-G1-019		Capital tau	271, 138, 22, 22 (open pop-up)	**A** <math><mi>Τ</mi></math> **B** <math><mi>x</mi><mi>Τ</mi></math> **C** <math><mi>Τ</mi></math> Empty editor → inserts Τ. The caret ends after the inserted element; typed characters follow it (typed “7” → Τ ∀).
TB-4-G1-020		Capital upsilon	293, 138, 22, 22 (open pop-up)	**A** <math><mi>Υ</mi></math> **B** <math><mi>x</mi><mi>Υ</mi></math> **C** <math><mi>Υ</mi></math> Empty editor → inserts Υ. The caret ends after the inserted element; typed characters follow it (typed “7” → Υ ∀).
TB-4-G1-021		Capital phi	315, 138, 22, 22 (open pop-up)	**A** <math><mi>Φ</mi></math> **B** <math><mi>x</mi><mi>Φ</mi></math> **C** <math><mi>Φ</mi></math> Empty editor → inserts Φ. The caret ends after the inserted element; typed characters follow it (typed “7” → Φ ∀).
TB-4-G1-022		Capital chi	337, 138, 22, 22 (open pop-up)	**A** <math><mi>Χ</mi></math> **B** <math><mi>x</mi><mi>Χ</mi></math> **C** <math><mi>Χ</mi></math> Empty editor → inserts Χ. The caret ends after the inserted element; typed characters follow it (typed “7” → Χ ∀).

TB-4-G1-023	Ψ	Capital psi	359, 138, 22, 22 (open pop-up)	**A** <math><mi>Ψ</mi></math> **B** <math><mi>x</mi><mi>Ψ</mi></math> **C** <math><mi>Ψ</mi></math> Empty editor → inserts Ψ. The caret ends after the inserted element; typed characters follow it (typed “7” → Ψ ∨).
TB-4-G1-024	Ω	Capital omega	381, 138, 22, 22 (open pop-up)	**A** <math><mi>Ω</mi></math> **B** <math><mi>x</mi><mi>Ω</mi></math> **C** <math><mi>Ω</mi></math> Empty editor → inserts Ω. The caret ends after the inserted element; typed characters follow it (typed “7” → Ω ∨).

“More” group 2 (52 controls) — expand button at 279.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-G2-001	𝔸	Double struck capital a	282, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">A</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">A</mi></math> **C** <math><mi mathvariant="double-struck">A</mi></math> Empty editor → inserts A. The caret ends after the inserted element; typed characters follow it (typed “7” → A ∨).
TB-4-G2-002	𝔻	Double struck capital d	304, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">D</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">D</mi></math> **C** <math><mi mathvariant="double-struck">D</mi></math> Empty editor → inserts D. The caret ends after the inserted element; typed characters follow it (typed “7” → D ∨).
TB-4-G2-003	𝔾	Double struck capital g	326, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">G</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">G</mi></math> **C** <math><mi mathvariant="double-struck">G</mi></math> Empty editor → inserts G. The caret ends after the inserted element; typed characters follow it (typed “7” → G ∨).
TB-4-G2-004	𝕁	Double struck capital j	348, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">J</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">J</mi></math> **C** <math><mi mathvariant="double-struck">J</mi></math> Empty editor → inserts J. The caret ends after the inserted element; typed characters follow it (typed “7” → J ∨).
TB-4-G2-005	𝕄	Double struck capital m	370, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">M</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">M</mi></math> **C** <math><mi mathvariant="double-struck">M</mi></math> Empty editor → inserts M. The caret ends after the inserted element; typed characters follow it (typed “7” → M ∨).
TB-4-G2-006	ℙ	Prime numbers (double-struck capital p)	392, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℙ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℙ</mi></math> **C** <math><mi mathvariant="normal">ℙ</mi></math> Empty editor → inserts ℙ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℙ ∨).
TB-4-G2-007	𝕊	Double struck capital s	414, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">S</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">S</mi></math> **C** <math><mi mathvariant="double-struck">S</mi></math> Empty editor → inserts S. The caret ends after the inserted element; typed characters follow it (typed “7” → S ∨).
TB-4-G2-008	𝕍	Double struck capital v	436, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">V</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">V</mi></math> **C** <math><mi mathvariant="double-struck">V</mi></math> Empty editor → inserts V. The caret ends after the inserted element; typed characters follow it (typed “7” → V ∨).
TB-4-G2-009	𝕐	Double struck capital y	458, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">Y</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">Y</mi></math> **C** <math><mi mathvariant="double-struck">Y</mi></math> Empty editor → inserts Y. The caret ends after the inserted element; typed characters follow it (typed “7” → Y ∨).
TB-4-G2-010	𝕒	Double struck a	480, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">a</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">a</mi></math> **C** <math><mi mathvariant="double-struck">a</mi></math> Empty editor → inserts a. The caret ends after the inserted element; typed characters follow it (typed “7” → a ∨).
TB-4-G2-011	𝕕	Double struck d	502, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">d</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">d</mi></math> **C** <math><mi mathvariant="double-struck">d</mi></math> Empty editor → inserts d. The caret ends after the inserted element; typed characters follow it (typed “7” → d ∨).
TB-4-G2-012	𝕘	Double struck g	524, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">g</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">g</mi></math> **C** <math><mi mathvariant="double-struck">g</mi></math> Empty editor → inserts g. The caret ends after the inserted element; typed characters follow it (typed “7” → g ∨).

TB-4-G2-013		Double struck j	546, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">j</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">j</mi></math> **C** <math><mi mathvariant="double-struck">j</mi></math> Empty editor → inserts j. The caret ends after the inserted element; typed characters follow it (typed “7” → j ∀).
TB-4-G2-014		Double struck m	568, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">m</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">m</mi></math> **C** <math><mi mathvariant="double-struck">m</mi></math> Empty editor → inserts m. The caret ends after the inserted element; typed characters follow it (typed “7” → m ∀).
TB-4-G2-015		Double struck p	590, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">p</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">p</mi></math> **C** <math><mi mathvariant="double-struck">p</mi></math> Empty editor → inserts p. The caret ends after the inserted element; typed characters follow it (typed “7” → p ∀).
TB-4-G2-016		Double struck s	612, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">s</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">s</mi></math> **C** <math><mi mathvariant="double-struck">s</mi></math> Empty editor → inserts s. The caret ends after the inserted element; typed characters follow it (typed “7” → s ∀).
TB-4-G2-017		Double struck v	634, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">v</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">v</mi></math> **C** <math><mi mathvariant="double-struck">v</mi></math> Empty editor → inserts v. The caret ends after the inserted element; typed characters follow it (typed “7” → v ∀).
TB-4-G2-018		Double struck y	656, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">y</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">y</mi></math> **C** <math><mi mathvariant="double-struck">y</mi></math> Empty editor → inserts y. The caret ends after the inserted element; typed characters follow it (typed “7” → y ∀).
TB-4-G2-019		Double struck capital b	282, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">B</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">B</mi></math> **C** <math><mi mathvariant="double-struck">B</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B ∀).
TB-4-G2-020		Double struck capital e	304, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">E</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">E</mi></math> **C** <math><mi mathvariant="double-struck">E</mi></math> Empty editor → inserts E. The caret ends after the inserted element; typed characters follow it (typed “7” → E ∀).
TB-4-G2-021		Double struck capital h	326, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℍ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℍ</mi></math> **C** <math><mi mathvariant="normal">ℍ</mi></math> Empty editor → inserts ℍ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℍ ∀).
TB-4-G2-022		Double struck capital k	348, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">K</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">K</mi></math> **C** <math><mi mathvariant="double-struck">K</mi></math> Empty editor → inserts K. The caret ends after the inserted element; typed characters follow it (typed “7” → K ∀).
TB-4-G2-023		Natural numbers (double-struck capital n)	370, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℕ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℕ</mi></math> **C** <math><mi mathvariant="normal">ℕ</mi></math> Empty editor → inserts ℕ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℕ ∀).
TB-4-G2-024		Rational numbers (double-struck capital q)	392, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℚ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℚ</mi></math> **C** <math><mi mathvariant="normal">ℚ</mi></math> Empty editor → inserts ℚ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℚ ∀).
TB-4-G2-025		Double struck capital t	414, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">T</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">T</mi></math> **C** <math><mi mathvariant="double-struck">T</mi></math> Empty editor → inserts T. The caret ends after the inserted element; typed characters follow it (typed “7” → T ∀).
TB-4-G2-026		Double struck capital w	436, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">W</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">W</mi></math> **C** <math><mi mathvariant="double-struck">W</mi></math> Empty editor → inserts W. The caret ends after the inserted element; typed characters follow it (typed “7” → W ∀).
TB-4-G2-027		Integer numbers (double-struck capital z)	458, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℤ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℤ</mi></math> **C** <math><mi mathvariant="normal">ℤ</mi></math> Empty editor → inserts ℤ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℤ ∀).

TB-4-G2-028		Double struck b	480, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">b</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">b</mi></math> **C** <math><mi mathvariant="double-struck">b</mi></math> Empty editor → inserts b. The caret ends after the inserted element; typed characters follow it (typed “7” → b ∀).
TB-4-G2-029		Double struck e	502, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">e</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">e</mi></math> **C** <math><mi mathvariant="double-struck">e</mi></math> Empty editor → inserts e. The caret ends after the inserted element; typed characters follow it (typed “7” → e ∀).
TB-4-G2-030		Double struck h	524, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">h</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">h</mi></math> **C** <math><mi mathvariant="double-struck">h</mi></math> Empty editor → inserts h. The caret ends after the inserted element; typed characters follow it (typed “7” → h ∀).
TB-4-G2-031		Double struck k	546, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">k</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">k</mi></math> **C** <math><mi mathvariant="double-struck">k</mi></math> Empty editor → inserts k. The caret ends after the inserted element; typed characters follow it (typed “7” → k ∀).
TB-4-G2-032		Double struck n	568, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">n</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">n</mi></math> **C** <math><mi mathvariant="double-struck">n</mi></math> Empty editor → inserts n. The caret ends after the inserted element; typed characters follow it (typed “7” → n ∀).
TB-4-G2-033		Double struck q	590, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">q</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">q</mi></math> **C** <math><mi mathvariant="double-struck">q</mi></math> Empty editor → inserts q. The caret ends after the inserted element; typed characters follow it (typed “7” → q ∀).
TB-4-G2-034		Double struck t	612, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">t</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">t</mi></math> **C** <math><mi mathvariant="double-struck">t</mi></math> Empty editor → inserts t. The caret ends after the inserted element; typed characters follow it (typed “7” → t ∀).
TB-4-G2-035		Double struck w	634, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">w</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">w</mi></math> **C** <math><mi mathvariant="double-struck">w</mi></math> Empty editor → inserts w. The caret ends after the inserted element; typed characters follow it (typed “7” → w ∀).
TB-4-G2-036		Double struck z	656, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">z</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">z</mi></math> **C** <math><mi mathvariant="double-struck">z</mi></math> Empty editor → inserts z. The caret ends after the inserted element; typed characters follow it (typed “7” → z ∀).
TB-4-G2-037		Complex numbers (double-struck capital c)	282, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℂ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℂ</mi></math> **C** <math><mi mathvariant="normal">ℂ</mi></math> Empty editor → inserts ℂ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℂ ∀).
TB-4-G2-038		Double struck capital f	304, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">F</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">F</mi></math> **C** <math><mi mathvariant="double-struck">F</mi></math> Empty editor → inserts F. The caret ends after the inserted element; typed characters follow it (typed “7” → F ∀).
TB-4-G2-039		Imaginary numbers (double-struck capital i)	326, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">𝕀</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">𝕀</mi></math> **C** <math><mi mathvariant="normal">𝕀</mi></math> Empty editor → inserts 𝕀. The caret ends after the inserted element; typed characters follow it (typed “7” → 𝕀 ∀).
TB-4-G2-040		Double struck capital l	348, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">L</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">L</mi></math> **C** <math><mi mathvariant="double-struck">L</mi></math> Empty editor → inserts L. The caret ends after the inserted element; typed characters follow it (typed “7” → L ∀).
TB-4-G2-041		Double struck capital o	370, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">O</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">O</mi></math> **C** <math><mi mathvariant="double-struck">O</mi></math> Empty editor → inserts O. The caret ends after the inserted element; typed characters follow it (typed “7” → O ∀).
TB-4-G2-042		Real numbers (double-struck capital r)	392, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">ℝ</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">ℝ</mi></math> **C** <math><mi mathvariant="normal">ℝ</mi></math> Empty editor → inserts ℝ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℝ ∀).

TB-4-G2-043		Double struck capital u	414, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">U</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">U</mi></math> **C** <math><mi mathvariant="double-struck">U</mi></math> Empty editor → inserts U. The caret ends after the inserted element; typed characters follow it (typed “7” → U ∨).
TB-4-G2-044		Double struck capital x	436, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">X</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">X</mi></math> **C** <math><mi mathvariant="double-struck">X</mi></math> Empty editor → inserts X. The caret ends after the inserted element; typed characters follow it (typed “7” → X ∨).
TB-4-G2-045		Double struck c	480, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">c</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">c</mi></math> **C** <math><mi mathvariant="double-struck">c</mi></math> Empty editor → inserts c. The caret ends after the inserted element; typed characters follow it (typed “7” → c ∨).
TB-4-G2-046		Double struck f	502, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">f</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">f</mi></math> **C** <math><mi mathvariant="double-struck">f</mi></math> Empty editor → inserts f. The caret ends after the inserted element; typed characters follow it (typed “7” → f ∨).
TB-4-G2-047		Double struck i	524, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">i</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">i</mi></math> **C** <math><mi mathvariant="double-struck">i</mi></math> Empty editor → inserts i. The caret ends after the inserted element; typed characters follow it (typed “7” → i ∨).
TB-4-G2-048		Double struck l	546, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">l</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">l</mi></math> **C** <math><mi mathvariant="double-struck">l</mi></math> Empty editor → inserts l. The caret ends after the inserted element; typed characters follow it (typed “7” → l ∨).
TB-4-G2-049		Double struck o	568, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">o</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">o</mi></math> **C** <math><mi mathvariant="double-struck">o</mi></math> Empty editor → inserts o. The caret ends after the inserted element; typed characters follow it (typed “7” → o ∨).
TB-4-G2-050		Double struck r	590, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">r</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">r</mi></math> **C** <math><mi mathvariant="double-struck">r</mi></math> Empty editor → inserts r. The caret ends after the inserted element; typed characters follow it (typed “7” → r ∨).
TB-4-G2-051		Double struck u	612, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">u</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">u</mi></math> **C** <math><mi mathvariant="double-struck">u</mi></math> Empty editor → inserts u. The caret ends after the inserted element; typed characters follow it (typed “7” → u ∨).
TB-4-G2-052		Double struck x	634, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="double-struck">x</mi></math> **B** <math><mi>x</mi><mi mathvariant="double-struck">x</mi></math> **C** <math><mi mathvariant="double-struck">x</mi></math> Empty editor → inserts x. The caret ends after the inserted element; typed characters follow it (typed “7” → x ∨).

“More” group 3 (52 controls) — expand button at 312.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-G3-001		Fraktur capital a	315, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">A</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">A</mi></math> **C** <math><mi mathvariant="fraktur">A</mi></math> Empty editor → inserts A. The caret ends after the inserted element; typed characters follow it (typed “7” → A ∨).
TB-4-G3-002		Fraktur capital d	337, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">D</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">D</mi></math> **C** <math><mi mathvariant="fraktur">D</mi></math> Empty editor → inserts D. The caret ends after the inserted element; typed characters follow it (typed “7” → D ∨).
TB-4-G3-003		Fraktur capital g	359, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">G</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">G</mi></math> **C** <math><mi mathvariant="fraktur">G</mi></math> Empty editor → inserts G. The caret ends after the inserted element; typed characters follow it (typed “7” → G ∨).
TB-4-G3-004		Fraktur capital j	381, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">J</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">J</mi></math> **C** <math><mi mathvariant="fraktur">J</mi></math> Empty editor → inserts J. The caret ends after the inserted element; typed characters follow it (typed “7” → J ∨).

TB-4-G3-005		Fraktur capital m	403, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">M</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">M</mi></math> **C** <math><mi mathvariant="fraktur">M</mi></math> Empty editor → inserts M. The caret ends after the inserted element; typed characters follow it (typed “7” → M ∀).
TB-4-G3-006		Fraktur capital p	425, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">P</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">P</mi></math> **C** <math><mi mathvariant="fraktur">P</mi></math> Empty editor → inserts P. The caret ends after the inserted element; typed characters follow it (typed “7” → P ∀).
TB-4-G3-007		Fraktur capital s	447, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">S</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">S</mi></math> **C** <math><mi mathvariant="fraktur">S</mi></math> Empty editor → inserts S. The caret ends after the inserted element; typed characters follow it (typed “7” → S ∀).
TB-4-G3-008		Fraktur capital v	469, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">V</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">V</mi></math> **C** <math><mi mathvariant="fraktur">V</mi></math> Empty editor → inserts V. The caret ends after the inserted element; typed characters follow it (typed “7” → V ∀).
TB-4-G3-009		Fraktur capital y	491, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">Y</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">Y</mi></math> **C** <math><mi mathvariant="fraktur">Y</mi></math> Empty editor → inserts Y. The caret ends after the inserted element; typed characters follow it (typed “7” → Y ∀).
TB-4-G3-010		Fraktur a	513, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">a</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">a</mi></math> **C** <math><mi mathvariant="fraktur">a</mi></math> Empty editor → inserts a. The caret ends after the inserted element; typed characters follow it (typed “7” → a ∀).
TB-4-G3-011		Fraktur d	535, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">d</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">d</mi></math> **C** <math><mi mathvariant="fraktur">d</mi></math> Empty editor → inserts d. The caret ends after the inserted element; typed characters follow it (typed “7” → d ∀).
TB-4-G3-012		Fraktur g	557, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">g</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">g</mi></math> **C** <math><mi mathvariant="fraktur">g</mi></math> Empty editor → inserts g. The caret ends after the inserted element; typed characters follow it (typed “7” → g ∀).
TB-4-G3-013		Fraktur j	579, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">j</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">j</mi></math> **C** <math><mi mathvariant="fraktur">j</mi></math> Empty editor → inserts j. The caret ends after the inserted element; typed characters follow it (typed “7” → j ∀).
TB-4-G3-014		Fraktur m	601, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">m</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">m</mi></math> **C** <math><mi mathvariant="fraktur">m</mi></math> Empty editor → inserts m. The caret ends after the inserted element; typed characters follow it (typed “7” → m ∀).
TB-4-G3-015		Fraktur p	623, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">p</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">p</mi></math> **C** <math><mi mathvariant="fraktur">p</mi></math> Empty editor → inserts p. The caret ends after the inserted element; typed characters follow it (typed “7” → p ∀).
TB-4-G3-016		Fraktur s	645, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">s</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">s</mi></math> **C** <math><mi mathvariant="fraktur">s</mi></math> Empty editor → inserts s. The caret ends after the inserted element; typed characters follow it (typed “7” → s ∀).
TB-4-G3-017		Fraktur v	667, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">v</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">v</mi></math> **C** <math><mi mathvariant="fraktur">v</mi></math> Empty editor → inserts v. The caret ends after the inserted element; typed characters follow it (typed “7” → v ∀).
TB-4-G3-018		Fraktur y	689, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">y</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">y</mi></math> **C** <math><mi mathvariant="fraktur">y</mi></math> Empty editor → inserts y. The caret ends after the inserted element; typed characters follow it (typed “7” → y ∀).
TB-4-G3-019		Fraktur capital b	315, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">B</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">B</mi></math> **C** <math><mi mathvariant="fraktur">B</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B ∀).

TB-4-G3-020		Fraktur capital e	337, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">E</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">E</mi></math> **C** <math><mi mathvariant="fraktur">E</mi></math> Empty editor → inserts E. The caret ends after the inserted element; typed characters follow it (typed “7” → E ∨).
TB-4-G3-021		Fraktur capital h	359, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">H</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">H</mi></math> **C** <math><mi mathvariant="fraktur">H</mi></math> Empty editor → inserts H. The caret ends after the inserted element; typed characters follow it (typed “7” → H ∨).
TB-4-G3-022		Fraktur capital k	381, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">K</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">K</mi></math> **C** <math><mi mathvariant="fraktur">K</mi></math> Empty editor → inserts K. The caret ends after the inserted element; typed characters follow it (typed “7” → K ∨).
TB-4-G3-023		Fraktur capital n	403, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">N</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">N</mi></math> **C** <math><mi mathvariant="fraktur">N</mi></math> Empty editor → inserts N. The caret ends after the inserted element; typed characters follow it (typed “7” → N ∨).
TB-4-G3-024		Fraktur capital q	425, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">Q</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">Q</mi></math> **C** <math><mi mathvariant="fraktur">Q</mi></math> Empty editor → inserts Q. The caret ends after the inserted element; typed characters follow it (typed “7” → Q ∨).
TB-4-G3-025		Fraktur capital t	447, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">T</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">T</mi></math> **C** <math><mi mathvariant="fraktur">T</mi></math> Empty editor → inserts T. The caret ends after the inserted element; typed characters follow it (typed “7” → T ∨).
TB-4-G3-026		Fraktur capital w	469, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">W</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">W</mi></math> **C** <math><mi mathvariant="fraktur">W</mi></math> Empty editor → inserts W. The caret ends after the inserted element; typed characters follow it (typed “7” → W ∨).
TB-4-G3-027		Fraktur capital z	491, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">Z</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">Z</mi></math> **C** <math><mi mathvariant="fraktur">Z</mi></math> Empty editor → inserts Z. The caret ends after the inserted element; typed characters follow it (typed “7” → Z ∨).
TB-4-G3-028		Fraktur b	513, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">b</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">b</mi></math> **C** <math><mi mathvariant="fraktur">b</mi></math> Empty editor → inserts b. The caret ends after the inserted element; typed characters follow it (typed “7” → b ∨).
TB-4-G3-029		Fraktur e	535, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">e</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">e</mi></math> **C** <math><mi mathvariant="fraktur">e</mi></math> Empty editor → inserts e. The caret ends after the inserted element; typed characters follow it (typed “7” → e ∨).
TB-4-G3-030		Fraktur h	557, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">h</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">h</mi></math> **C** <math><mi mathvariant="fraktur">h</mi></math> Empty editor → inserts h. The caret ends after the inserted element; typed characters follow it (typed “7” → h ∨).
TB-4-G3-031		Fraktur k	579, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">k</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">k</mi></math> **C** <math><mi mathvariant="fraktur">k</mi></math> Empty editor → inserts k. The caret ends after the inserted element; typed characters follow it (typed “7” → k ∨).
TB-4-G3-032		Fraktur n	601, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">n</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">n</mi></math> **C** <math><mi mathvariant="fraktur">n</mi></math> Empty editor → inserts n. The caret ends after the inserted element; typed characters follow it (typed “7” → n ∨).
TB-4-G3-033		Fraktur q	623, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">q</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">q</mi></math> **C** <math><mi mathvariant="fraktur">q</mi></math> Empty editor → inserts q. The caret ends after the inserted element; typed characters follow it (typed “7” → q ∨).
TB-4-G3-034		Fraktur t	645, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">t</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">t</mi></math> **C** <math><mi mathvariant="fraktur">t</mi></math> Empty editor → inserts t. The caret ends after the inserted element; typed characters follow it (typed “7” → t ∨).

TB-4-G3-035		Fraktur w	667, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">w</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">w</mi></math> **C** <math><mi mathvariant="fraktur">w</mi></math> Empty editor → inserts w. The caret ends after the inserted element; typed characters follow it (typed “7” → w ∀).
TB-4-G3-036		Fraktur z	689, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">z</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">z</mi></math> **C** <math><mi mathvariant="fraktur">z</mi></math> Empty editor → inserts z. The caret ends after the inserted element; typed characters follow it (typed “7” → z ∀).
TB-4-G3-037		Fraktur capital c	315, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">C</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">C</mi></math> **C** <math><mi mathvariant="fraktur">C</mi></math> Empty editor → inserts C. The caret ends after the inserted element; typed characters follow it (typed “7” → C ∀).
TB-4-G3-038		Fraktur capital f	337, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">F</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">F</mi></math> **C** <math><mi mathvariant="fraktur">F</mi></math> Empty editor → inserts F. The caret ends after the inserted element; typed characters follow it (typed “7” → F ∀).
TB-4-G3-039		Fraktur capital i	359, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">I</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">I</mi></math> **C** <math><mi mathvariant="fraktur">I</mi></math> Empty editor → inserts I. The caret ends after the inserted element; typed characters follow it (typed “7” → I ∀).
TB-4-G3-040		Fraktur capital l	381, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">L</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">L</mi></math> **C** <math><mi mathvariant="fraktur">L</mi></math> Empty editor → inserts L. The caret ends after the inserted element; typed characters follow it (typed “7” → L ∀).
TB-4-G3-041		Fraktur capital o	403, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">O</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">O</mi></math> **C** <math><mi mathvariant="fraktur">O</mi></math> Empty editor → inserts O. The caret ends after the inserted element; typed characters follow it (typed “7” → O ∀).
TB-4-G3-042		Fraktur capital r	425, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">R</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">R</mi></math> **C** <math><mi mathvariant="fraktur">R</mi></math> Empty editor → inserts R. The caret ends after the inserted element; typed characters follow it (typed “7” → R ∀).
TB-4-G3-043		Fraktur capital u	447, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">U</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">U</mi></math> **C** <math><mi mathvariant="fraktur">U</mi></math> Empty editor → inserts U. The caret ends after the inserted element; typed characters follow it (typed “7” → U ∀).
TB-4-G3-044		Fraktur capital x	469, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">X</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">X</mi></math> **C** <math><mi mathvariant="fraktur">X</mi></math> Empty editor → inserts X. The caret ends after the inserted element; typed characters follow it (typed “7” → X ∀).
TB-4-G3-045		Fraktur c	513, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">c</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">c</mi></math> **C** <math><mi mathvariant="fraktur">c</mi></math> Empty editor → inserts c. The caret ends after the inserted element; typed characters follow it (typed “7” → c ∀).
TB-4-G3-046		Fraktur f	535, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">f</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">f</mi></math> **C** <math><mi mathvariant="fraktur">f</mi></math> Empty editor → inserts f. The caret ends after the inserted element; typed characters follow it (typed “7” → f ∀).
TB-4-G3-047		Fraktur i	557, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">i</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">i</mi></math> **C** <math><mi mathvariant="fraktur">i</mi></math> Empty editor → inserts i. The caret ends after the inserted element; typed characters follow it (typed “7” → i ∀).
TB-4-G3-048		Fraktur l	579, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">l</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">l</mi></math> **C** <math><mi mathvariant="fraktur">l</mi></math> Empty editor → inserts l. The caret ends after the inserted element; typed characters follow it (typed “7” → l ∀).
TB-4-G3-049		Fraktur o	601, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">o</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">o</mi></math> **C** <math><mi mathvariant="fraktur">o</mi></math> Empty editor → inserts o. The caret ends after the inserted element; typed characters follow it (typed “7” → o ∀).

TB-4-G3-050		Fraktur r	623, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">r</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">r</mi></math> **C** <math><mi mathvariant="fraktur">r</mi></math> Empty editor → inserts r. The caret ends after the inserted element; typed characters follow it (typed “7” → r ∀).
TB-4-G3-051		Fraktur u	645, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">u</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">u</mi></math> **C** <math><mi mathvariant="fraktur">u</mi></math> Empty editor → inserts u. The caret ends after the inserted element; typed characters follow it (typed “7” → u ∀).
TB-4-G3-052		Fraktur x	667, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="fraktur">x</mi></math> **B** <math><mi>x</mi><mi mathvariant="fraktur">x</mi></math> **C** <math><mi mathvariant="fraktur">x</mi></math> Empty editor → inserts x. The caret ends after the inserted element; typed characters follow it (typed “7” → x ∀).

“More” group 4 (52 controls) — expand button at 345.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-G4-001		Script capital a	325, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">A</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">A</mi></math> **C** <math><mi mathvariant="script">A</mi></math> Empty editor → inserts A. The caret ends after the inserted element; typed characters follow it (typed “7” → A ∀).
TB-4-G4-002		Script capital d	347, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">D</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">D</mi></math> **C** <math><mi mathvariant="script">D</mi></math> Empty editor → inserts D. The caret ends after the inserted element; typed characters follow it (typed “7” → D ∀).
TB-4-G4-003		Script capital g	369, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">G</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">G</mi></math> **C** <math><mi mathvariant="script">G</mi></math> Empty editor → inserts G. The caret ends after the inserted element; typed characters follow it (typed “7” → G ∀).
TB-4-G4-004		Script capital j	391, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">J</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">J</mi></math> **C** <math><mi mathvariant="script">J</mi></math> Empty editor → inserts J. The caret ends after the inserted element; typed characters follow it (typed “7” → J ∀).
TB-4-G4-005		Script capital m	413, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">M</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">M</mi></math> **C** <math><mi mathvariant="script">M</mi></math> Empty editor → inserts M. The caret ends after the inserted element; typed characters follow it (typed “7” → M ∀).
TB-4-G4-006		Script capital p	435, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">P</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">P</mi></math> **C** <math><mi mathvariant="script">P</mi></math> Empty editor → inserts P. The caret ends after the inserted element; typed characters follow it (typed “7” → P ∀).
TB-4-G4-007		Script capital s	457, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">S</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">S</mi></math> **C** <math><mi mathvariant="script">S</mi></math> Empty editor → inserts S. The caret ends after the inserted element; typed characters follow it (typed “7” → S ∀).
TB-4-G4-008		Script capital v	479, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">V</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">V</mi></math> **C** <math><mi mathvariant="script">V</mi></math> Empty editor → inserts V. The caret ends after the inserted element; typed characters follow it (typed “7” → V ∀).
TB-4-G4-009		Script capital y	501, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">Y</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">Y</mi></math> **C** <math><mi mathvariant="script">Y</mi></math> Empty editor → inserts Y. The caret ends after the inserted element; typed characters follow it (typed “7” → Y ∀).
TB-4-G4-010		Script a	523, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">a</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">a</mi></math> **C** <math><mi mathvariant="script">a</mi></math> Empty editor → inserts a. The caret ends after the inserted element; typed characters follow it (typed “7” → a ∀).
TB-4-G4-011		Script d	545, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">d</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">d</mi></math> **C** <math><mi mathvariant="script">d</mi></math> Empty editor → inserts d. The caret ends after the inserted element; typed characters follow it (typed “7” → d ∀).

TB-4-G4-012		Script g	567, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">g</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">g</mi></math> **C** <math><mi mathvariant="script">g</mi></math> Empty editor → inserts g. The caret ends after the inserted element; typed characters follow it (typed “7” → g ∨).
TB-4-G4-013		Script j	589, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">j</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">j</mi></math> **C** <math><mi mathvariant="script">j</mi></math> Empty editor → inserts j. The caret ends after the inserted element; typed characters follow it (typed “7” → j ∨).
TB-4-G4-014		Script m	611, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">m</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">m</mi></math> **C** <math><mi mathvariant="script">m</mi></math> Empty editor → inserts m. The caret ends after the inserted element; typed characters follow it (typed “7” → m ∨).
TB-4-G4-015		Script p	633, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">p</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">p</mi></math> **C** <math><mi mathvariant="script">p</mi></math> Empty editor → inserts p. The caret ends after the inserted element; typed characters follow it (typed “7” → p ∨).
TB-4-G4-016		Script s	655, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">s</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">s</mi></math> **C** <math><mi mathvariant="script">s</mi></math> Empty editor → inserts s. The caret ends after the inserted element; typed characters follow it (typed “7” → s ∨).
TB-4-G4-017		Script v	677, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">v</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">v</mi></math> **C** <math><mi mathvariant="script">v</mi></math> Empty editor → inserts v. The caret ends after the inserted element; typed characters follow it (typed “7” → v ∨).
TB-4-G4-018		Script y	699, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">y</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">y</mi></math> **C** <math><mi mathvariant="script">y</mi></math> Empty editor → inserts y. The caret ends after the inserted element; typed characters follow it (typed “7” → y ∨).
TB-4-G4-019		Script capital b	325, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">B</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">B</mi></math> **C** <math><mi mathvariant="script">B</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B ∨).
TB-4-G4-020		Script capital e	347, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">E</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">E</mi></math> **C** <math><mi mathvariant="script">E</mi></math> Empty editor → inserts E. The caret ends after the inserted element; typed characters follow it (typed “7” → E ∨).
TB-4-G4-021		Script capital h	369, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">H</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">H</mi></math> **C** <math><mi mathvariant="script">H</mi></math> Empty editor → inserts H. The caret ends after the inserted element; typed characters follow it (typed “7” → H ∨).
TB-4-G4-022		Script capital k	391, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">K</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">K</mi></math> **C** <math><mi mathvariant="script">K</mi></math> Empty editor → inserts K. The caret ends after the inserted element; typed characters follow it (typed “7” → K ∨).
TB-4-G4-023		Script capital n	413, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">N</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">N</mi></math> **C** <math><mi mathvariant="script">N</mi></math> Empty editor → inserts N. The caret ends after the inserted element; typed characters follow it (typed “7” → N ∨).
TB-4-G4-024		Script capital q	435, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">Q</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">Q</mi></math> **C** <math><mi mathvariant="script">Q</mi></math> Empty editor → inserts Q. The caret ends after the inserted element; typed characters follow it (typed “7” → Q ∨).
TB-4-G4-025		Script capital t	457, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">T</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">T</mi></math> **C** <math><mi mathvariant="script">T</mi></math> Empty editor → inserts T. The caret ends after the inserted element; typed characters follow it (typed “7” → T ∨).
TB-4-G4-026		Script capital w	479, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">W</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">W</mi></math> **C** <math><mi mathvariant="script">W</mi></math> Empty editor → inserts W. The caret ends after the inserted element; typed characters follow it (typed “7” → W ∨).

TB-4-G4-027		Script capital z	501, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">Z</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">Z</mi></math> **C** <math><mi mathvariant="script">Z</mi></math> Empty editor → inserts Z. The caret ends after the inserted element; typed characters follow it (typed “7” → Z ∀).
TB-4-G4-028		Script b	523, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">b</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">b</mi></math> **C** <math><mi mathvariant="script">b</mi></math> Empty editor → inserts b. The caret ends after the inserted element; typed characters follow it (typed “7” → b ∀).
TB-4-G4-029		Script e	545, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">e</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">e</mi></math> **C** <math><mi mathvariant="script">e</mi></math> Empty editor → inserts e. The caret ends after the inserted element; typed characters follow it (typed “7” → e ∀).
TB-4-G4-030		Script h	567, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">h</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">h</mi></math> **C** <math><mi mathvariant="script">h</mi></math> Empty editor → inserts h. The caret ends after the inserted element; typed characters follow it (typed “7” → h ∀).
TB-4-G4-031		Script k	589, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">k</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">k</mi></math> **C** <math><mi mathvariant="script">k</mi></math> Empty editor → inserts k. The caret ends after the inserted element; typed characters follow it (typed “7” → k ∀).
TB-4-G4-032		Script n	611, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">n</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">n</mi></math> **C** <math><mi mathvariant="script">n</mi></math> Empty editor → inserts n. The caret ends after the inserted element; typed characters follow it (typed “7” → n ∀).
TB-4-G4-033		Script q	633, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">q</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">q</mi></math> **C** <math><mi mathvariant="script">q</mi></math> Empty editor → inserts q. The caret ends after the inserted element; typed characters follow it (typed “7” → q ∀).
TB-4-G4-034		Script t	655, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">t</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">t</mi></math> **C** <math><mi mathvariant="script">t</mi></math> Empty editor → inserts t. The caret ends after the inserted element; typed characters follow it (typed “7” → t ∀).
TB-4-G4-035		Script w	677, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">w</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">w</mi></math> **C** <math><mi mathvariant="script">w</mi></math> Empty editor → inserts w. The caret ends after the inserted element; typed characters follow it (typed “7” → w ∀).
TB-4-G4-036		Script z	699, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">z</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">z</mi></math> **C** <math><mi mathvariant="script">z</mi></math> Empty editor → inserts z. The caret ends after the inserted element; typed characters follow it (typed “7” → z ∀).
TB-4-G4-037		Script capital c	325, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">C</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">C</mi></math> **C** <math><mi mathvariant="script">C</mi></math> Empty editor → inserts C. The caret ends after the inserted element; typed characters follow it (typed “7” → C ∀).
TB-4-G4-038		Script capital f	347, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">F</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">F</mi></math> **C** <math><mi mathvariant="script">F</mi></math> Empty editor → inserts F. The caret ends after the inserted element; typed characters follow it (typed “7” → F ∀).
TB-4-G4-039		Script capital i	369, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">I</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">I</mi></math> **C** <math><mi mathvariant="script">I</mi></math> Empty editor → inserts I. The caret ends after the inserted element; typed characters follow it (typed “7” → I ∀).
TB-4-G4-040		Script capital l	391, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">L</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">L</mi></math> **C** <math><mi mathvariant="script">L</mi></math> Empty editor → inserts L. The caret ends after the inserted element; typed characters follow it (typed “7” → L ∀).
TB-4-G4-041		Script capital o	413, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">O</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">O</mi></math> **C** <math><mi mathvariant="script">O</mi></math> Empty editor → inserts O. The caret ends after the inserted element; typed characters follow it (typed “7” → O ∀).

TB-4-G4-042		Script capital r	435, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">R</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">R</mi></math> **C** <math><mi mathvariant="script">R</mi></math> Empty editor → inserts R. The caret ends after the inserted element; typed characters follow it (typed “7” → R ∀).
TB-4-G4-043		Script capital u	457, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">U</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">U</mi></math> **C** <math><mi mathvariant="script">U</mi></math> Empty editor → inserts U. The caret ends after the inserted element; typed characters follow it (typed “7” → U ∀).
TB-4-G4-044		Script capital x	479, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">X</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">X</mi></math> **C** <math><mi mathvariant="script">X</mi></math> Empty editor → inserts X. The caret ends after the inserted element; typed characters follow it (typed “7” → X ∀).
TB-4-G4-045		Script c	523, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">c</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">c</mi></math> **C** <math><mi mathvariant="script">c</mi></math> Empty editor → inserts c. The caret ends after the inserted element; typed characters follow it (typed “7” → c ∀).
TB-4-G4-046		Script f	545, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">f</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">f</mi></math> **C** <math><mi mathvariant="script">f</mi></math> Empty editor → inserts f. The caret ends after the inserted element; typed characters follow it (typed “7” → f ∀).
TB-4-G4-047		Script i	567, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">i</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">i</mi></math> **C** <math><mi mathvariant="script">i</mi></math> Empty editor → inserts i. The caret ends after the inserted element; typed characters follow it (typed “7” → i ∀).
TB-4-G4-048		Script l	589, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">l</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">l</mi></math> **C** <math><mi mathvariant="script">l</mi></math> Empty editor → inserts l. The caret ends after the inserted element; typed characters follow it (typed “7” → l ∀).
TB-4-G4-049		Script o	611, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">o</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">o</mi></math> **C** <math><mi mathvariant="script">o</mi></math> Empty editor → inserts o. The caret ends after the inserted element; typed characters follow it (typed “7” → o ∀).
TB-4-G4-050		Script r	633, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">r</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">r</mi></math> **C** <math><mi mathvariant="script">r</mi></math> Empty editor → inserts r. The caret ends after the inserted element; typed characters follow it (typed “7” → r ∀).
TB-4-G4-051		Script u	655, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">u</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">u</mi></math> **C** <math><mi mathvariant="script">u</mi></math> Empty editor → inserts u. The caret ends after the inserted element; typed characters follow it (typed “7” → u ∀).
TB-4-G4-052		Script x	677, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="script">x</mi></math> **B** <math><mi>x</mi><mi mathvariant="script">x</mi></math> **C** <math><mi mathvariant="script">x</mi></math> Empty editor → inserts x. The caret ends after the inserted element; typed characters follow it (typed “7” → x ∀).

“More” group 5 (5 controls) — expand button at 378.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-G5-001		Alef	381, 94, 22, 22 (open pop-up)	**A** <math><mo>ℵ</mo></math> **B** <math><mi>x</mi><mo>ℵ</mo></math> **C** <math><mo>ℵ</mo></math> Empty editor → inserts ℵ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℵ ∀).
TB-4-G5-002		Script capital l	403, 94, 22, 22 (open pop-up)	**A** <math><mi>ℒ</mi></math> **B** <math><mi>x</mi><mi>ℒ</mi></math> **C** <math><mi>ℒ</mi></math> Empty editor → inserts ℒ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℒ ∀).
TB-4-G5-003		Script capital p	381, 116, 22, 22 (open pop-up)	**A** <math><mi>℘</mi></math> **B** <math><mi>x</mi><mi>℘</mi></math> **C** <math><mi>℘</mi></math> Empty editor → inserts ℘. The caret ends after the inserted element; typed characters follow it (typed “7” → ℘ ∀).

TB-4-G5-004	ℤ	Z transform	403, 116, 22, 22 (open pop-up)	**A** <math><mo>Ƶ</mo></math> **B** <math><mi>x</mi><mo>Ƶ</mo></math> **C** <math><mo>Ƶ</mo></math> Empty editor → inserts ℤ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℤ ∀).
TB-4-G5-005	ℱ	Script capital f	381, 138, 22, 22 (open pop-up)	**A** <math><mi>ℱ</mi></math> **B** <math><mi>x</mi><mi>ℱ</mi></math> **C** <math><mi>ℱ</mi></math> Empty editor → inserts ℱ. The caret ends after the inserted element; typed characters follow it (typed “7” → ℱ ∀).

“More” group 6 (118 controls) — expand button at 457.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-4-G6-001	H	Hydrogen	325, 94, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">H</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">H</mi></math> **C** <math><mi mathvariant="normal">H</mi></math> Empty editor → inserts H. The caret ends after the inserted element; typed characters follow it (typed “7” → H ∀).
TB-4-G6-002	He	Helium	699, 94, 22, 22 (open pop-up)	**A** <math><mi>He</mi></math> **B** <math><mi>x</mi><mi>He</mi></math> **C** <math><mi>He</mi><mi>x</mi></math> Empty editor → inserts He. The caret ends after the inserted element; typed characters follow it (typed “7” → He ∀).
TB-4-G6-003	Li	Lithium	325, 116, 22, 22 (open pop-up)	**A** <math><mi>Li</mi></math> **B** <math><mi>x</mi><mi>Li</mi></math> **C** <math><mi>Li</mi><mi>x</mi></math> Empty editor → inserts Li. The caret ends after the inserted element; typed characters follow it (typed “7” → Li ∀).
TB-4-G6-004	Be	Beryllium	347, 116, 22, 22 (open pop-up)	**A** <math><mi>Be</mi></math> **B** <math><mi>x</mi><mi>Be</mi></math> **C** <math><mi>Be</mi><mi>x</mi></math> Empty editor → inserts Be. The caret ends after the inserted element; typed characters follow it (typed “7” → Be ∀).
TB-4-G6-005	B	Boron	589, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">B</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">B</mi></math> **C** <math><mi mathvariant="normal">B</mi></math> Empty editor → inserts B. The caret ends after the inserted element; typed characters follow it (typed “7” → B ∀).
TB-4-G6-006	C	Carbon	611, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">C</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">C</mi></math> **C** <math><mi mathvariant="normal">C</mi></math> Empty editor → inserts C. The caret ends after the inserted element; typed characters follow it (typed “7” → C ∀).
TB-4-G6-007	N	Nitrogen	633, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">N</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">N</mi></math> **C** <math><mi mathvariant="normal">N</mi></math> Empty editor → inserts N. The caret ends after the inserted element; typed characters follow it (typed “7” → N ∀).
TB-4-G6-008	O	Oxygen	655, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">O</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">O</mi></math> **C** <math><mi mathvariant="normal">O</mi></math> Empty editor → inserts O. The caret ends after the inserted element; typed characters follow it (typed “7” → O ∀).
TB-4-G6-009	F	Fluorine	677, 116, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">F</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">F</mi></math> **C** <math><mi mathvariant="normal">F</mi></math> Empty editor → inserts F. The caret ends after the inserted element; typed characters follow it (typed “7” → F ∀).
TB-4-G6-010	Ne	Neon	699, 116, 22, 22 (open pop-up)	**A** <math><mi>Ne</mi></math> **B** <math><mi>x</mi><mi>Ne</mi></math> **C** <math><mi>Ne</mi><mi>x</mi></math> Empty editor → inserts Ne. The caret ends after the inserted element; typed characters follow it (typed “7” → Ne ∀).
TB-4-G6-011	Na	Sodium	325, 138, 22, 22 (open pop-up)	**A** <math><mi>Na</mi></math> **B** <math><mi>x</mi><mi>Na</mi></math> **C** <math><mi>Na</mi><mi>x</mi></math> Empty editor → inserts Na. The caret ends after the inserted element; typed characters follow it (typed “7” → Na ∀).
TB-4-G6-012	Mg	Magnesium	347, 138, 22, 22 (open pop-up)	**A** <math><mi>Mg</mi></math> **B** <math><mi>x</mi><mi>Mg</mi></math> **C** <math><mi>Mg</mi><mi>x</mi></math> Empty editor → inserts Mg. The caret ends after the inserted element; typed characters follow it (typed “7” → Mg ∀).

TB-4-G6-013	Al	Aluminium	589, 138, 22, 22 (open pop-up)	**A** <math><mi>Al</mi></math> **B** <math><mi>x</mi><mi>Al</mi></math> **C** <math><mi>Al</mi><mi>x</mi></math> Empty editor → inserts Al. The caret ends after the inserted element; typed characters follow it (typed “7” → Al ∀).
TB-4-G6-014	Si	Silicon	611, 138, 22, 22 (open pop-up)	**A** <math><mi>Si</mi></math> **B** <math><mi>x</mi><mi>Si</mi></math> **C** <math><mi>Si</mi><mi>x</mi></math> Empty editor → inserts Si. The caret ends after the inserted element; typed characters follow it (typed “7” → Si ∀).
TB-4-G6-015	P	Phosphorus	633, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">P</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">P</mi></math> **C** <math><mi mathvariant="normal">P</mi></math> Empty editor → inserts P. The caret ends after the inserted element; typed characters follow it (typed “7” → P ∀).
TB-4-G6-016	S	Sulfur	655, 138, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">S</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">S</mi></math> **C** <math><mi mathvariant="normal">S</mi></math> Empty editor → inserts S. The caret ends after the inserted element; typed characters follow it (typed “7” → S ∀).
TB-4-G6-017	Cl	Chlorine	677, 138, 22, 22 (open pop-up)	**A** <math><mi>Cl</mi></math> **B** <math><mi>x</mi><mi>Cl</mi></math> **C** <math><mi>Cl</mi><mi>x</mi></math> Empty editor → inserts Cl. The caret ends after the inserted element; typed characters follow it (typed “7” → Cl ∀).
TB-4-G6-018	Ar	Argon	699, 138, 22, 22 (open pop-up)	**A** <math><mi>Ar</mi></math> **B** <math><mi>x</mi><mi>Ar</mi></math> **C** <math><mi>Ar</mi><mi>x</mi></math> Empty editor → inserts Ar. The caret ends after the inserted element; typed characters follow it (typed “7” → Ar ∀).
TB-4-G6-019	K	Potassium	325, 160, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">K</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">K</mi></math> **C** <math><mi mathvariant="normal">K</mi></math> Empty editor → inserts K. The caret ends after the inserted element; typed characters follow it (typed “7” → K ∀).
TB-4-G6-020	Ca	Calcium	347, 160, 22, 22 (open pop-up)	**A** <math><mi>Ca</mi></math> **B** <math><mi>x</mi><mi>Ca</mi></math> **C** <math><mi>Ca</mi><mi>x</mi></math> Empty editor → inserts Ca. The caret ends after the inserted element; typed characters follow it (typed “7” → Ca ∀).
TB-4-G6-021	Sc	Scandium	369, 160, 22, 22 (open pop-up)	**A** <math><mi>Sc</mi></math> **B** <math><mi>x</mi><mi>Sc</mi></math> **C** <math><mi>Sc</mi><mi>x</mi></math> Empty editor → inserts Sc. The caret ends after the inserted element; typed characters follow it (typed “7” → Sc ∀).
TB-4-G6-022	Ti	Titanium	391, 160, 22, 22 (open pop-up)	**A** <math><mi>Ti</mi></math> **B** <math><mi>x</mi><mi>Ti</mi></math> **C** <math><mi>Ti</mi><mi>x</mi></math> Empty editor → inserts Ti. The caret ends after the inserted element; typed characters follow it (typed “7” → Ti ∀).
TB-4-G6-023	V	Vanadium	413, 160, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">V</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">V</mi></math> **C** <math><mi mathvariant="normal">V</mi></math> Empty editor → inserts V. The caret ends after the inserted element; typed characters follow it (typed “7” → V ∀).
TB-4-G6-024	Cr	Chromium	435, 160, 22, 22 (open pop-up)	**A** <math><mi>Cr</mi></math> **B** <math><mi>x</mi><mi>Cr</mi></math> **C** <math><mi>Cr</mi><mi>x</mi></math> Empty editor → inserts Cr. The caret ends after the inserted element; typed characters follow it (typed “7” → Cr ∀).
TB-4-G6-025	Mn	Manganese	457, 160, 22, 22 (open pop-up)	**A** <math><mi>Mn</mi></math> **B** <math><mi>x</mi><mi>Mn</mi></math> **C** <math><mi>Mn</mi><mi>x</mi></math> Empty editor → inserts Mn. The caret ends after the inserted element; typed characters follow it (typed “7” → Mn ∀).
TB-4-G6-026	Fe	Iron	479, 160, 22, 22 (open pop-up)	**A** <math><mi>Fe</mi></math> **B** <math><mi>x</mi><mi>Fe</mi></math> **C** <math><mi>Fe</mi><mi>x</mi></math> Empty editor → inserts Fe. The caret ends after the inserted element; typed characters follow it (typed “7” → Fe ∀).
TB-4-G6-027	Co	Cobalt	501, 160, 22, 22 (open pop-up)	**A** <math><mi>Co</mi></math> **B** <math><mi>x</mi><mi>Co</mi></math> **C** <math><mi>Co</mi><mi>x</mi></math> Empty editor → inserts Co. The caret ends after the inserted element; typed characters follow it (typed “7” → Co ∀).

TB-4-G6-028	Ni	Nickel	523, 160, 22, 22 (open pop-up)	**A** <math><mi>Ni</mi></math> **B** <math><mi>x</mi><mi>Ni</mi></math> **C** <math><mi>Ni</mi><mi>x</mi></math> Empty editor → inserts Ni. The caret ends after the inserted element; typed characters follow it (typed “7” → Ni ∨).
TB-4-G6-029	Cu	Copper	545, 160, 22, 22 (open pop-up)	**A** <math><mi>Cu</mi></math> **B** <math><mi>x</mi><mi>Cu</mi></math> **C** <math><mi>Cu</mi><mi>x</mi></math> Empty editor → inserts Cu. The caret ends after the inserted element; typed characters follow it (typed “7” → Cu ∨).
TB-4-G6-030	Zn	Zinc	567, 160, 22, 22 (open pop-up)	**A** <math><mi>Zn</mi></math> **B** <math><mi>x</mi><mi>Zn</mi></math> **C** <math><mi>Zn</mi><mi>x</mi></math> Empty editor → inserts Zn. The caret ends after the inserted element; typed characters follow it (typed “7” → Zn ∨).
TB-4-G6-031	Ga	Gallium	589, 160, 22, 22 (open pop-up)	**A** <math><mi>Ga</mi></math> **B** <math><mi>x</mi><mi>Ga</mi></math> **C** <math><mi>Ga</mi><mi>x</mi></math> Empty editor → inserts Ga. The caret ends after the inserted element; typed characters follow it (typed “7” → Ga ∨).
TB-4-G6-032	Ge	Germanium	611, 160, 22, 22 (open pop-up)	**A** <math><mi>Ge</mi></math> **B** <math><mi>x</mi><mi>Ge</mi></math> **C** <math><mi>Ge</mi><mi>x</mi></math> Empty editor → inserts Ge. The caret ends after the inserted element; typed characters follow it (typed “7” → Ge ∨).
TB-4-G6-033	As	Arsenic	633, 160, 22, 22 (open pop-up)	**A** <math><mi>As</mi></math> **B** <math><mi>x</mi><mi>As</mi></math> **C** <math><mi>As</mi><mi>x</mi></math> Empty editor → inserts As. The caret ends after the inserted element; typed characters follow it (typed “7” → As ∨).
TB-4-G6-034	Se	Selenium	655, 160, 22, 22 (open pop-up)	**A** <math><mi>Se</mi></math> **B** <math><mi>x</mi><mi>Se</mi></math> **C** <math><mi>Se</mi><mi>x</mi></math> Empty editor → inserts Se. The caret ends after the inserted element; typed characters follow it (typed “7” → Se ∨).
TB-4-G6-035	Br	Bromine	677, 160, 22, 22 (open pop-up)	**A** <math><mi>Br</mi></math> **B** <math><mi>x</mi><mi>Br</mi></math> **C** <math><mi>Br</mi><mi>x</mi></math> Empty editor → inserts Br. The caret ends after the inserted element; typed characters follow it (typed “7” → Br ∨).
TB-4-G6-036	Kr	Krypton	699, 160, 22, 22 (open pop-up)	**A** <math><mi>Kr</mi></math> **B** <math><mi>x</mi><mi>Kr</mi></math> **C** <math><mi>Kr</mi><mi>x</mi></math> Empty editor → inserts Kr. The caret ends after the inserted element; typed characters follow it (typed “7” → Kr ∨).
TB-4-G6-037	Rb	Rubidium	325, 182, 22, 22 (open pop-up)	**A** <math><mi>Rb</mi></math> **B** <math><mi>x</mi><mi>Rb</mi></math> **C** <math><mi>Rb</mi><mi>x</mi></math> Empty editor → inserts Rb. The caret ends after the inserted element; typed characters follow it (typed “7” → Rb ∨).
TB-4-G6-038	Sr	Strontium	347, 182, 22, 22 (open pop-up)	**A** <math><mi>Sr</mi></math> **B** <math><mi>x</mi><mi>Sr</mi></math> **C** <math><mi>Sr</mi><mi>x</mi></math> Empty editor → inserts Sr. The caret ends after the inserted element; typed characters follow it (typed “7” → Sr ∨).
TB-4-G6-039	Y	Yttrium	369, 182, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">Y</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">Y</mi></math> **C** <math><mi mathvariant="normal">Y</mi><mi>x</mi></math> Empty editor → inserts Y. The caret ends after the inserted element; typed characters follow it (typed “7” → Y ∨).
TB-4-G6-040	Zr	Zirconium	391, 182, 22, 22 (open pop-up)	**A** <math><mi>Zr</mi></math> **B** <math><mi>x</mi><mi>Zr</mi></math> **C** <math><mi>Zr</mi><mi>x</mi></math> Empty editor → inserts Zr. The caret ends after the inserted element; typed characters follow it (typed “7” → Zr ∨).
TB-4-G6-041	Nb	Niobium	413, 182, 22, 22 (open pop-up)	**A** <math><mi>Nb</mi></math> **B** <math><mi>x</mi><mi>Nb</mi></math> **C** <math><mi>Nb</mi><mi>x</mi></math> Empty editor → inserts Nb. The caret ends after the inserted element; typed characters follow it (typed “7” → Nb ∨).
TB-4-G6-042	Mo	Molybdenum	435, 182, 22, 22 (open pop-up)	**A** <math><mi>Mo</mi></math> **B** <math><mi>x</mi><mi>Mo</mi></math> **C** <math><mi>Mo</mi><mi>x</mi></math> Empty editor → inserts Mo. The caret ends after the inserted element; typed characters follow it (typed “7” → Mo ∨).

TB-4-G6-043	Tc	Technetium	457, 182, 22, 22 (open pop-up)	**A** <math><mi>Tc</mi></math> **B** <math><mi>x</mi><mi>Tc</mi></math> **C** <math><mi>Tc</mi><mi>x</mi></math> Empty editor → inserts Tc. The caret ends after the inserted element; typed characters follow it (typed “7” → Tc ∨).
TB-4-G6-044	Ru	Ruthenium	479, 182, 22, 22 (open pop-up)	**A** <math><mi>Ru</mi></math> **B** <math><mi>x</mi><mi>Ru</mi></math> **C** <math><mi>Ru</mi><mi>x</mi></math> Empty editor → inserts Ru. The caret ends after the inserted element; typed characters follow it (typed “7” → Ru ∨).
TB-4-G6-045	Rh	Rhodium	501, 182, 22, 22 (open pop-up)	**A** <math><mi>Rh</mi></math> **B** <math><mi>x</mi><mi>Rh</mi></math> **C** <math><mi>Rh</mi><mi>x</mi></math> Empty editor → inserts Rh. The caret ends after the inserted element; typed characters follow it (typed “7” → Rh ∨).
TB-4-G6-046	Pd	Palladium	523, 182, 22, 22 (open pop-up)	**A** <math><mi>Pd</mi></math> **B** <math><mi>x</mi><mi>Pd</mi></math> **C** <math><mi>Pd</mi><mi>x</mi></math> Empty editor → inserts Pd. The caret ends after the inserted element; typed characters follow it (typed “7” → Pd ∨).
TB-4-G6-047	Ag	Silver	545, 182, 22, 22 (open pop-up)	**A** <math><mi>Ag</mi></math> **B** <math><mi>x</mi><mi>Ag</mi></math> **C** <math><mi>Ag</mi><mi>x</mi></math> Empty editor → inserts Ag. The caret ends after the inserted element; typed characters follow it (typed “7” → Ag ∨).
TB-4-G6-048	Cd	Cadmium	567, 182, 22, 22 (open pop-up)	**A** <math><mi>Cd</mi></math> **B** <math><mi>x</mi><mi>Cd</mi></math> **C** <math><mi>Cd</mi><mi>x</mi></math> Empty editor → inserts Cd. The caret ends after the inserted element; typed characters follow it (typed “7” → Cd ∨).
TB-4-G6-049	In	Indium	589, 182, 22, 22 (open pop-up)	**A** <math><mi>In</mi></math> **B** <math><mi>x</mi><mi>In</mi></math> **C** <math><mi>In</mi><mi>x</mi></math> Empty editor → inserts In. The caret ends after the inserted element; typed characters follow it (typed “7” → In ∨).
TB-4-G6-050	Sn	Tin	611, 182, 22, 22 (open pop-up)	**A** <math><mi>Sn</mi></math> **B** <math><mi>x</mi><mi>Sn</mi></math> **C** <math><mi>Sn</mi><mi>x</mi></math> Empty editor → inserts Sn. The caret ends after the inserted element; typed characters follow it (typed “7” → Sn ∨).
TB-4-G6-051	Sb	Antimony	633, 182, 22, 22 (open pop-up)	**A** <math><mi>Sb</mi></math> **B** <math><mi>x</mi><mi>Sb</mi></math> **C** <math><mi>Sb</mi><mi>x</mi></math> Empty editor → inserts Sb. The caret ends after the inserted element; typed characters follow it (typed “7” → Sb ∨).
TB-4-G6-052	Te	Tellurium	655, 182, 22, 22 (open pop-up)	**A** <math><mi>Te</mi></math> **B** <math><mi>x</mi><mi>Te</mi></math> **C** <math><mi>Te</mi><mi>x</mi></math> Empty editor → inserts Te. The caret ends after the inserted element; typed characters follow it (typed “7” → Te ∨).
TB-4-G6-053	I	Iodine	677, 182, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">I</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">I</mi></math> **C** <math><mi mathvariant="normal">I</mi></math> Empty editor → inserts I. The caret ends after the inserted element; typed characters follow it (typed “7” → I ∨).
TB-4-G6-054	Xe	Xenon	699, 182, 22, 22 (open pop-up)	**A** <math><mi>Xe</mi></math> **B** <math><mi>x</mi><mi>Xe</mi></math> **C** <math><mi>Xe</mi><mi>x</mi></math> Empty editor → inserts Xe. The caret ends after the inserted element; typed characters follow it (typed “7” → Xe ∨).
TB-4-G6-055	Cs	Caesium	325, 204, 22, 22 (open pop-up)	**A** <math><mi>Cs</mi></math> **B** <math><mi>x</mi><mi>Cs</mi></math> **C** <math><mi>Cs</mi><mi>x</mi></math> Empty editor → inserts Cs. The caret ends after the inserted element; typed characters follow it (typed “7” → Cs ∨).
TB-4-G6-056	Ba	Barium	347, 204, 22, 22 (open pop-up)	**A** <math><mi>Ba</mi></math> **B** <math><mi>x</mi><mi>Ba</mi></math> **C** <math><mi>Ba</mi><mi>x</mi></math> Empty editor → inserts Ba. The caret ends after the inserted element; typed characters follow it (typed “7” → Ba ∨).
TB-4-G6-057	Hf	Hafnium	391, 204, 22, 22 (open pop-up)	**A** <math><mi>Hf</mi></math> **B** <math><mi>x</mi><mi>Hf</mi></math> **C** <math><mi>Hf</mi><mi>x</mi></math> Empty editor → inserts Hf. The caret ends after the inserted element; typed characters follow it (typed “7” → Hf ∨).

TB-4-G6-058	Ta	Tantalum	413, 204, 22, 22 (open pop-up)	**A** <math><mi>Ta</mi></math> **B** <math><mi>x</mi><mi>Ta</mi></math> **C** <math><mi>Ta</mi><mi>x</mi></math> Empty editor → inserts Ta. The caret ends after the inserted element; typed characters follow it (typed “7” → Ta ∨).
TB-4-G6-059	W	Tungsten	435, 204, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">W</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">W</mi></math> **C** <math><mi mathvariant="normal">W</mi><mi>x</mi></math> Empty editor → inserts W. The caret ends after the inserted element; typed characters follow it (typed “7” → W ∨).
TB-4-G6-060	Re	Rhenium	457, 204, 22, 22 (open pop-up)	**A** <math><mi>Re</mi></math> **B** <math><mi>x</mi><mi>Re</mi></math> **C** <math><mi>Re</mi><mi>x</mi></math> Empty editor → inserts Re. The caret ends after the inserted element; typed characters follow it (typed “7” → Re ∨).
TB-4-G6-061	Os	Osmium	479, 204, 22, 22 (open pop-up)	**A** <math><mi>Os</mi></math> **B** <math><mi>x</mi><mi>Os</mi></math> **C** <math><mi>Os</mi><mi>x</mi></math> Empty editor → inserts Os. The caret ends after the inserted element; typed characters follow it (typed “7” → Os ∨).
TB-4-G6-062	Ir	Iridium	501, 204, 22, 22 (open pop-up)	**A** <math><mi>Ir</mi></math> **B** <math><mi>x</mi><mi>Ir</mi></math> **C** <math><mi>Ir</mi><mi>x</mi></math> Empty editor → inserts Ir. The caret ends after the inserted element; typed characters follow it (typed “7” → Ir ∨).
TB-4-G6-063	Pt	Platinum	523, 204, 22, 22 (open pop-up)	**A** <math><mi>Pt</mi></math> **B** <math><mi>x</mi><mi>Pt</mi></math> **C** <math><mi>Pt</mi><mi>x</mi></math> Empty editor → inserts Pt. The caret ends after the inserted element; typed characters follow it (typed “7” → Pt ∨).
TB-4-G6-064	Au	Gold	545, 204, 22, 22 (open pop-up)	**A** <math><mi>Au</mi></math> **B** <math><mi>x</mi><mi>Au</mi></math> **C** <math><mi>Au</mi><mi>x</mi></math> Empty editor → inserts Au. The caret ends after the inserted element; typed characters follow it (typed “7” → Au ∨).
TB-4-G6-065	Hg	Mercury	567, 204, 22, 22 (open pop-up)	**A** <math><mi>Hg</mi></math> **B** <math><mi>x</mi><mi>Hg</mi></math> **C** <math><mi>Hg</mi><mi>x</mi></math> Empty editor → inserts Hg. The caret ends after the inserted element; typed characters follow it (typed “7” → Hg ∨).
TB-4-G6-066	Tl	Thallium	589, 204, 22, 22 (open pop-up)	**A** <math><mi>Tl</mi></math> **B** <math><mi>x</mi><mi>Tl</mi></math> **C** <math><mi>Tl</mi><mi>x</mi></math> Empty editor → inserts Tl. The caret ends after the inserted element; typed characters follow it (typed “7” → Tl ∨).
TB-4-G6-067	Pb	Lead	611, 204, 22, 22 (open pop-up)	**A** <math><mi>Pb</mi></math> **B** <math><mi>x</mi><mi>Pb</mi></math> **C** <math><mi>Pb</mi><mi>x</mi></math> Empty editor → inserts Pb. The caret ends after the inserted element; typed characters follow it (typed “7” → Pb ∨).
TB-4-G6-068	Bi	Bismuth	633, 204, 22, 22 (open pop-up)	**A** <math><mi>Bi</mi></math> **B** <math><mi>x</mi><mi>Bi</mi></math> **C** <math><mi>Bi</mi><mi>x</mi></math> Empty editor → inserts Bi. The caret ends after the inserted element; typed characters follow it (typed “7” → Bi ∨).
TB-4-G6-069	Po	Polonium	655, 204, 22, 22 (open pop-up)	**A** <math><mi>Po</mi></math> **B** <math><mi>x</mi><mi>Po</mi></math> **C** <math><mi>Po</mi><mi>x</mi></math> Empty editor → inserts Po. The caret ends after the inserted element; typed characters follow it (typed “7” → Po ∨).
TB-4-G6-070	At	Astatine	677, 204, 22, 22 (open pop-up)	**A** <math><mi>At</mi></math> **B** <math><mi>x</mi><mi>At</mi></math> **C** <math><mi>At</mi><mi>x</mi></math> Empty editor → inserts At. The caret ends after the inserted element; typed characters follow it (typed “7” → At ∨).
TB-4-G6-071	Rn	Radon	699, 204, 22, 22 (open pop-up)	**A** <math><mi>Rn</mi></math> **B** <math><mi>x</mi><mi>Rn</mi></math> **C** <math><mi>Rn</mi><mi>x</mi></math> Empty editor → inserts Rn. The caret ends after the inserted element; typed characters follow it (typed “7” → Rn ∨).
TB-4-G6-072	Fr	Francium	325, 226, 22, 22 (open pop-up)	**A** <math><mi>Fr</mi></math> **B** <math><mi>x</mi><mi>Fr</mi></math> **C** <math><mi>Fr</mi><mi>x</mi></math> Empty editor → inserts Fr. The caret ends after the inserted element; typed characters follow it (typed “7” → Fr ∨).

TB-4-G6-073	Ra	Radium	347, 226, 22, 22 (open pop-up)	**A** <math><mi>Ra</mi></math> **B** <math><mi>x</mi><mi>Ra</mi></math> **C** <math><mi>Ra</mi><mi>x</mi></math> Empty editor → inserts Ra. The caret ends after the inserted element; typed characters follow it (typed “7” → Ra ∀).
TB-4-G6-074	Rf	Rutherfordium	391, 226, 22, 22 (open pop-up)	**A** <math><mi>Rf</mi></math> **B** <math><mi>x</mi><mi>Rf</mi></math> **C** <math><mi>Rf</mi><mi>x</mi></math> Empty editor → inserts Rf. The caret ends after the inserted element; typed characters follow it (typed “7” → Rf ∀).
TB-4-G6-075	Db	Dubnium	413, 226, 22, 22 (open pop-up)	**A** <math><mi>Db</mi></math> **B** <math><mi>x</mi><mi>Db</mi></math> **C** <math><mi>Db</mi><mi>x</mi></math> Empty editor → inserts Db. The caret ends after the inserted element; typed characters follow it (typed “7” → Db ∀).
TB-4-G6-076	Sg	Seaborgium	435, 226, 22, 22 (open pop-up)	**A** <math><mi>Sg</mi></math> **B** <math><mi>x</mi><mi>Sg</mi></math> **C** <math><mi>Sg</mi><mi>x</mi></math> Empty editor → inserts Sg. The caret ends after the inserted element; typed characters follow it (typed “7” → Sg ∀).
TB-4-G6-077	Bh	Bohrium	457, 226, 22, 22 (open pop-up)	**A** <math><mi>Bh</mi></math> **B** <math><mi>x</mi><mi>Bh</mi></math> **C** <math><mi>Bh</mi><mi>x</mi></math> Empty editor → inserts Bh. The caret ends after the inserted element; typed characters follow it (typed “7” → Bh ∀).
TB-4-G6-078	Hs	Hassium	479, 226, 22, 22 (open pop-up)	**A** <math><mi>Hs</mi></math> **B** <math><mi>x</mi><mi>Hs</mi></math> **C** <math><mi>Hs</mi><mi>x</mi></math> Empty editor → inserts Hs. The caret ends after the inserted element; typed characters follow it (typed “7” → Hs ∀).
TB-4-G6-079	Mt	Meitnerium	501, 226, 22, 22 (open pop-up)	**A** <math><mi>Mt</mi></math> **B** <math><mi>x</mi><mi>Mt</mi></math> **C** <math><mi>Mt</mi><mi>x</mi></math> Empty editor → inserts Mt. The caret ends after the inserted element; typed characters follow it (typed “7” → Mt ∀).
TB-4-G6-080	Ds	Darmstadtium	523, 226, 22, 22 (open pop-up)	**A** <math><mi>Ds</mi></math> **B** <math><mi>x</mi><mi>Ds</mi></math> **C** <math><mi>Ds</mi><mi>x</mi></math> Empty editor → inserts Ds. The caret ends after the inserted element; typed characters follow it (typed “7” → Ds ∀).
TB-4-G6-081	Rg	Roentgenium	545, 226, 22, 22 (open pop-up)	**A** <math><mi>Rg</mi></math> **B** <math><mi>x</mi><mi>Rg</mi></math> **C** <math><mi>Rg</mi><mi>x</mi></math> Empty editor → inserts Rg. The caret ends after the inserted element; typed characters follow it (typed “7” → Rg ∀).
TB-4-G6-082	Cn	Copernicium	567, 226, 22, 22 (open pop-up)	**A** <math><mi>Cn</mi></math> **B** <math><mi>x</mi><mi>Cn</mi></math> **C** <math><mi>Cn</mi><mi>x</mi></math> Empty editor → inserts Cn. The caret ends after the inserted element; typed characters follow it (typed “7” → Cn ∀).
TB-4-G6-083	Nh	Nihonium	589, 226, 22, 22 (open pop-up)	**A** <math><mi>Nh</mi></math> **B** <math><mi>x</mi><mi>Nh</mi></math> **C** <math><mi>Nh</mi><mi>x</mi></math> Empty editor → inserts Nh. The caret ends after the inserted element; typed characters follow it (typed “7” → Nh ∀).
TB-4-G6-084	Fl	Flerovium	611, 226, 22, 22 (open pop-up)	**A** <math><mi>Fl</mi></math> **B** <math><mi>x</mi><mi>Fl</mi></math> **C** <math><mi>Fl</mi><mi>x</mi></math> Empty editor → inserts Fl. The caret ends after the inserted element; typed characters follow it (typed “7” → Fl ∀).
TB-4-G6-085	Mc	Moscovium	633, 226, 22, 22 (open pop-up)	**A** <math><mi>Mc</mi></math> **B** <math><mi>x</mi><mi>Mc</mi></math> **C** <math><mi>Mc</mi><mi>x</mi></math> Empty editor → inserts Mc. The caret ends after the inserted element; typed characters follow it (typed “7” → Mc ∀).
TB-4-G6-086	Lv	Livermorium	655, 226, 22, 22 (open pop-up)	**A** <math><mi>Lv</mi></math> **B** <math><mi>x</mi><mi>Lv</mi></math> **C** <math><mi>Lv</mi><mi>x</mi></math> Empty editor → inserts Lv. The caret ends after the inserted element; typed characters follow it (typed “7” → Lv ∀).
TB-4-G6-087	Ts	Tennessine	677, 226, 22, 22 (open pop-up)	**A** <math><mi>Ts</mi></math> **B** <math><mi>x</mi><mi>Ts</mi></math> **C** <math><mi>Ts</mi><mi>x</mi></math> Empty editor → inserts Ts. The caret ends after the inserted element; typed characters follow it (typed “7” → Ts ∀).

TB-4-G6-088	Og	Oganesson	699, 226, 22, 22 (open pop-up)	**A** <math><mi>Og</mi></math> **B** <math><mi>x</mi><mi>Og</mi></math> **C** <math><mi>Og</mi><mi>x</mi></math> Empty editor → inserts Og. The caret ends after the inserted element; typed characters follow it (typed “7” → Og ∨).
TB-4-G6-089	La	Lanthanum	391, 248, 22, 22 (open pop-up)	**A** <math><mi>La</mi></math> **B** <math><mi>x</mi><mi>La</mi></math> **C** <math><mi>La</mi><mi>x</mi></math> Empty editor → inserts La. The caret ends after the inserted element; typed characters follow it (typed “7” → La ∨).
TB-4-G6-090	Ce	Cerium	413, 248, 22, 22 (open pop-up)	**A** <math><mi>Ce</mi></math> **B** <math><mi>x</mi><mi>Ce</mi></math> **C** <math><mi>Ce</mi><mi>x</mi></math> Empty editor → inserts Ce. The caret ends after the inserted element; typed characters follow it (typed “7” → Ce ∨).
TB-4-G6-091	Pr	Praseodymium	435, 248, 22, 22 (open pop-up)	**A** <math><mi>Pr</mi></math> **B** <math><mi>x</mi><mi>Pr</mi></math> **C** <math><mi>Pr</mi><mi>x</mi></math> Empty editor → inserts Pr. The caret ends after the inserted element; typed characters follow it (typed “7” → Pr ∨).
TB-4-G6-092	Nd	Neodymium	457, 248, 22, 22 (open pop-up)	**A** <math><mi>Nd</mi></math> **B** <math><mi>x</mi><mi>Nd</mi></math> **C** <math><mi>Nd</mi><mi>x</mi></math> Empty editor → inserts Nd. The caret ends after the inserted element; typed characters follow it (typed “7” → Nd ∨).
TB-4-G6-093	Pm	Promethium	479, 248, 22, 22 (open pop-up)	**A** <math><mi>Pm</mi></math> **B** <math><mi>x</mi><mi>Pm</mi></math> **C** <math><mi>Pm</mi><mi>x</mi></math> Empty editor → inserts Pm. The caret ends after the inserted element; typed characters follow it (typed “7” → Pm ∨).
TB-4-G6-094	Sm	Samarium	501, 248, 22, 22 (open pop-up)	**A** <math><mi>Sm</mi></math> **B** <math><mi>x</mi><mi>Sm</mi></math> **C** <math><mi>Sm</mi><mi>x</mi></math> Empty editor → inserts Sm. The caret ends after the inserted element; typed characters follow it (typed “7” → Sm ∨).
TB-4-G6-095	Eu	Europium	523, 248, 22, 22 (open pop-up)	**A** <math><mi>Eu</mi></math> **B** <math><mi>x</mi><mi>Eu</mi></math> **C** <math><mi>Eu</mi><mi>x</mi></math> Empty editor → inserts Eu. The caret ends after the inserted element; typed characters follow it (typed “7” → Eu ∨).
TB-4-G6-096	Gd	Gadolinium	545, 248, 22, 22 (open pop-up)	**A** <math><mi>Gd</mi></math> **B** <math><mi>x</mi><mi>Gd</mi></math> **C** <math><mi>Gd</mi><mi>x</mi></math> Empty editor → inserts Gd. The caret ends after the inserted element; typed characters follow it (typed “7” → Gd ∨).
TB-4-G6-097	Tb	Terbium	567, 248, 22, 22 (open pop-up)	**A** <math><mi>Tb</mi></math> **B** <math><mi>x</mi><mi>Tb</mi></math> **C** <math><mi>Tb</mi><mi>x</mi></math> Empty editor → inserts Tb. The caret ends after the inserted element; typed characters follow it (typed “7” → Tb ∨).
TB-4-G6-098	Dy	Dysprosium	589, 248, 22, 22 (open pop-up)	**A** <math><mi>Dy</mi></math> **B** <math><mi>x</mi><mi>Dy</mi></math> **C** <math><mi>Dy</mi><mi>x</mi></math> Empty editor → inserts Dy. The caret ends after the inserted element; typed characters follow it (typed “7” → Dy ∨).
TB-4-G6-099	Ho	Holmium	611, 248, 22, 22 (open pop-up)	**A** <math><mi>Ho</mi></math> **B** <math><mi>x</mi><mi>Ho</mi></math> **C** <math><mi>Ho</mi><mi>x</mi></math> Empty editor → inserts Ho. The caret ends after the inserted element; typed characters follow it (typed “7” → Ho ∨).
TB-4-G6-100	Er	Erbium	633, 248, 22, 22 (open pop-up)	**A** <math><mi>Er</mi></math> **B** <math><mi>x</mi><mi>Er</mi></math> **C** <math><mi>Er</mi><mi>x</mi></math> Empty editor → inserts Er. The caret ends after the inserted element; typed characters follow it (typed “7” → Er ∨).
TB-4-G6-101	Tm	Thulium	655, 248, 22, 22 (open pop-up)	**A** <math><mi>Tm</mi></math> **B** <math><mi>x</mi><mi>Tm</mi></math> **C** <math><mi>Tm</mi><mi>x</mi></math> Empty editor → inserts Tm. The caret ends after the inserted element; typed characters follow it (typed “7” → Tm ∨).
TB-4-G6-102	Yb	Ytterbium	677, 248, 22, 22 (open pop-up)	**A** <math><mi>Yb</mi></math> **B** <math><mi>x</mi><mi>Yb</mi></math> **C** <math><mi>Yb</mi><mi>x</mi></math> Empty editor → inserts Yb. The caret ends after the inserted element; typed characters follow it (typed “7” → Yb ∨).

TB-4-G6-103	Lu	Lutetium	699, 248, 22, 22 (open pop-up)	**A** <math><mi>Lu</mi></math> **B** <math><mi>x</mi><mi>Lu</mi></math> **C** <math><mi>Lu</mi><mi>x</mi></math> Empty editor → inserts Lu. The caret ends after the inserted element; typed characters follow it (typed “7” → Lu ∨).
TB-4-G6-104	Ac	Actinium	391, 270, 22, 22 (open pop-up)	**A** <math><mi>Ac</mi></math> **B** <math><mi>x</mi><mi>Ac</mi></math> **C** <math><mi>Ac</mi><mi>x</mi></math> Empty editor → inserts Ac. The caret ends after the inserted element; typed characters follow it (typed “7” → Ac ∨).
TB-4-G6-105	Th	Thorium	413, 270, 22, 22 (open pop-up)	**A** <math><mi>Th</mi></math> **B** <math><mi>x</mi><mi>Th</mi></math> **C** <math><mi>Th</mi><mi>x</mi></math> Empty editor → inserts Th. The caret ends after the inserted element; typed characters follow it (typed “7” → Th ∨).
TB-4-G6-106	Pa	Protactinium	435, 270, 22, 22 (open pop-up)	**A** <math><mi>Pa</mi></math> **B** <math><mi>x</mi><mi>Pa</mi></math> **C** <math><mi>Pa</mi><mi>x</mi></math> Empty editor → inserts Pa. The caret ends after the inserted element; typed characters follow it (typed “7” → Pa ∨).
TB-4-G6-107	U	Uranium	457, 270, 22, 22 (open pop-up)	**A** <math><mi mathvariant="normal">U</mi></math> **B** <math><mi>x</mi><mi mathvariant="normal">U</mi></math> **C** <math><mi mathvariant="normal">U</mi></math> Empty editor → inserts U. The caret ends after the inserted element; typed characters follow it (typed “7” → U ∨).
TB-4-G6-108	Np	Neptunium	479, 270, 22, 22 (open pop-up)	**A** <math><mi>Np</mi></math> **B** <math><mi>x</mi><mi>Np</mi></math> **C** <math><mi>Np</mi><mi>x</mi></math> Empty editor → inserts Np. The caret ends after the inserted element; typed characters follow it (typed “7” → Np ∨).
TB-4-G6-109	Pu	Plutonium	501, 270, 22, 22 (open pop-up)	**A** <math><mi>Pu</mi></math> **B** <math><mi>x</mi><mi>Pu</mi></math> **C** <math><mi>Pu</mi><mi>x</mi></math> Empty editor → inserts Pu. The caret ends after the inserted element; typed characters follow it (typed “7” → Pu ∨).
TB-4-G6-110	Am	Americium	523, 270, 22, 22 (open pop-up)	**A** <math><mi>Am</mi></math> **B** <math><mi>x</mi><mi>Am</mi></math> **C** <math><mi>Am</mi><mi>x</mi></math> Empty editor → inserts Am. The caret ends after the inserted element; typed characters follow it (typed “7” → Am ∨).
TB-4-G6-111	Cm	Curium	545, 270, 22, 22 (open pop-up)	**A** <math><mi>Cm</mi></math> **B** <math><mi>x</mi><mi>Cm</mi></math> **C** <math><mi>Cm</mi><mi>x</mi></math> Empty editor → inserts Cm. The caret ends after the inserted element; typed characters follow it (typed “7” → Cm ∨).
TB-4-G6-112	Bk	Barkelium	567, 270, 22, 22 (open pop-up)	**A** <math><mi>Bk</mi></math> **B** <math><mi>x</mi><mi>Bk</mi></math> **C** <math><mi>Bk</mi><mi>x</mi></math> Empty editor → inserts Bk. The caret ends after the inserted element; typed characters follow it (typed “7” → Bk ∨).
TB-4-G6-113	Cf	Californium	589, 270, 22, 22 (open pop-up)	**A** <math><mi>Cf</mi></math> **B** <math><mi>x</mi><mi>Cf</mi></math> **C** <math><mi>Cf</mi><mi>x</mi></math> Empty editor → inserts Cf. The caret ends after the inserted element; typed characters follow it (typed “7” → Cf ∨).
TB-4-G6-114	Es	Einsteinium	611, 270, 22, 22 (open pop-up)	**A** <math><mi>Es</mi></math> **B** <math><mi>x</mi><mi>Es</mi></math> **C** <math><mi>Es</mi><mi>x</mi></math> Empty editor → inserts Es. The caret ends after the inserted element; typed characters follow it (typed “7” → Es ∨).
TB-4-G6-115	Fm	Fermium	633, 270, 22, 22 (open pop-up)	**A** <math><mi>Fm</mi></math> **B** <math><mi>x</mi><mi>Fm</mi></math> **C** <math><mi>Fm</mi><mi>x</mi></math> Empty editor → inserts Fm. The caret ends after the inserted element; typed characters follow it (typed “7” → Fm ∨).
TB-4-G6-116	Md	Mendelevium	655, 270, 22, 22 (open pop-up)	**A** <math><mi>Md</mi></math> **B** <math><mi>x</mi><mi>Md</mi></math> **C** <math><mi>Md</mi><mi>x</mi></math> Empty editor → inserts Md. The caret ends after the inserted element; typed characters follow it (typed “7” → Md ∨).
TB-4-G6-117	No	Nobelium	677, 270, 22, 22 (open pop-up)	**A** <math><mi>No</mi></math> **B** <math><mi>x</mi><mi>No</mi></math> **C** <math><mi>No</mi><mi>x</mi></math> Empty editor → inserts No. The caret ends after the inserted element; typed characters follow it (typed “7” → No ∨).

TB-4-G6-118	Lr	Lawrencium	699, 270, 22, 22 (open pop-up)	**A** <math><mi>Lr</mi></math> **B** <math><mi>x</mi><mi>Lr</mi></math> **C** <math><mi>Lr</mi><mi>x</mi></math> Empty editor → inserts Lr. The caret ends after the inserted element; typed characters follow it (typed “7” → Lr ∀).

C.5 Tab 5 — Matrices and elementary (Alt+5)

Main panel (20 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-5-001		Table	3, 28, 33, 33	Opens the size picker (MPK-001…009). Results: Table → ; matrices → …
TB-5-002		Matrix with vertical bars	36, 28, 33, 33	Opens the size picker (MPK-001…009). Results: Table → ; matrices → …
TB-5-003		Matrix with square brackets	3, 61, 33, 33	Opens the size picker (MPK-001…009). Results: Table → ; matrices → …
TB-5-004		Matrix with parenthesis	36, 61, 33, 33	Opens the size picker (MPK-001…009). Results: Table → ; matrices → …
TB-5-005		3 rows column	72, 28, 33, 33	**A** <math><mtable><mtr><mtd/></mtr><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></math> **B** <math><mi>x</mi><mtable><mtr><mtd/></mtr><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></math> **C** <math><mtable><mtr><mtd><mi>x</mi></mtd></mtr><mtr><mtd/></mtr><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></math> Empty editor → inserts table 3×1[□ ; □ ; □]. The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → table 3×1[7 ; □ ; □]).
TB-5-006		2 rows column with square brackets	105, 28, 33, 33	**A** <math><mfenced open="[" close="]"><mtable><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced open="[" close="]"><mtable><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced open="[" close="]"><mtable><mtr><mtd><mi>x</mi></mtd></mtr><mtr><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced [table 2×1[□ ; □]]. The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → fenced [table 2×1[7 ; □]]).
TB-5-007		2 rows column with parenthesis	138, 28, 33, 33	**A** <math><mfenced><mtable><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced><mtable><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced><mtable><mtr><mtd><mi>x</mi></mtd></mtr><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced (table 2×1[□ ; □]). The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → fenced (table 2×1[7 ; □])).
TB-5-008		3 column row	72, 61, 33, 33	**A** <math><mtable><mtr><mtd/><mtd/><mtd/></mtr></mtable></math> **B** <math><mi>x</mi><mtable><mtr><mtd/><mtd/><mtd/></mtr></mtable></math> **C** <math><mtable><mtr><mtd><mi>x</mi></mtd><mtd/><mtd/></mtr></mtable></math> Empty editor → inserts table 1×3[□ | □ | □]. The caret is placed inside the table; the next typed character goes there (typed “7” → table 1×3[7 | □ | □]).
TB-5-009		2 columns row with square brackets	105, 61, 33, 33	**A** <math><mfenced open="[" close="]"><mtable><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced open="[" close="]"><mtable><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced open="[" close="]"><mtable><mtr><mtd><mi>x</mi></mtd><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced [table 1×2[□ | □]]. The caret is placed inside the table; the next typed character goes there (typed “7” → fenced [table 1×2[7 | □]]).
TB-5-010		2 columns row with parenthesis	138, 61, 33, 33	**A** <math><mfenced><mtable><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced><mtable><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced><mtable><mtr><mtd><mi>x</mi></mtd><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced (table 1×2[□ | □]). The caret is placed inside the table; the next typed character goes there (typed “7” → fenced (table 1×2[7 | □])).
TB-5-011		2 rows column with left curly brackets	174, 28, 33, 33	**A** <math><mfenced open="{" close=""><mtable columnalign="left"><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced open="{" close=""><mtable columnalign="left"><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced open="{" close=""><mtable columnalign="left"><mtr><mtd><mi>x</mi></mtd></mtr><mtr><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced { table 2×1[□ ; □] ø. The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → fenced { table 2×1[7 ; □] ø).

TB-5-012		Piecewise function	207, 28, 33, 33	**A** <math><mfenced open="{" close=""><mtable columnspacing="1.4ex" columnalign="left"><mtr><mtd/><mtd/></mtr><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced open="{" close=""><mtable columnspacing="1.4ex" columnalign="left"><mtr><mtd/><mtd/></mtr><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced open="{" close=""><mtable columnspacing="1.4ex" columnalign="left"><mtr><mtd><mi>x</mi></mtd><mtd/></mtr><mtr><mtd/><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced { table 2×2[□ | □ ; □ | □] ⌀. The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → fenced { table 2×2[7 | □ ; □ | □] ⌀).
TB-5-013		2 rows column with right curly brackets	174, 61, 33, 33	**A** <math><mfenced open="" close="}"><mtable columnalign="right"><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **B** <math><mi>x</mi><mfenced open="" close="}"><mtable columnalign="right"><mtr><mtd/></mtr><mtr><mtd/></mtr></mtable></mfenced></math> **C** <math><mfenced open="" close="}"><mtable columnalign="right"><mtr><mtd><mi>x</mi></mtd></mtr><mtr><mtd/></mtr></mtable></mfenced></math> Empty editor → inserts fenced ⌀ table 2×1[□ ; □] }. The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → fenced ⌀ table 2×1[7 ; □] }).
TB-5-014		Aligned equations	207, 61, 33, 33	**A** <math><mtable columnspacing="2px" columnalign="right center left"><mtr><mtd/><mtd><mo>=</mo></mtd><mtd/></mtr><mtr><mtd/><mtd><mo>=</mo></mtd><mtd/></mtr></mtable></math> **B** <math><mi>x</mi><mtable columnspacing="2px" columnalign="right center left"><mtr><mtd/><mtd><mo>=</mo></mtd><mtd/></mtr><mtr><mtd/><mtd><mo>=</mo></mtd><mtd/></mtr></mtable></math> **C** <math><mtable columnspacing="2px" columnalign="right center left"><mtr><mtd><mi>x</mi></mtd><mtd><mo>=</mo></mtd><mtd/></mtr><mtr><mtd/><mtd><mo>=</mo></mtd><mtd/></mtr></mtable></math> Empty editor → inserts table 2×3[□ | = | □ ; □ | = | □]. The caret is placed in the slot 1 of the table; the next typed character goes there (typed “7” → table 2×3[7 | = | □ ; □ | = | □]).
TB-5-015		Vertical ellipsis	243, 28, 33, 33	**A** <math><mo>⋮</mo></math> **B** <math><mi>x</mi><mo>⋮</mo></math> **C** <math><mo>⋮</mo></math> Empty editor → inserts ⋮. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋮ 7).
TB-5-016		Up right diagonal ellipsis	276, 28, 33, 33	**A** <math><mo>⋰</mo></math> **B** <math><mi>x</mi><mo>⋰</mo></math> **C** <math><mo>⋰</mo></math> Empty editor → inserts ⋰. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋰ 7).
TB-5-017		Midline horizontal ellipsis	243, 61, 33, 33	**A** <math><mo>⋯</mo></math> **B** <math><mi>x</mi><mo>⋯</mo></math> **C** <math><mo>⋯</mo></math> Empty editor → inserts ⋯. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋯ 7).
TB-5-018		Down right diagonal ellipsis	276, 61, 33, 33	**A** <math><mo>⋱</mo></math> **B** <math><mi>x</mi><mo>⋱</mo></math> **C** <math><mo>⋱</mo></math> Empty editor → inserts ⋱. The caret ends after the inserted element; typed characters follow it (typed “7” → ⋱ 7).
TB-5-019		3 rows stack with line and plus sign	312, 28, 33, 33	**A** <math><mstack charalign="center" stackalign="right"><msrow/><msrow><mo>+</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> **B** <math><mi>x</mi><mstack charalign="center" stackalign="right"><msrow/><msrow><mo>+</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> **C** <math><mstack charalign="center" stackalign="right"><mi>x</mi><msrow><mo>+</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> Empty editor → inserts mstack(msrow() msrow(+ □) msline() msrow()). The caret is placed in the slot 1 of the elementary stack; the next typed character goes there (typed “7” → mstack(7 msrow(+ □) msline() msrow())).
TB-5-020		Long division (used in the United States, Great Britain, and elsewhere)	312, 61, 33, 33	**A** <math><mlongdiv charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/></msgroup></mlongdiv></math> **B** <math><mi>x</mi><mlongdiv charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/></msgroup></mlongdiv></math> **C** <math><mlongdiv charalign="center" charspacing="0px" stackalign="left"><mi>x</mi><mrow/><msgroup><msrow/></msgroup></mlongdiv></math> Empty editor → inserts mlongdiv(□ □ msgroup(msrow())). The caret is placed in the slot 1 of the long division; the next typed character goes there (typed “7” → mlongdiv(7 □ msgroup(msrow()))).

“More” group 1 (6 controls) — expand button at 345.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-5-G1-001		3 rows stack with line	348, 94, 33, 33 (open pop-up)	**A** <math><mstack charalign="center" stackalign="right"><msrow/><msrow/><msline/><msrow/></mstack></math> **B** <math><mi>x</mi><mstack charalign="center" stackalign="right"><msrow/><msrow/><msline/><msrow/></mstack></math> **C** <math><mstack charalign="center" stackalign="right"><mi>x</mi><msrow/><msline/><msrow/></mstack></math> Empty editor → inserts mstack(msrow() msrow() msline() msrow()). The caret is placed in the slot 1 of the elementary stack; the next typed character goes there (typed “7” → mstack(7 msrow() msline() msrow())).

TB-5-G1-002		3 rows stack with line and minus sign	381, 94, 33, 33 (open pop-up)	**A** <math><mstack charalign="center" stackalign="right"><msrow/><msrow><mo>-</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> **B** <math><mi>x</mi><mstack charalign="center" stackalign="right"><msrow/><msrow><mo>-</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> **C** <math><mstack charalign="center" stackalign="right"><mi>x</mi><msrow><mo>-</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> Empty editor → inserts mstack(msrow() msrow(- □) msline() msrow()). The caret is placed in the slot 1 of the elementary stack; the next typed character goes there (typed “7” → mstack(7 msrow(- □) msline() msrow())).
TB-5-G1-003		3 rows stack with line and multiplication sign	414, 94, 33, 33 (open pop-up)	**A** <math><mstack charalign="center" stackalign="right"><msrow/><msrow><mo>×</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> **B** <math><mi>x</mi><mstack charalign="center" stackalign="right"><msrow/><msrow><mo>×</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> **C** <math><mstack charalign="center" stackalign="right"><mi>x</mi><msrow><mo>×</mo><maction actiontype="argument"><mrow/></maction></msrow><msline/><msrow/></mstack></math> Empty editor → inserts mstack(msrow() msrow(× □) msline() msrow()). The caret is placed in the slot 1 of the elementary stack; the next typed character goes there (typed “7” → mstack(7 msrow(× □) msline() msrow())).
TB-5-G1-004		Long division (used in Spain, Brazil and elsewhere)	348, 127, 33, 33 (open pop-up)	**A** <math><mlongdiv longdivstyle="shortstackedrightright" charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/></msgroup></mlongdiv></math> **B** <math><mi>x</mi><mlongdiv longdivstyle="shortstackedrightright" charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/></msgroup></mlongdiv></math> **C** <math><mlongdiv longdivstyle="shortstackedrightright" charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><mi>x</mi></msgroup></mlongdiv></math> Empty editor → inserts mlongdiv(□ □ msgroup(msrow())). The caret is placed in the slot 3 of the long division; the next typed character goes there (typed “7” → mlongdiv(□ □ msgroup(7))).
TB-5-G1-005		Long division with remainder (used in Spain, Brazil and elsewhere)	381, 127, 33, 33 (open pop-up)	**A** <math><mlongdiv longdivstyle="shortstackedrightright" charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/><msrow/></msgroup></mlongdiv></math> **B** <math><mi>x</mi><mlongdiv longdivstyle="shortstackedrightright" charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/><msrow/></msgroup></mlongdiv></math> **C** <math><mlongdiv longdivstyle="shortstackedrightright" charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><mi>x</mi><msrow/></msgroup></mlongdiv></math> Empty editor → inserts mlongdiv(□ □ msgroup(msrow() msrow())). The caret is placed in the slot 3 of the long division; the next typed character goes there (typed “7” → mlongdiv(□ □ msgroup(7 msrow()))).
TB-5-G1-006		Long division with remainder (used in the United States, Great Britain, and elsewhere)	414, 127, 33, 33 (open pop-up)	**A** <math><mlongdiv charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/><msrow/></msgroup></mlongdiv></math> **B** <math><mi>x</mi><mlongdiv charalign="center" charspacing="0px" stackalign="left"><mrow/><mrow/><msgroup><msrow/><msrow/></msgroup></mlongdiv></math> **C** <math><mlongdiv charalign="center" charspacing="0px" stackalign="left"><mi>x</mi><mrow/><msgroup><msrow/><msrow/></msgroup></mlongdiv></math> Empty editor → inserts mlongdiv(□ □ msgroup(msrow() msrow())). The caret is placed in the slot 1 of the long division; the next typed character goes there (typed “7” → mlongdiv(7 □ msgroup(msrow() msrow()))).

C.6 Tab 6 — Scripts and layout (Alt+6)

Main panel (24 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-6-001		Big fraction	3, 28, 33, 33	**A** <math><mfrac><mrow/><mrow/></mfrac></math> **B** <math><mi>x</mi><mfrac><mrow/><mrow/></mfrac></math> **C** <math><mfrac><mi>x</mi><mrow/></mfrac></math> Empty editor → inserts frac(□ ⁄ □). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → frac(7 ⁄ □)).
TB-6-002		Small fraction	36, 28, 33, 33	**A** <math><mstyle displaystyle="false"><mfrac><mrow/><mrow/></mfrac></mstyle></math> **B** <math><mi>x</mi><mstyle displaystyle="false"><mfrac><mrow/><mrow/></mfrac></mstyle></math> **C** <math><mstyle displaystyle="false"><mfrac><mi>x</mi><mrow/></mfrac></mstyle></math> Empty editor → inserts mstyle[displaystyle=false](frac(□ ⁄ □)). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → mstyle[displaystyle=false](frac(7 ⁄ □))).
TB-6-003		Bevelled big fraction	3, 61, 33, 33	**A** <math><mfrac bevelled="true"><mrow/><mrow/></mfrac></math> **B** <math><mi>x</mi><mfrac bevelled="true"><mrow/><mrow/></mfrac></math> **C** <math><mfrac bevelled="true"><mi>x</mi><mrow/></mfrac></math> Empty editor → inserts bevelledFrac(□ ⁄ □). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → bevelledFrac(7 ⁄ □)).
TB-6-004		Bevelled small fraction	36, 61, 33, 33	**A** <math><mstyle displaystyle="false"><mfrac bevelled="true"><mrow/><mrow/></mfrac></mstyle></math> **B** <math><mi>x</mi><mstyle displaystyle="false"><mfrac bevelled="true"><mrow/><mrow/></mfrac></mstyle></math> **C** <math><mstyle displaystyle="false"><mfrac bevelled="true"><mi>x</mi><mrow/></mfrac></mstyle></math> Empty editor → inserts mstyle[displaystyle=false](bevelledFrac(□ ⁄ □)). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → mstyle[displaystyle=false](bevelledFrac(7 ⁄ □))).

TB-6-005		Square root	72, 28, 33, 33	**A** <math><msqrt/></math> **B** <math><mi>x</mi><msqrt/></math> **C** <math><msqrt><mi>x</mi></msqrt></math> Empty editor → inserts √(□). The caret is placed inside the square root; the next typed character goes there (typed “7” → √(7)).
TB-6-006		Root	72, 61, 33, 33	**A** <math><mroot><mrow/><mrow/></mroot></math> **B** <math><mi>x</mi><mroot><mrow/><mrow/></mroot></math> **C** <math><mroot><mi>x</mi><mrow/></mroot></math> Empty editor → inserts root(radicand □, index □). The caret is placed in the radicand of the root; the next typed character goes there (typed “7” → root(radicand 7, index □)).
TB-6-007		Superscript (Ctrl+Up)	108, 28, 33, 33	**A** <math><msup><maction actiontype="argument"><mrow/></maction><mrow/></msup></math> **B** <math><msup><mi>x</mi><mrow/></msup></math> **C** <math><msup><mi>x</mi><mrow/></msup></math> Empty editor → inserts □^{□}. The caret is placed in the base of the superscript; the next typed character goes there (typed “7” → 7^{□}).
TB-6-008		Superscript and subscript	141, 28, 33, 33	**A** <math><msubsup><maction actiontype="argument"><mrow/></maction><mrow/><mrow/></msubsup></math> **B** <math><msubsup><mi>x</mi><mrow/><mrow/></msubsup></math> **C** <math><msubsup><mi>x</mi><mrow/><mrow/></msubsup></math> Empty editor → inserts □_{□}^{□}. The caret is placed in the base of the subscript+superscript; the next typed character goes there (typed “7” → 7_{□}^{□}).
TB-6-009		Subscript (Ctrl+Down)	174, 28, 33, 33	**A** <math><msub><maction actiontype="argument"><mrow/></maction><mrow/></msub></math> **B** <math><msub><mi>x</mi><mrow/></msub></math> **C** <math><msub><mi>x</mi><mrow/></msub></math> Empty editor → inserts □_{□}. The caret is placed in the base of the subscript; the next typed character goes there (typed “7” → 7_{□}).
TB-6-010		Left superscript	108, 61, 33, 33	**A** <math><mmultiscripts><maction actiontype="argument"><mrow/></maction><mprescripts/><none/><mrow/></mmultiscripts></math> **B** <math><mi>x</mi><mmultiscripts><maction actiontype="argument"><mrow/></maction><mprescripts/><none/><mrow/></mmultiscripts></math> **C** <math><mmultiscripts><mi>x</mi><mprescripts/><none/><mrow/></mmultiscripts></math> Empty editor → inserts multiscripts(□; ⟨pre⟩; ⌀; □). The caret is placed in the slot 1 of the left/right scripts; the next typed character goes there (typed “7” → multiscripts(7; ⟨pre⟩; ⌀; □)).
TB-6-011		Left subscript and superscript	141, 61, 33, 33	**A** <math><mmultiscripts><maction actiontype="argument"><mrow/></maction><mprescripts/><mrow/><mrow/></mmultiscripts></math> **B** <math><mi>x</mi><mmultiscripts><maction actiontype="argument"><mrow/></maction><mprescripts/><mrow/><mrow/></mmultiscripts></math> **C** <math><mmultiscripts><mi>x</mi><mprescripts/><mrow/><mrow/></mmultiscripts></math> Empty editor → inserts multiscripts(□; ⟨pre⟩; □; □). The caret is placed in the slot 1 of the left/right scripts; the next typed character goes there (typed “7” → multiscripts(7; ⟨pre⟩; □; □)).
TB-6-012		Left subscript	174, 61, 33, 33	**A** <math><mmultiscripts><maction actiontype="argument"><mrow/></maction><mprescripts/><mrow/><none/></mmultiscripts></math> **B** <math><mi>x</mi><mmultiscripts><maction actiontype="argument"><mrow/></maction><mprescripts/><mrow/><none/></mmultiscripts></math> **C** <math><mmultiscripts><mi>x</mi><mprescripts/><mrow/><none/></mmultiscripts></math> Empty editor → inserts multiscripts(□; ⟨pre⟩; □; ⌀). The caret is placed in the slot 1 of the left/right scripts; the next typed character goes there (typed “7” → multiscripts(7; ⟨pre⟩; □; ⌀)).
TB-6-013		Element over	210, 28, 33, 33	**A** <math><mover><mrow/><mrow/></mover></math> **B** <math><mi>x</mi><mover><mrow/><mrow/></mover></math> **C** <math><mover><mi>x</mi><mrow/></mover></math> Empty editor → inserts over(□; □). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; □)).
TB-6-014		Element under	243, 28, 33, 33	**A** <math><munder><mrow/><mrow/></munder></math> **B** <math><mi>x</mi><munder><mrow/><mrow/></munder></math> **C** <math><munder><mi>x</mi><mrow/></munder></math> Empty editor → inserts under(□; □). The caret is placed in the base of the under-script; the next typed character goes there (typed “7” → under(7; □)).
TB-6-015		Elements under and over	210, 61, 33, 33	**A** <math><munderover><mrow/><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mrow/><mrow/><mrow/></munderover></math> **C** <math><munderover><mi>x</mi><mrow/><mrow/></munderover></math> Empty editor → inserts underover(□; □; □). The caret is placed in the base of the under+over scripts; the next typed character goes there (typed “7” → underover(7; □; □)).
TB-6-016		Under script with brace	279, 28, 33, 33	**A** <math><munder><munder><mrow/><mo>⏟</mo></munder><mrow/></munder></math> **B** <math><mi>x</mi><munder><munder><mrow/><mo>⏟</mo></munder><mrow/></munder></math> **C** <math><munder><munder><mi>x</mi><mo>⏟</mo></munder><mrow/></munder></math> Empty editor → inserts under(under(□; ⏟); □). The caret is placed in the base of the under-script; the next typed character goes there (typed “7” → under(under(7; ⏟); □)).
TB-6-017		Over script with brace	279, 61, 33, 33	**A** <math><mover><mover><mrow/><mo>⏞</mo></mover><mrow/></mover></math> **B** <math><mi>x</mi><mover><mover><mrow/><mo>⏞</mo></mover><mrow/></mover></math> **C** <math><mover><mover><mi>x</mi><mo>⏞</mo></mover><mrow/></mover></math> Empty editor → inserts over(over(□; ⏞); □). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(over(7; ⏞); □)).
TB-6-018		Big operator with under and over scripts	315, 28, 33, 33	**A** <math><munderover><mo largeop="true"></mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo largeop="true"></mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mrow><mi>x</mi><mo largeop="true"></mo></mrow><mrow/><mrow/></munderover></math> Empty editor → inserts underover(; □; □). The caret is placed in the base of the under+over scripts; the next typed character goes there (typed “7” → underover(7; □; □)).

TB-6-019		Big operator with subscript and superscript	348, 28, 33, 33	**A** <math><msubsup><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/><mrow/></msubsup></math> **B** <math><mi>x</mi><msubsup><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/><mrow/></msubsup></math> **C** <math><msubsup><maction actiontype="argument"><mo largeop="true"></mo></maction><mi>x</mi><mrow/></msubsup></math> Empty editor → inserts _{□}^{□}. The caret is placed in the subscript of the subscript+superscript; the next typed character goes there (typed “7” → _{7}^{□}).
TB-6-020		Big operator with under script	315, 61, 33, 33	**A** <math><munder><mo largeop="true"></mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo largeop="true"></mo><mrow/></munder></math> **C** <math><munder><mrow><mi>x</mi><mo largeop="true"></mo></mrow><mrow/></munder></math> Empty editor → inserts under(; □). The caret is placed in the base of the under-script; the next typed character goes there (typed “7” → under(7; □)).
TB-6-021		Big operator with subscript	348, 61, 33, 33	**A** <math><msub><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/></msub></math> **B** <math><mi>x</mi><msub><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/></msub></math> **C** <math><msub><maction actiontype="argument"><mo largeop="true"></mo></maction><mi>x</mi></msub></math> Empty editor → inserts _{□}. The caret is placed in the subscript of the subscript; the next typed character goes there (typed “7” → _{7}).
TB-6-022		Digit space	384, 28, 33, 33	**A** <math><mo> </mo></math> **B** <math><mi>x</mi><mo> </mo></math> **C** <math><mo> </mo></math> Empty editor → inserts . The caret ends after the inserted element; typed characters follow it (typed “7” → 7).
TB-6-023		Thinner space	417, 28, 33, 33	**A** <math><mo> </mo></math> **B** <math><mi>x</mi><mo> </mo></math> **C** <math><mo> </mo></math> Empty editor → inserts . The caret ends after the inserted element; typed characters follow it (typed “7” → 7).
TB-6-024		Back space	384, 61, 33, 33	**A** <math><mspace width="-0.2em"/></math> **B** <math><mi>x</mi><mspace width="-0.2em"/></math> **C** <math><mspace width="-0.2em"/></math> Empty editor → inserts space[width=-0.2em]. The caret ends after the inserted element; typed characters follow it (typed “7” → space[width=-0.2em] 7).

C.7 Tab 7 — Decorations (Alt+7)

Main panel (28 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-7-001		Parentheses	3, 28, 33, 33	**A** <math><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced><mrow/></mfenced></math> **C** <math><mfenced><mi>x</mi></mfenced></math> Empty editor → inserts fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced (7)).
TB-7-002		Vertical bars	36, 28, 33, 33	**A** <math><mfenced open="|" close="|"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="|" close="|"><mrow/></mfenced></math> **C** <math><mfenced open="|" close="|"><mi>x</mi></mfenced></math> Empty editor → inserts fenced	□ |. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced | 7 |).
TB-7-003		Angle brackets	69, 28, 33, 33	**A** <math><mfenced open="<" close=">"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="<" close=">"><mrow/></mfenced></math> **C** <math><mfenced open="<" close=">"><mi>x</mi></mfenced></math> Empty editor → inserts fenced < □ >. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced < 7 >).
TB-7-004		Square brackets	3, 61, 33, 33	**A** <math><mfenced open="[" close="]"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="[" close="]"><mrow/></mfenced></math> **C** <math><mfenced open="[" close="]"><mi>x</mi></mfenced></math> Empty editor → inserts fenced [□]. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced [7]).
TB-7-005		Double vertical bars	36, 61, 33, 33	**A** <math><mfenced open="||" close="||"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="||" close="||"><mrow/></mfenced></math> **C** <math><mfenced open="||" close="||"><mi>x</mi></mfenced></math> Empty editor → inserts fenced || □ ||. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced || 7 ||).
TB-7-006		Curly brackets	69, 61, 33, 33	**A** <math><mfenced open="{" close="}"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="{" close="}"><mrow/></mfenced></math> **C** <math><mfenced open="{" close="}"><mi>x</mi></mfenced></math> Empty editor → inserts fenced { □ }. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced { 7 }).
TB-7-007		Top curly bracket	113, 28, 33, 33	**A** <math><mover><mrow/><mo>⏞</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>⏞</mo></mover></math> **C** <math><mover><mi>x</mi><mo>⏞</mo></mover></math> Empty editor → inserts over(□; ⏜). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ⏜)).

TB-7-008		Top parenthesis	146, 28, 33, 33	**A** <math><mover><mrow/><mo>⏜</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>⏜</mo></mover></math> **C** <math><mover><mi>x</mi><mo>⏜</mo></mover></math> Empty editor → inserts over(□; ⏜). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ⏜)).
TB-7-009		Bottom curly bracket	113, 61, 33, 33	**A** <math><munder><mrow/><mo>⏟</mo></munder></math> **B** <math><mi>x</mi><munder><mrow/><mo>⏟</mo></munder></math> **C** <math><munder><mi>x</mi><mo>⏟</mo></munder></math> Empty editor → inserts under(□; ⏟). The caret is placed in the base of the under-script; the next typed character goes there (typed “7” → under(7; ⏟)).
TB-7-010		Bottom parenthesis	146, 61, 33, 33	**A** <math><munder><mrow/><mo>⏝</mo></munder></math> **B** <math><mi>x</mi><munder><mrow/><mo>⏝</mo></munder></math> **C** <math><munder><mi>x</mi><mo>⏝</mo></munder></math> Empty editor → inserts under(□; ⏝). The caret is placed in the base of the under-script; the next typed character goes there (typed “7” → under(7; ⏝)).
TB-7-011		Vector accent	182, 28, 22, 22	**A** <math><mover><mrow/><mo>⇀</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>⇀</mo></mover></math> **C** <math><mover><mi>x</mi><mo>⇀</mo></mover></math> Empty editor → inserts over(□; ⇀). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ⇀)).
TB-7-012		Arrow accent	204, 28, 22, 22	**A** <math><mover><mrow/><mo>→</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>→</mo></mover></math> **C** <math><mover><mi>x</mi><mo>→</mo></mover></math> Empty editor → inserts over(□; →). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; →)).
TB-7-013		Left right arrow accent	226, 28, 22, 22	**A** <math><mover><mrow/><mo>↔</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>↔</mo></mover></math> **C** <math><mover><mi>x</mi><mo>↔</mo></mover></math> Empty editor → inserts over(□; ↔). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ↔)).
TB-7-014		Bar accent	182, 50, 22, 22	**A** <math><mover><mrow/><mo>¯</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>¯</mo></mover></math> **C** <math><mover><mi>x</mi><mo>¯</mo></mover></math> Empty editor → inserts over(□; ¯). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ¯)).
TB-7-015		Hat accent	204, 50, 22, 22	**A** <math><mover><mrow/><mo>^</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>^</mo></mover></math> **C** <math><mover><mi>x</mi><mo>^</mo></mover></math> Empty editor → inserts over(□; ^). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ^)).
TB-7-016		Tilde	226, 50, 22, 22	**A** <math><mover><mrow/><mo>~</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>~</mo></mover></math> **C** <math><mover><mi>x</mi><mo>~</mo></mover></math> Empty editor → inserts over(□; ~). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ~)).
TB-7-017		Diaeresis	182, 72, 22, 22	**A** <math><mover><mrow/><mo>¨</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>¨</mo></mover></math> **C** <math><mover><mi>x</mi><mo>¨</mo></mover></math> Empty editor → inserts over(□; ¨). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ¨)).
TB-7-018		Dot accent	204, 72, 22, 22	**A** <math><mover><mrow/><mo>˙</mo></mover></math> **B** <math><mi>x</mi><mover><mrow/><mo>˙</mo></mover></math> **C** <math><mover><mi>x</mi><mo>˙</mo></mover></math> Empty editor → inserts over(□; ˙). The caret is placed in the base of the over-script; the next typed character goes there (typed “7” → over(7; ˙)).
TB-7-019		Enclose top	251, 28, 33, 33	**A** <math><menclose notation="top"/></math> **B** <math><mi>x</mi><menclose notation="top"/></math> **C** <math><menclose notation="top"><mi>x</mi></menclose></math> Empty editor → inserts enclose[top](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[top](7)).
TB-7-020		Enclose left	284, 28, 33, 33	**A** <math><menclose notation="left"/></math> **B** <math><mi>x</mi><menclose notation="left"/></math> **C** <math><menclose notation="left"><mi>x</mi></menclose></math> Empty editor → inserts enclose[left](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[left](7)).
TB-7-021		Enclose box	317, 28, 33, 33	**A** <math><menclose notation="box"/></math> **B** <math><mi>x</mi><menclose notation="box"/></math> **C** <math><menclose notation="box"><mi>x</mi></menclose></math> Empty editor → inserts enclose[box](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[box](7)).
TB-7-022		Enclose bottom	251, 61, 33, 33	**A** <math><menclose notation="bottom"/></math> **B** <math><mi>x</mi><menclose notation="bottom"/></math> **C** <math><menclose notation="bottom"><mi>x</mi></menclose></math> Empty editor → inserts enclose[bottom](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[bottom](7)).
TB-7-023		Enclose right	284, 61, 33, 33	**A** <math><menclose notation="right"/></math> **B** <math><mi>x</mi><menclose notation="right"/></math> **C** <math><menclose notation="right"><mi>x</mi></menclose></math> Empty editor → inserts enclose[right](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[right](7)).

TB-7-024		Enclose circle	317, 61, 33, 33	**A** <math><menclose notation="circle"/></math> **B** <math><mi>x</mi><menclose notation="circle"/></math> **C** <math><menclose notation="circle"><mi>x</mi></menclose></math> Empty editor → inserts enclose[circle](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[circle](7)).
TB-7-025		Up diagonal strike	361, 28, 33, 33	**A** <math><menclose notation="updiagonalstrike"/></math> **B** <math><mi>x</mi><menclose notation="updiagonalstrike"/></math> **C** <math><menclose notation="updiagonalstrike"><mi>x</mi></menclose></math> Empty editor → inserts enclose[updiagonalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[updiagonalstrike](7)).
TB-7-026		Horizontal strike	394, 28, 33, 33	**A** <math><menclose notation="horizontalstrike"/></math> **B** <math><mi>x</mi><menclose notation="horizontalstrike"/></math> **C** <math><menclose notation="horizontalstrike"><mi>x</mi></menclose></math> Empty editor → inserts enclose[horizontalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[horizontalstrike](7)).
TB-7-027		Down diagonal strike	361, 61, 33, 33	**A** <math><menclose notation="downdiagonalstrike"/></math> **B** <math><mi>x</mi><menclose notation="downdiagonalstrike"/></math> **C** <math><menclose notation="downdiagonalstrike"><mi>x</mi></menclose></math> Empty editor → inserts enclose[downdiagonalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[downdiagonalstrike](7)).
TB-7-028		Up and down diagonal strike	394, 61, 33, 33	**A** <math><menclose notation="downdiagonalstrike updiagonalstrike"/></math> **B** <math><mi>x</mi><menclose notation="downdiagonalstrike updiagonalstrike"/></math> **C** <math><menclose notation="downdiagonalstrike updiagonalstrike"><mi>x</mi></menclose></math> Empty editor → inserts enclose[downdiagonalstrike updiagonalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[downdiagonalstrike updiagonalstrike](7)).

“More” group 1 (3 controls) — expand button at 103.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-7-G1-001		Floor	106, 94, 33, 33 (open pop-up)	**A** <math><mfenced open="⌊" close="⌋"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="⌊" close="⌋"><mrow/></mfenced></math> **C** <math><mfenced open="⌊" close="⌋"><mi>x</mi></mfenced></math> Empty editor → inserts fenced ⌊ □ ⌋. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced ⌊ 7 ⌋).
TB-7-G1-002		Angle brackets with bar	139, 94, 33, 33 (open pop-up)	**A** <math><mfenced open="<" close=">" separators="|"><mrow/><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="<" close=">" separators="|"><mrow/><mrow/></mfenced></math> **C** <math><mfenced open="<" close=">" separators="|"><mi>x</mi><mrow/></mfenced></math> Empty editor → inserts fenced[sep |] < □, □ >. The caret is placed in the slot 1 of the fence; the next typed character goes there (typed “7” → fenced[sep |] < 7, □ >).
TB-7-G1-003		Ceiling	106, 127, 33, 33 (open pop-up)	**A** <math><mfenced open="⌈" close="⌉"><mrow/></mfenced></math> **B** <math><mi>x</mi><mfenced open="⌈" close="⌉"><mrow/></mfenced></math> **C** <math><mfenced open="⌈" close="⌉"><mi>x</mi></mfenced></math> Empty editor → inserts fenced ⌈ □ ⌉. The caret is placed inside the fence; the next typed character goes there (typed “7” → fenced ⌈ 7 ⌉).

“More” group 2 (2 controls) — expand button at 351.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-7-G2-001		Enclose actuarial	354, 94, 33, 33 (open pop-up)	**A** <math><menclose notation="actuarial"/></math> **B** <math><mi>x</mi><menclose notation="actuarial"/></math> **C** <math><menclose notation="actuarial"><mi>x</mi></menclose></math> Empty editor → inserts enclose[actuarial](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[actuarial](7)).
TB-7-G2-002		Enclose rounded box	354, 127, 33, 33 (open pop-up)	**A** <math><menclose notation="roundedbox"/></math> **B** <math><mi>x</mi><menclose notation="roundedbox"/></math> **C** <math><menclose notation="roundedbox"><mi>x</mi></menclose></math> Empty editor → inserts enclose[roundedbox](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[roundedbox](7)).

“More” group 3 (3 controls) — expand button at 427.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-7-G3-001		Vertical strike	430, 94, 33, 33 (open pop-up)	**A** `<math><menclose notation="verticalstrike"/></math>` **B** `<math><mi>x</mi><menclose notation="verticalstrike"/></math>` **C** `<math><menclose notation="verticalstrike"><mi>x</mi></menclose></math>` Empty editor → inserts enclose[verticalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[verticalstrike](7)).
TB-7-G3-002		Long division	463, 94, 33, 33 (open pop-up)	**A** `<math><menclose/></math>` **B** `<math><mi>x</mi><menclose/></math>` **C** `<math><menclose><mi>x</mi></menclose></math>` Empty editor → inserts enclose[longdiv](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[longdiv](7)).
TB-7-G3-003		Horizontal and vertical strikes	430, 127, 33, 33 (open pop-up)	**A** `<math><menclose notation="verticalstrike horizontalstrike"/></math>` **B** `<math><mi>x</mi><menclose notation="verticalstrike horizontalstrike"/></math>` **C** `<math><menclose notation="verticalstrike horizontalstrike"><mi>x</mi></menclose></math>` Empty editor → inserts enclose[verticalstrike horizontalstrike](□). The caret is placed inside the enclosure; the next typed character goes there (typed “7” → enclose[verticalstrike horizontalstrike](7)).

C.8 Tab 8 — Big operators (Alt+8)

Main panel (14 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-8-001		Summation with under and over scripts	3, 28, 33, 33	**A** `<math><munderover><mo>∑</mo><mrow/><mrow/></munderover></math>` **B** `<math><mi>x</mi><munderover><mo>∑</mo><mrow/><mrow/></munderover></math>` **C** `<math><munderover><mo>∑</mo><mrow/><mrow/></munderover><mi>x</mi></math>` Empty editor → inserts underover(∑; □; □). The caret ends after the inserted element; typed characters follow it (typed “7” → underover(∑; □; □) 7).
TB-8-002		Summation with subscript and superscript	36, 28, 33, 33	**A** `<math><mstyle displaystyle="false"><munderover><mo>∑</mo><mrow/><mrow/></munderover></mstyle></math>` **B** `<math><mi>x</mi><mstyle displaystyle="false"><munderover><mo>∑</mo><mrow/><mrow/></munderover></mstyle></math>` **C** `<math><mstyle displaystyle="false"><munderover><mo>∑</mo><mi>x</mi><mrow/></munderover></mstyle></math>` Empty editor → inserts mstyle[displaystyle=false](underover(∑; □; □)). The caret is placed in the under-script of the under+over scripts; the next typed character goes there (typed “7” → mstyle[displaystyle=false](underover(∑; 7; □))).
TB-8-003		Summation with under scripts	3, 61, 33, 33	**A** `<math><munder><mo>∑</mo><mrow/></munder></math>` **B** `<math><mi>x</mi><munder><mo>∑</mo><mrow/></munder></math>` **C** `<math><munder><mo>∑</mo><mrow/></munder><mi>x</mi></math>` Empty editor → inserts under(∑; □). The caret ends after the inserted element; typed characters follow it (typed “7” → under(∑; □) 7).
TB-8-004		Summation with subscript	36, 61, 33, 33	**A** `<math><mstyle displaystyle="false"><munder><mo>∑</mo><mrow/></munder></mstyle></math>` **B** `<math><mi>x</mi><mstyle displaystyle="false"><munder><mo>∑</mo><mrow/></munder></mstyle></math>` **C** `<math><mstyle displaystyle="false"><munder><mo>∑</mo><mi>x</mi></munder></mstyle></math>` Empty editor → inserts mstyle[displaystyle=false](under(∑; □)). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → mstyle[displaystyle=false](under(∑; 7))).
TB-8-005		Product with under and over scripts	72, 28, 33, 33	**A** `<math><munderover><mo>∏</mo><mrow/><mrow/></munderover></math>` **B** `<math><mi>x</mi><munderover><mo>∏</mo><mrow/><mrow/></munderover></math>` **C** `<math><munderover><mo>∏</mo><mrow/><mrow/></munderover><mi>x</mi></math>` Empty editor → inserts underover(∏; □; □). The caret ends after the inserted element; typed characters follow it (typed “7” → underover(∏; □; □) 7).
TB-8-006		Product with subscript and superscript	105, 28, 33, 33	**A** `<math><mstyle displaystyle="false"><munderover><mo>∏</mo><mrow/><mrow/></munderover></mstyle></math>` **B** `<math><mi>x</mi><mstyle displaystyle="false"><munderover><mo>∏</mo><mrow/><mrow/></munderover></mstyle></math>` **C** `<math><mstyle displaystyle="false"><munderover><mo>∏</mo><mi>x</mi><mrow/></munderover></mstyle></math>` Empty editor → inserts mstyle[displaystyle=false](underover(∏; □; □)). The caret is placed in the under-script of the under+over scripts; the next typed character goes there (typed “7” → mstyle[displaystyle=false](underover(∏; 7; □))).
TB-8-007		Product with under scripts	72, 61, 33, 33	**A** `<math><munder><mo>∏</mo><mrow/></munder></math>` **B** `<math><mi>x</mi><munder><mo>∏</mo><mrow/></munder></math>` **C** `<math><munder><mo>∏</mo><mrow/></munder><mi>x</mi></math>` Empty editor → inserts under(∏; □). The caret ends after the inserted element; typed characters follow it (typed “7” → under(∏; □) 7).

TB-8-008		Product with subscript	105, 61, 33, 33	**A** <math><mstyle displaystyle="false"><munder><mo>∏</mo><mrow/></munder></mstyle></math> **B** <math><mi>x</mi><mstyle displaystyle="false"><munder><mo>∏</mo><mrow/></munder></mstyle></math> **C** <math><mstyle displaystyle="false"><munder><mo>∏</mo><mi>x</mi></munder></mstyle></math> Empty editor → inserts mstyle[displaystyle=false](under(∏; □)). The caret is placed in the under-script of the under-script; the next typed character goes there (typed “7” → mstyle[displaystyle=false](under(∏; 7))).
TB-8-009		Big operator with under and over scripts	141, 28, 33, 33	**A** <math><munderover><mo largeop="true"></mo><mrow/><mrow/></munderover></math> **B** <math><mi>x</mi><munderover><mo largeop="true"></mo><mrow/><mrow/></munderover></math> **C** <math><munderover><mrow><mi>x</mi><mo largeop="true"></mo></mrow><mrow/><mrow/></munderover></math> Empty editor → inserts underover(; □; □). The caret is placed in the base of the under+over scripts; the next typed character goes there (typed “7” → underover(7; □; □)).
TB-8-010		Big operator with subscript and superscript	174, 28, 33, 33	**A** <math><msubsup><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/><mrow/></msubsup></math> **B** <math><mi>x</mi><msubsup><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/><mrow/></msubsup></math> **C** <math><msubsup><maction actiontype="argument"><mo largeop="true"></mo></maction><mi>x</mi><mrow/></msubsup></math> Empty editor → inserts _{□}^{□}. The caret is placed in the subscript of the subscript+superscript; the next typed character goes there (typed “7” → _{7}^{□}).
TB-8-011		Big operator with under script	141, 61, 33, 33	**A** <math><munder><mo largeop="true"></mo><mrow/></munder></math> **B** <math><mi>x</mi><munder><mo largeop="true"></mo><mrow/></munder></math> **C** <math><munder><mrow><mi>x</mi><mo largeop="true"></mo></mrow><mrow/></munder></math> Empty editor → inserts under(; □). The caret is placed in the base of the under-script; the next typed character goes there (typed “7” → under(7; □)).
TB-8-012		Big operator with subscript	174, 61, 33, 33	**A** <math><msub><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/></msub></math> **B** <math><mi>x</mi><msub><maction actiontype="argument"><mo largeop="true"></mo></maction><mrow/></msub></math> **C** <math><msub><maction actiontype="argument"><mo largeop="true"></mo></maction><mi>x</mi></msub></math> Empty editor → inserts _{□}. The caret is placed in the subscript of the subscript; the next typed character goes there (typed “7” → _{7}).
TB-8-013		Big intersection	210, 28, 33, 33	**A** <math><mo largeop="true">∩</mo></math> **B** <math><mi>x</mi><mo largeop="true">∩</mo></math> **C** <math><mo largeop="true">∩</mo></math> Empty editor → inserts ∩. The caret ends after the inserted element; typed characters follow it (typed “7” → ∩ 7).
TB-8-014		Big union	210, 61, 33, 33	**A** <math><mo largeop="true">∪</mo></math> **B** <math><mi>x</mi><mo largeop="true">∪</mo></math> **C** <math><mo largeop="true">∪</mo></math> Empty editor → inserts ∪. The caret ends after the inserted element; typed characters follow it (typed “7” → ∪ 7).

“More” group 1 (5 controls) — expand button at 243.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-8-G1-001		Big square cap	246, 94, 33, 33 (open pop-up)	**A** <math><mo largeop="true">⊓</mo></math> **B** <math><mi>x</mi><mo largeop="true">⊓</mo></math> **C** <math><mo largeop="true">⊓</mo></math> Empty editor → inserts ⊓. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊓ 7).
TB-8-G1-002		Product	279, 94, 33, 33 (open pop-up)	**A** <math><mo>∏</mo></math> **B** <math><mi>x</mi><mo>∏</mo></math> **C** <math><mo>∏</mo></math> Empty editor → inserts ∏. The caret ends after the inserted element; typed characters follow it (typed “7” → ∏ 7).
TB-8-G1-003		Sum	312, 94, 33, 33 (open pop-up)	**A** <math><mo>∑</mo></math> **B** <math><mi>x</mi><mo>∑</mo></math> **C** <math><mo>∑</mo></math> Empty editor → inserts ∑. The caret ends after the inserted element; typed characters follow it (typed “7” → ∑ 7).
TB-8-G1-004		Big square cup	246, 127, 33, 33 (open pop-up)	**A** <math><mo largeop="true">⊔</mo></math> **B** <math><mi>x</mi><mo largeop="true">⊔</mo></math> **C** <math><mo largeop="true">⊔</mo></math> Empty editor → inserts ⊔. The caret ends after the inserted element; typed characters follow it (typed “7” → ⊔ 7).
TB-8-G1-005		Coproduct	279, 127, 33, 33 (open pop-up)	**A** <math><mo>∐</mo></math> **B** <math><mi>x</mi><mo>∐</mo></math> **C** <math><mo>∐</mo></math> Empty editor → inserts ∐. The caret ends after the inserted element; typed characters follow it (typed “7” → ∐ 7).

C.9 Tab 9 — Calculus (Alt+9)

Main panel (24 controls)

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-9-001		Definite integral	3, 28, 33, 33	**A** <math><msubsup><mo>∫</mo><mrow/><mrow/></msubsup></math> **B** <math><mi>x</mi><msubsup><mo>∫</mo><mrow/><mrow/></msubsup></math> **C** <math><msubsup><mo>∫</mo><mrow/><mrow/></msubsup><mi>x</mi></math> Empty editor → inserts ∫_{□}^{□}. The caret ends after the inserted element; typed characters follow it (typed “7” → ∫_{□}^{□} 7).
TB-9-002		Definite integral with differential	36, 28, 33, 33	**A** <math><msubsup><mo>∫</mo><mrow/><mrow/></msubsup><maction actiontype="argument"><mrow/></maction><mo>d</mo><maction actiontype="argument"><mrow/></maction></math> **B** <math><mi>x</mi><msubsup><mo>∫</mo><mrow/><mrow/></msubsup><maction actiontype="argument"><mrow/></maction><mo>d</mo><maction actiontype="argument"><mrow/></maction></math> **C** <math><msubsup><mo>∫</mo><mrow/><mrow/></msubsup><mi>x</mi><mo>d</mo><maction actiontype="argument"><mrow/></maction></math> Empty editor → inserts ∫_{□}^{□} □ d □. The caret ends after the inserted element; typed characters follow it (typed “7” → ∫_{□}^{□} 7 d □).
TB-9-003		Integral with subscript	3, 61, 33, 33	**A** <math><msub><mo>∫</mo><mrow/></msub></math> **B** <math><mi>x</mi><msub><mo>∫</mo><mrow/></msub></math> **C** <math><msub><mo>∫</mo><mrow/></msub><mi>x</mi></math> Empty editor → inserts ∫_{□}. The caret ends after the inserted element; typed characters follow it (typed “7” → ∫_{□} 7).
TB-9-004		Integral with subscript and differential	36, 61, 33, 33	**A** <math><msub><mo>∫</mo><mrow/></msub><maction actiontype="argument"><mrow/></maction><mo>d</mo><maction actiontype="argument"><mrow/></maction></math> **B** <math><mi>x</mi><msub><mo>∫</mo><mrow/></msub><maction actiontype="argument"><mrow/></maction><mo>d</mo><maction actiontype="argument"><mrow/></maction></math> **C** <math><msub><mo>∫</mo><mrow/></msub><mi>x</mi><mo>d</mo><maction actiontype="argument"><mrow/></maction></math> Empty editor → inserts ∫_{□} □ d □. The caret ends after the inserted element; typed characters follow it (typed “7” → ∫_{□} 7 d □).
TB-9-005		Differential	72, 28, 33, 33	**A** <math><mo>d</mo></math> **B** <math><mi>x</mi><mo>d</mo></math> **C** <math><mo>d</mo></math> Empty editor → inserts d. The caret ends after the inserted element; typed characters follow it (typed “7” → d 7).
TB-9-006		Derivative	105, 28, 33, 33	**A** <math><mfrac><mrow><mo>d</mo><maction actiontype="argument"><mrow/></maction></mrow><mrow><mo>d</mo><maction actiontype="argument"><mrow/></maction></mrow></mfrac></math> **B** <math><mi>x</mi><mfrac><mrow><mo>d</mo><maction actiontype="argument"><mrow/></maction></mrow><mrow><mo>d</mo><maction actiontype="argument"><mrow/></maction></mrow></mfrac></math> **C** <math><mfrac><mrow><mo>d</mo><mi>x</mi></mrow><mrow><mo>d</mo><maction actiontype="argument"><mrow/></maction></mrow></mfrac></math> Empty editor → inserts frac((d □) ⁄ (d □)). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → frac((d 7) ⁄ (d □))).
TB-9-007		Partial differential	72, 61, 33, 33	**A** <math><mo>∂</mo></math> **B** <math><mi>x</mi><mo>∂</mo></math> **C** <math><mo>∂</mo></math> Empty editor → inserts ∂. The caret ends after the inserted element; typed characters follow it (typed “7” → ∂ 7).
TB-9-008		Partial derivative	105, 61, 33, 33	**A** <math><mfrac><mrow><mo>∂</mo><maction actiontype="argument"><mrow/></maction></mrow><mrow><mo>∂</mo><maction actiontype="argument"><mrow/></maction></mrow></mfrac></math> **B** <math><mi>x</mi><mfrac><mrow><mo>∂</mo><maction actiontype="argument"><mrow/></maction></mrow><mrow><mo>∂</mo><maction actiontype="argument"><mrow/></maction></mrow></mfrac></math> **C** <math><mfrac><mrow><mo>∂</mo><mi>x</mi></mrow><mrow><mo>∂</mo><maction actiontype="argument"><mrow/></maction></mrow></mfrac></math> Empty editor → inserts frac((∂ □) ⁄ (∂ □)). The caret is placed in the numerator of the fraction; the next typed character goes there (typed “7” → frac((∂ 7) ⁄ (∂ □))).
TB-9-009		Limit to infinity	141, 28, 33, 33	**A** <math><munder><mi>lim</mi><mrow><maction actiontype="argument"><mrow/></maction><mo>→</mo><mo>∞</mo></mrow></munder></math> **B** <math><mi>x</mi><munder><mi>lim</mi><mrow><maction actiontype="argument"><mrow/></maction><mo>→</mo><mo>∞</mo></mrow></munder></math> **C** <math><munder><mi>lim</mi><mrow><maction actiontype="argument"><mrow/></maction><mo>→</mo><mo>∞</mo></mrow></munder><mi>x</mi></math> Empty editor → inserts under(lim; (□ → ∞)). The caret ends after the inserted element; typed characters follow it (typed “7” → under(lim; (□ → ∞)) 7).
TB-9-010		Limit with under script	141, 61, 33, 33	**A** <math><munder><mi>lim</mi><mrow/></munder></math> **B** <math><mi>x</mi><munder><mi>lim</mi><mrow/></munder></math> **C** <math><munder><mi>lim</mi><mrow/></munder><mi>x</mi></math> Empty editor → inserts under(lim; □). The caret ends after the inserted element; typed characters follow it (typed “7” → under(lim; □) 7).
TB-9-011		Curl	177, 28, 33, 33	**A** <math><mo>∇</mo><mo>×</mo><maction actiontype="argument"><mrow/></maction></math> **B** <math><mi>x</mi><mo>∇</mo><mo>×</mo><maction actiontype="argument"><mrow/></maction></math> **C** <math><mo>∇</mo><mo>×</mo><mi>x</mi></math> Empty editor → inserts ∇ × □. The caret ends after the inserted element; typed characters follow it (typed “7” → ∇ × 7).
TB-9-012		Gradient	210, 28, 33, 33	**A** <math><mo>∇</mo><maction actiontype="argument"><mrow/></maction></math> **B** <math><mi>x</mi><mo>∇</mo><maction actiontype="argument"><mrow/></maction></math> **C** <math><mo>∇</mo><mi>x</mi></math> Empty editor → inserts ∇ □. The caret ends after the inserted element; typed characters follow it (typed “7” → ∇ 7).

TB-9-013		Divergence	177, 61, 33, 33	**A** <math><mo>∇</mo><mo>·</mo><maction actiontype="argument"><mrow/></maction></math> **B** <math><mi>x</mi><mo>∇</mo><mo>·</mo><maction actiontype="argument"><mrow/></maction></math> **C** <math><mo>∇</mo><mo>·</mo><mi>x</mi></math> Empty editor → inserts ∇ · □. The caret ends after the inserted element; typed characters follow it (typed “7” → ∇ · 7).
TB-9-014		Laplacian	210, 61, 33, 33	**A** <math><mo>∆</mo><maction actiontype="argument"><mrow/></maction></math> **B** <math><mi>x</mi><mo>∆</mo><maction actiontype="argument"><mrow/></maction></math> **C** <math><mo>∆</mo><mi>x</mi></math> Empty editor → inserts Δ □. The caret ends after the inserted element; typed characters follow it (typed “7” → Δ 7).
TB-9-015		Integral	246, 28, 33, 33	**A** <math><mo>∫</mo></math> **B** <math><mi>x</mi><mo>∫</mo></math> **C** <math><mo>∫</mo></math> Empty editor → inserts ∫. The caret ends after the inserted element; typed characters follow it (typed “7” → ∫ 7).
TB-9-016		Double integral	279, 28, 33, 33	**A** <math><mo>∬</mo></math> **B** <math><mi>x</mi><mo>∬</mo></math> **C** <math><mo>∬</mo></math> Empty editor → inserts ∬. The caret ends after the inserted element; typed characters follow it (typed “7” → ∬ 7).
TB-9-017		Contour integral	246, 61, 33, 33	**A** <math><mo>∮</mo></math> **B** <math><mi>x</mi><mo>∮</mo></math> **C** <math><mo>∮</mo></math> Empty editor → inserts ∮. The caret ends after the inserted element; typed characters follow it (typed “7” → ∮ 7).
TB-9-018		Surface integral	279, 61, 33, 33	**A** <math><mo>∯</mo></math> **B** <math><mi>x</mi><mo>∯</mo></math> **C** <math><mo>∯</mo></math> Empty editor → inserts ∯. The caret ends after the inserted element; typed characters follow it (typed “7” → ∯ 7).
TB-9-019	sin	Sine	323, 28, 33, 33	**A** <math><mi>sin</mi><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><mi>sin</mi><mfenced><mrow/></mfenced></math> **C** <math><mi>sin</mi><mfenced><mi>x</mi></mfenced></math> Empty editor → inserts sin fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → sin fenced (7)).
TB-9-020	cos	Cosine	356, 28, 33, 33	**A** <math><mi>cos</mi><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><mi>cos</mi><mfenced><mrow/></mfenced></math> **C** <math><mi>cos</mi><mfenced><mi>x</mi></mfenced></math> Empty editor → inserts cos fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → cos fenced (7)).
TB-9-021	tan	Tangent	389, 28, 33, 33	**A** <math><mi>tan</mi><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><mi>tan</mi><mfenced><mrow/></mfenced></math> **C** <math><mi>tan</mi><mfenced><mi>x</mi></mfenced></math> Empty editor → inserts tan fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → tan fenced (7)).
TB-9-022	log	Logarithm	323, 61, 33, 33	**A** <math><mi>log</mi><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><mi>log</mi><mfenced><mrow/></mfenced></math> **C** <math><mi>log</mi><mfenced><mi>x</mi></mfenced></math> Empty editor → inserts log fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → log fenced (7)).
TB-9-023	$\log_{\square}$	Logarithm to base n	356, 61, 33, 33	**A** <math><msub><mi>log</mi><mrow/></msub><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><msub><mi>log</mi><mrow/></msub><mfenced><mrow/></mfenced></math> **C** <math><msub><mi>log</mi><mi>x</mi></msub><mfenced><mrow/></mfenced></math> Empty editor → inserts log_{□} fenced (□). The caret is placed in the subscript of the subscript; the next typed character goes there (typed “7” → log_{7} fenced (□)).
TB-9-024	ln	Natural logarithm	389, 61, 33, 33	**A** <math><mi>ln</mi><mfenced><mrow/></mfenced></math> **B** <math><mi>x</mi><mi>ln</mi><mfenced><mrow/></mfenced></math> **C** <math><mi>ln</mi><mfenced><mi>x</mi></mfenced></math> Empty editor → inserts ln fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → ln fenced (7)).

“More” group 1 (2 controls) — expand button at 313.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-9-G1-001		Triple integral	316, 94, 33, 33 (open pop-up)	**A** <math><mo>∭</mo></math> **B** <math><mi>x</mi><mo>∭</mo></math> **C** <math><mo>∭</mo></math> Empty editor → inserts ∭. The caret ends after the inserted element; typed characters follow it (typed “7” → ∭ 7).
TB-9-G1-002		Volume integral	316, 127, 33, 33 (open pop-up)	**A** <math><mo>∰</mo></math> **B** <math><mi>x</mi><mo>∰</mo></math> **C** <math><mo>∰</mo></math> Empty editor → inserts ⨌. The caret ends after the inserted element; typed characters follow it (typed “7” → ⨌ 7).

“More” group 2 (6 controls) — expand button at 422.5, 30.9, 7×60

ID	Icon	Tooltip / aria-label	x,y,w,h (px)	Expected result (MathML) — A: empty editor · B: after “x” · C: “x” selected
TB-9-G2-001		Cosecant	425, 94, 33, 33 (open pop-up)	**A** `<math><mi>csc</mi><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mi>csc</mi><mfenced><mrow/></mfenced></math>` **C** `<math><mi>csc</mi><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts csc fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → csc fenced (7)).
TB-9-G2-002		Secant	458, 94, 33, 33 (open pop-up)	**A** `<math><mi>sec</mi><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mi>sec</mi><mfenced><mrow/></mfenced></math>` **C** `<math><mi>sec</mi><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts sec fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → sec fenced (7)).
TB-9-G2-003		Cotangent	491, 94, 33, 33 (open pop-up)	**A** `<math><mi>cot</mi><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><mi>cot</mi><mfenced><mrow/></mfenced></math>` **C** `<math><mi>cot</mi><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts cot fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → cot fenced (7)).
TB-9-G2-004		Arcsine	425, 127, 33, 33 (open pop-up)	**A** `<math><msup><mi>sin</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><msup><mi>sin</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mrow/></mfenced></math>` **C** `<math><msup><mi>sin</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts sin^{(- 1)} fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → sin^{(- 1)} fenced (7)).
TB-9-G2-005		Arccosine	458, 127, 33, 33 (open pop-up)	**A** `<math><msup><mi>cos</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><msup><mi>cos</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mrow/></mfenced></math>` **C** `<math><msup><mi>cos</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts cos^{(- 1)} fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → cos^{(- 1)} fenced (7)).
TB-9-G2-006		Arctangent	491, 127, 33, 33 (open pop-up)	**A** `<math><msup><mi>tan</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mrow/></mfenced></math>` **B** `<math><mi>x</mi><msup><mi>tan</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mrow/></mfenced></math>` **C** `<math><msup><mi>tan</mi><mrow><mo>-</mo><mn>1</mn></mrow></msup><mfenced><mi>x</mi></mfenced></math>` Empty editor → inserts tan^{(- 1)} fenced (□). The caret is placed inside the fence; the next typed character goes there (typed “7” → tan^{(- 1)} fenced (7)).

Appendix D — MathML profile produced by the editor

Derived from 6606 documents produced by the reference.

Element	Occurrences	Attributes seen
`mi`	7004	`mathvariant, mathcolor`
`math`	6606	`dir, style, xmlns:wrs`
`mtd`	5139	
`mo`	2677	`largeop, lspace, rspace, mathvariant`
`mn`	1833	`mathvariant`
`mrow`	1164	
`mtr`	1143	
`mfenced`	623	`open, close, separators, wrs:valign`
`mtable`	481	`columnalign, columnspacing, align, frame, rowlines, columnlines, wrs:columnalign, wrs:decimalseparators, rowalign, equalrows, equalcolumns, framespacing`
`msrow`	250	
`mover`	208	
`maction`	177	`actiontype`
`munder`	176	
`munderover`	137	
`menclose`	130	`notation`
`mfrac`	112	`bevelled, mathcolor`
`msline`	99	`length, position`
`mstack`	94	`charalign, stackalign, charspacing`
`mspace`	85	`width, linebreak, height, depth`
`mstyle`	64	`displaystyle, mathsize, indentalign`
`msub`	62	
`msup`	49	
`none`	45	
`mlongdiv`	40	`charalign, charspacing, stackalign, longdivstyle`
`msgroup`	40	
`msubsup`	40	
`msqrt`	24	
`mmultiscripts`	24	
`mprescripts`	24	
`mroot`	20	
`mscarries`	17	`location`
`mpadded`	10	`lspace, voffset`
`mtext`	5	

Attribute	On elements	Values observed

actiontype	maction	argument
align	mtable	top · baseline · center · axis 2 · bottom
bevelled	mfrac	true
charalign	mstack, mlongdiv	center
charspacing	mlongdiv, mstack	0px · 1px · 2px · 3px · 4px · 5px · 6px · 7px · 8px · 9px · 10px · 12px · 15px · 20px
close	mfenced	] · | · } · ⌋ · > · || · ⌋ ·] ·) · [· < · { · ⌊ · ⌈
columnalign	mtable	left · right · right center left · left center right · right left · right right left · right center · right right · center left · center center · center right · left left · center
columnlines	mtable	none solid none · solid none
columnspacing	mtable	1.4ex · 2px · 0px · 0.5em · 1ex · 5
depth	mspace	2ex · 2px · 0.5em · 1ex · 5
dir	math	rtl
displaystyle	mstyle	false · true
equalcolumns	mtable	true
equalrows	mtable	true
frame	mtable	solid
framespacing	mtable	2px 0.5ex · 0.5em 0.5ex · 1ex 0.5ex · 5 0.5ex · 0.4em 2px · 0.4em 0.5em · 0.4em 1ex · 0.4em 5
height	mspace	3ex · 2px · 0.5em · 1ex · 5
indentalign	mstyle	left · center · right
largeop	mo	true
length	msline	2 · 3
linebreak	mspace	newline
location	mscarries	nw · ne · w · e · sw · s · se
longdivstyle	mlongdiv	shortstackedrightright
lspace	mpadded, mo	+1px · 0.5em · 5 · -1px · 2px · 1ex
mathcolor	mi, mfrac	#FF0000 · #007F00 · #00AA55 · #FF8800 · #0000FF · #FF00FF · #00FFFF · #00FF00 · #FFFF00 · #7F0000 · #E5C304 · #AE761E · #7F007F · #00007F
mathsize	mstyle	8px · 12px · 16px · 24px · 36px · 48px · 72px
mathvariant	mi, mn, mo	normal · fraktur · script · double-struck · bold-italic · bold · italic
notation	menclose	updiagonalstrike · top · left · box · bottom · right · circle · horizontalstrike · downdiagonalstrike · downdiagonalstrike updiagonalstrike · actuarial · roundedbox · verticalstrike · verticalstrike horizontalstrike
open	mfenced	[· | · { · ⌋ · < · || · ⌊ · ⌈ · (·] · > · } · ⌋ ·]
position	msline	1
rowalign	mtable	baseline top · baseline center · baseline bottom · baseline baseline · baseline axis
rowlines	mtable	none solid none
rspace	mo	0.5em · 5 · 2px · 1ex
separators	mfenced	| · . · ; · : ·
stackalign	mstack, mlongdiv	right · left
style	math	font-family:Arial · font-family:stix · font-family:'Courier New' · font-family:Tahoma · font-family:'Times New Roman' · font-family:Verdana
voffset	mpadded	-1px · +1px

`width`	mspace	-0.2em · 1em · 2px · 0.5em · 1ex · 5
`wrs:columnalign`	mtable	right decimal · right relation · center decimal · center relation · decimal · relation
`wrs:decimalseparators`	mtable	dot dot · dot comma · dot apostrophe · dot space · dot · comma · apostrophe · space
`wrs:valign`	mfenced	middle · middle-baseline
`xmlns:wrs`	math	http://www.wiris.com/xml/mathml-extension

Appendix E — Recognised words, digit sets and number handling

Item	Specification
Reserved (function) words, Automatic italic on	**sin, cos, tan, log, ln** — exact lower-case run inside a longer typed run is also recognised (“arcsin” → a r c *sin*, “sinh” → *sin* h, “asin” → a *sin*, “cosec” → *cos* e c). **Not** recognised (typed as single italic letters): cot, sec, csc, cosec (as a word), lg, exp, lim, liminf, limsup, max, min, sup, inf, det, dim, ker, gcd, lcm, mod, deg, arg, Re, Im, sgn, Pr, hom, tr, rank, sen, tg, ctg, upper-case variants (Sin, SIN).
Number tokens	A run of ASCII digits is one `mn`; “.” and “,” are separate `mo` tokens (3.14 → 3 . 14; 1,000 → 1 , 000).
Digit sets	Western 0–9 (default); “Arabic-Indic numerals” button → U+0660…U+0669; “Eastern Arabic-Indic numerals” → U+06F0…U+06F9; the selection persists for later typing (Q-08). Locale parameters: `arabicIndicLanguages` (ar_eg, ar_sd, ar_sa), `easternArabicIndicLanguages` (fa, ps, ur), `europeanLanguages`.
Space	U+00A0 as `mo`.
Unicode input	Any BMP/astral character is accepted; classification: letters → `mi`, symbols/punctuation → `mo`.

Appendix F — Colour palette (25 swatches, row-major)

col 1	col 2	col 3	col 4	col 5
#FF0000	#FF00FF	#0000FF	#00FFFF	#00FF00
#FFFF00	#7F0000	#E5C304	#AE761E	#7F007F
#00007F	#007F7F	#007F00	#827F00	#000000 (no attribute)
#191919	#333333	#4C4C4C	#666666	#7F7F7F
#999999	#B2B2B2	#CCCCCC	#E5E5E5	#FFFFFF

Palette panel: 102 × 118 px at the button’s lower-left; swatches 13 × 13 px, pitch 18 px (columns) and 18 px (rows), first swatch at (6, 6) inside the panel; a “Color:” label and a hex input below the swatches.

Appendix G — Contextual controls per context

Every control listed was activated in the reference; results are in FR-DOC §19–§21 and the evidence files `s10b_ctx_*.json`, `s23_length_fields.json`, `s16_elementary.json`.

Matrix (parenthesised table) — 43 controls

#	Control	Kind	Options / recorded effect
matrix-01	setOpenBracket	drop-down	( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | ‖
matrix-02	setCloseBracket	drop-down	( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | ‖
matrix-03	setSeparator	drop-down	, . ; :
matrix-04	Baseline	toggle	click: no change
matrix-05	Middle	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mfenced[wrs:valign=middle] − mfenced
matrix-06	Middle-baseline	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mfenced[wrs:valign=middle-baseline] − mfenced
matrix-07	Align with axis	toggle	click: no change
matrix-08	Align with top	button	click: + mtable[align=top] − mtable
matrix-09	Align with baseline	button	click: + mtable[align=baseline] − mtable
matrix-10	Align with center	button	click: + mtable[align=center] − mtable
matrix-11	Align to row	button	click: + mtable[align=axis 2] − mtable
matrix-12	Align with bottom	button	click: + mtable[align=bottom] − mtable
matrix-13	Insert row below	button	click: + mtr, mtd ×2
matrix-14	Insert row above	button	click: + mtr, mtd ×2
matrix-15	Insert column at right	button	click: + mtd ×2
matrix-16	Insert column at left	button	click: + mtd ×2
matrix-17	Remove row	button	click: − mtr, mtd ×2, mi ×2
matrix-18	Remove column	button	click: − mtd ×2, mi ×2

matrix-19	Insert frame	button	`click: + mtable[frame=solid] – mtable`
matrix-20	Remove frame	button	`click: no change`
matrix-21	Activate solid lines	toggle	`click: no change`
matrix-22	Insert line below	button	`click: + mtable[rowlines=none solid none] – mtable`
matrix-23	Insert right line	button	`click: + mtable[columnlines=none solid none] – mtable`
matrix-24	Activate dash lines	button	`click: no change`
matrix-25	Remove line below	button	`click: no change`
matrix-26	Remove right line	button	`click: no change`
matrix-27	Align left	button	`click: + mtable[columnalign=center left] – mtable`
matrix-28	Align center	toggle	`click: + mtable[columnalign=center center] – mtable`
matrix-29	Align right	button	`click: + mtable[columnalign=center right] – mtable`
matrix-30	Decimal align	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=center decimal] – mtable`
matrix-31	Decimal align symbol	drop-down	`dot comma apostrophe space`
matrix-32	Align rows to top	button	`click: + mtable[rowalign=baseline top] – mtable`
matrix-33	Align rows to center	button	`click: + mtable[rowalign=baseline center] – mtable`
matrix-34	Align rows to bottom	button	`click: + mtable[rowalign=baseline bottom] – mtable`
matrix-35	Relation align	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=center relation] – mtable`
matrix-36	Align rows to baseline	toggle	`click: + mtable[rowalign=baseline baseline] – mtable`
matrix-37	Align rows to axis	button	`click: + mtable[rowalign=baseline axis] – mtable`
matrix-38	Force equal row heights	button	`click: + mtable[equalrows=true] – mtable`
matrix-39	Force equal column widths	button	`click: + mtable[equalcolumns=true] – mtable`

matrix-40	Column spacing at right	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
matrix-41	Horizontal frame spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
matrix-42	Row spacing below	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
matrix-43	Vertical frame spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change

Table without fence — 37 controls

#	Control	Kind	Options / recorded effect
table-01	Align with axis	toggle	click: no change
table-02	Align with top	button	click: + mtable[align=top] − mtable
table-03	Align with baseline	button	click: + mtable[align=baseline] − mtable
table-04	Align with center	button	click: + mtable[align=center] − mtable
table-05	Align to row	button	click: + mtable[align=axis 2] − mtable
table-06	Align with bottom	button	click: + mtable[align=bottom] − mtable
table-07	Insert row below	button	click: + mtr, mtd ×2
table-08	Insert row above	button	click: + mtr, mtd ×2
table-09	Insert column at right	button	click: + mtd ×2
table-10	Insert column at left	button	click: + mtd ×2
table-11	Remove row	button	click: − mtr, mtd ×2, mi ×2
table-12	Remove column	button	click: − mtd ×2, mi ×2
table-13	Insert frame	button	click: + mtable[frame=solid] − mtable
table-14	Remove frame	button	click: no change
table-15	Activate solid lines	toggle	click: no change
table-16	Insert line below	button	click: + mtable[rowlines=none solid none] − mtable
table-17	Insert right line	button	click: + mtable[columnlines=none solid none] − mtable
table-18	Activate dash lines	button	click: no change
table-19	Remove line below	button	click: no change
table-20	Remove right line	button	click: no change
table-21	Align left	button	click: + mtable[columnalign=center left] − mtable
table-22	Align center	toggle	click: + mtable[columnalign=center center] − mtable
table-23	Align right	button	click: + mtable[columnalign=center right] − mtable
table-24	Decimal align	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=center decimal] − mtable
table-25	Decimal align symbol	drop-down	dot comma apostrophe space

table-26	Align rows to top	button	`click: + mtable[rowalign=baseline top] − mtable`
table-27	Align rows to center	button	`click: + mtable[rowalign=baseline center] − mtable`
table-28	Align rows to bottom	button	`click: + mtable[rowalign=baseline bottom] − mtable`
table-29	Relation align	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=center relation] − mtable`
table-30	Align rows to baseline	toggle	`click: + mtable[rowalign=baseline baseline] − mtable`
table-31	Align rows to axis	button	`click: + mtable[rowalign=baseline axis] − mtable`
table-32	Force equal row heights	button	`click: + mtable[equalrows=true] − mtable`
table-33	Force equal column widths	button	`click: + mtable[equalcolumns=true] − mtable`

table-34	Column spacing at right	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

table-35	Horizontal frame spacing	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

table-36	Row spacing below	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

table-37	Vertical frame spacing	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

Piecewise function — 43 controls

#	Control	Kind	Options / recorded effect
piecewise-01	setOpenBracket	drop-down	`( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | ‖`
piecewise-02	setCloseBracket	drop-down	`( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | ‖`
piecewise-03	setSeparator	drop-down	`, . ; :`
piecewise-04	Baseline	toggle	`click: no change`
piecewise-05	Middle	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mfenced[close= open={ wrs:valign=middle] − mfenced[close= open={]`
piecewise-06	Middle-baseline	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mfenced[close= open={ wrs:valign=middle-baseline] − mfenced[close= open={]`
piecewise-07	Align with axis	toggle	`click: no change`
piecewise-08	Align with top	button	`click: + mtable[align=top columnalign=left] − mtable[columnalign=left]`
piecewise-09	Align with baseline	button	`click: + mtable[align=baseline columnalign=left] − mtable[columnalign=left]`
piecewise-10	Align with center	button	`click: + mtable[align=center columnalign=left] − mtable[columnalign=left]`

piecewise-11	Align to row	button	`click: + mtable[align=axis 2 columnalign=left] − mtable[columnalign=left]`
piecewise-12	Align with bottom	button	`click: + mtable[align=bottom columnalign=left] − mtable[columnalign=left]`
piecewise-13	Insert row below	button	`click: + mtr, mtd`
piecewise-14	Insert row above	button	`click: + mtr, mtd`
piecewise-15	Insert column at right	button	`click: + mtd ×2`
piecewise-16	Insert column at left	button	`click: + mtable[columnalign=left left], mtd ×2 − mtable[columnalign=left]`
piecewise-17	Remove row	button	`click: − mtr, mtd, mi`
piecewise-18	Remove column	button	`click: no change`
piecewise-19	Insert frame	button	`click: + mtable[columnalign=left frame=solid] − mtable[columnalign=left]`
piecewise-20	Remove frame	button	`click: no change`
piecewise-21	Activate solid lines	toggle	`click: no change`
piecewise-22	Insert line below	button	`click: + mtable[columnalign=left rowlines=none solid none] − mtable[columnalign=left]`
piecewise-23	Insert right line	button	`click: + mtable[columnalign=left columnlines=solid none] − mtable[columnalign=left]`
piecewise-24	Activate dash lines	button	`click: no change`
piecewise-25	Remove line below	button	`click: no change`
piecewise-26	Remove right line	button	`click: no change`
piecewise-27	Align left	toggle	`click: no change`
piecewise-28	Align center	button	`click: + mtable[columnalign=center] − mtable[columnalign=left]`
piecewise-29	Align right	button	`click: + mtable[columnalign=right] − mtable[columnalign=left]`
piecewise-30	Decimal align	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=decimal] − mtable[columnalign=left]`
piecewise-31	Decimal align symbol	drop-down	`dot comma apostrophe space`
piecewise-32	Align rows to top	button	`click: + mtable[columnalign=left rowalign=baseline top] − mtable[columnalign=left]`

piecewise-33	Align rows to center	button	click: + mtable[columnalign=left rowalign=baseline center] − mtable[columnalign=left]
piecewise-34	Align rows to bottom	button	click: + mtable[columnalign=left rowalign=baseline bottom] − mtable[columnalign=left]
piecewise-35	Relation align	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=relation] − mtable[columnalign=left]
piecewise-36	Align rows to baseline	toggle	click: + mtable[columnalign=left rowalign=baseline baseline] − mtable[columnalign=left]
piecewise-37	Align rows to axis	button	click: + mtable[columnalign=left rowalign=baseline axis] − mtable[columnalign=left]
piecewise-38	Force equal row heights	button	click: + mtable[columnalign=left equalrows=true] − mtable[columnalign=left]
piecewise-39	Force equal column widths	button	click: + mtable[columnalign=left equalcolumns=true] − mtable[columnalign=left]
piecewise-40	Column spacing at right	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
piecewise-41	Horizontal frame spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
piecewise-42	Row spacing below	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
piecewise-43	Vertical frame spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change

Aligned equations — 39 controls

#	Control	Kind	Options / recorded effect
aligned-01	Left spacing	length field	click: no change type "2px" + Enter: + mtr, mtd ×2, mi ×2, mn type "0.5em" + Enter: + mo[lspace=0.5em] − mo type "5" + Enter: + mo[lspace=5] − mo
aligned-02	Right spacing	length field	click: no change type "2px" + Enter: + mtable[align=top columnalign=right left], mtr, mtd ×2, mi ×2, mn − mtable[columnalign=right left] type "0.5em" + Enter: + mo[rspace=0.5em] − mo type "5" + Enter: + mo[rspace=5] − mo
aligned-03	Align with axis	toggle	click: no change
aligned-04	Align with top	button	click: + mtable[align=top columnalign=right left] − mtable[columnalign=right left]
aligned-05	Align with baseline	button	click: + mtable[align=baseline columnalign=right left] − mtable[columnalign=right left]
aligned-06	Align with center	button	click: + mtable[align=center columnalign=right left] − mtable[columnalign=right left]
aligned-07	Align to row	button	click: + mtable[align=axis 2 columnalign=right left] − mtable[columnalign=right left]

aligned-08	Align with bottom	button	`click: + mtable[align=bottom columnalign=right left] − mtable[columnalign=right left]`
aligned-09	Insert row below	button	`click: + mtr, mtd ×2`
aligned-10	Insert row above	button	`click: + mtr, mtd ×2`
aligned-11	Insert column at right	button	`click: + mtd ×2`
aligned-12	Insert column at left	button	`click: + mtable[columnalign=right right left], mtd ×2 − mtable[columnalign=right left]`
aligned-13	Remove row	button	`click: − mtr, mtd ×2, mi ×2, mo`
aligned-14	Remove column	button	`click: + mtable[columnalign=right] − mtable[columnalign=right left], mtd ×2, mi ×2, mo ×2`
aligned-15	Insert frame	button	`click: + mtable[columnalign=right left frame=solid] − mtable[columnalign=right left]`
aligned-16	Remove frame	button	`click: no change`
aligned-17	Activate solid lines	toggle	`click: no change`
aligned-18	Insert line below	button	`click: + mtable[columnalign=right left rowlines=none solid none] − mtable[columnalign=right left]`
aligned-19	Insert right line	button	`click: + mtable[columnalign=right left columnlines=none solid none] − mtable[columnalign=right left]`
aligned-20	Activate dash lines	button	`click: no change`
aligned-21	Remove line below	button	`click: no change`
aligned-22	Remove right line	button	`click: no change`
aligned-23	Align left	toggle	`click: no change`
aligned-24	Align center	button	`click: + mtable[columnalign=right center] − mtable[columnalign=right left]`
aligned-25	Align right	button	`click: + mtable[columnalign=right right] − mtable[columnalign=right left]`
aligned-26	Decimal align	button	`click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=right decimal] − mtable[columnalign=right left]`
aligned-27	Decimal align symbol	drop-down	`dot comma apostrophe space`
aligned-28	Align rows to top	button	`click: + mtable[columnalign=right left rowalign=baseline top] − mtable[columnalign=right left]`
aligned-29	Align rows to center	button	`click: + mtable[columnalign=right left rowalign=baseline center] − mtable[columnalign=right left]`

aligned-30	Align rows to bottom	button	click: + mtable[columnalign=right left rowalign=baseline bottom] − mtable[columnalign=right left]
aligned-31	Relation align	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mtable[wrs:columnalign=right relation] − mtable[columnalign=right left]
aligned-32	Align rows to baseline	toggle	click: + mtable[columnalign=right left rowalign=baseline baseline] − mtable[columnalign=right left]
aligned-33	Align rows to axis	button	click: + mtable[columnalign=right left rowalign=baseline axis] − mtable[columnalign=right left]
aligned-34	Force equal row heights	button	click: + mtable[columnalign=right left equalrows=true] − mtable[columnalign=right left]
aligned-35	Force equal column widths	button	click: + mtable[columnalign=right left equalcolumns=true] − mtable[columnalign=right left]
aligned-36	Column spacing at right	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
aligned-37	Horizontal frame spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
aligned-38	Row spacing below	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change
aligned-39	Vertical frame spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: no change type "5" + Enter: no change

Fenced expression — 8 controls

#	Control	Kind	Options / recorded effect
fence_in-01	setOpenBracket	drop-down	( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | ‖
fence_in-02	setCloseBracket	drop-down	( ) [ ] ⟨ ⟩ { } ⌊ ⌋ ⌈ ⌉ ⟦ ⟧ | ‖
fence_in-03	setSeparator	drop-down	, . ; :
fence_in-04	Baseline	toggle	click: no change
fence_in-05	Middle	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mfenced[close=] open=[wrs:valign=middle] − mfenced[close=] open=[]
fence_in-06	Middle-baseline	button	click: + math[xmlns:wrs=http://www.wiris.com/xml/mathml-extension], mfenced[close=] open=[wrs:valign=middle-baseline] − mfenced[close=] open=[]
fence_in-07	Left spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: + mo[lspace=0.5em] − mo type "5" + Enter: + mo[lspace=5] − mo
fence_in-08	Right spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: + mo[rspace=0.5em] − mo type "5" + Enter: + mo[rspace=5] − mo

Operator in an expression — 2 controls

#	Control	Kind	Options / recorded effect
operator-01	Left spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: + mo[lspace=0.5em] − mo type "5" + Enter: + mo[lspace=5] − mo
operator-02	Right spacing	length field	click: no change type "2px" + Enter: no change type "0.5em" + Enter: + mo[rspace=0.5em] − mo type "5" + Enter: + mo[rspace=5] − mo

Multi-line expression — 6 controls

#	Control	Kind	Options / recorded effect
multiline-01	Left spacing	length field	click: no change type "2px" + Enter: + mstyle[indentalign=left], mi ×2, mspace[linebreak=newline], mn type "0.5em" + Enter: no change type "5" + Enter: no change
multiline-02	Right spacing	length field	click: no change type "2px" + Enter: + mtable[columnalign=right center left columnspacing=0px], mtr ×3, mtd ×9, mi ×2, mn − mspace[linebreak=newline] type "0.5em" + Enter: no change type "5" + Enter: no change
multiline-03	Align left	toggle	click: + mstyle[indentalign=left]
multiline-04	Relation align	button	click: + mtable[columnalign=right center left columnspacing=0px], mtr ×2, mtd ×6 − mspace[linebreak=newline]
multiline-05	Align center	button	click: + mstyle[indentalign=center]
multiline-06	Align right	button	click: + mstyle[indentalign=right]

Elementary stack (addition) — 17 controls

#	Control	Kind	Options / recorded effect
stack-01	Insert line	button	click: + msline
stack-02	Join the selected digits in a single column	button	click: no change
stack-03	Up diagonal strike	button	click: + menclose[notation=updiagonalstrike]
stack-04	Prepend row	button	click: + msrow
stack-05	Append row	button	click: + msrow
stack-06	Set carries location at north-west	button	click: no change
stack-07	Set carries location at north	button	click: no change
stack-08	Set carries location at north-east	button	click: no change
stack-09	Set carries location at west	button	click: no change
stack-10	Insert carries in this row	button	click: + msrow, mscarries
stack-11	Set carries location at east	button	click: no change
stack-12	Set carries location at south-west	button	click: no change
stack-13	Set carries location at south	button	click: no change

stack-14	Set carries location at south-east	button	`click: no change`
stack-15	Move to left	button	`click: + msline[length=2 position=1] − msline`
stack-16	Move to right	button	`click: + msrow, none ×2`
stack-17	Spacing between digits	drop-down	`medium 0px 1px 2px 3px 4px 5px 6px 7px 8px 9px 10px 12px 15px 20px 25px 30px`

Existing mspace element — 3 controls

#	Control	Kind	Options / recorded effect

mspace-01	Space width	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

mspace-02	Space height	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

mspace-03	Space depth	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: no change`
			`type "5" + Enter: no change`

Big operator ∑ — 2 controls

#	Control	Kind	Options / recorded effect

sum-01	Left spacing	length field	`click: no change`
			`type "2px" + Enter: + mo[lspace=2px] − mo`
			`type "0.5em" + Enter: + mo[lspace=0.5em] − mo`
			`type "5" + Enter: + mo[lspace=5] − mo`

sum-02	Right spacing	length field	`click: no change`
			`type "2px" + Enter: + mo[rspace=2px] − mo`
			`type "0.5em" + Enter: + mo[rspace=0.5em] − mo`
			`type "5" + Enter: + mo[rspace=5] − mo`

Selected single symbol — 4 controls

#	Control	Kind	Options / recorded effect
symbol_sel-01	Nudge up	button	`click: + mpadded[voffset=-1px]`
symbol_sel-02	Nudge right	button	`click: + mpadded[lspace=+1px]`
symbol_sel-03	Nudge left	button	`click: + mpadded[lspace=-1px]`
symbol_sel-04	Nudge down	button	`click: + mpadded[voffset=+1px]`

Selected number separator — 6 controls

#	Control	Kind	Options / recorded effect
numsep_sel-01	Nudge up	button	`click: + mpadded[voffset=-1px]`
numsep_sel-02	Nudge right	button	`click: + mpadded[lspace=+1px]`

numsep_sel-03	Nudge left	button	`click: + mpadded[lspace=-1px]`
numsep_sel-04	Nudge down	button	`click: + mpadded[voffset=+1px]`

numsep_sel-05	Left spacing	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: + mo[lspace=0.5em] − mo`
			`type "5" + Enter: + mo[lspace=5] − mo`

numsep_sel-06	Right spacing	length field	`click: no change`
			`type "2px" + Enter: no change`
			`type "0.5em" + Enter: + mo[rspace=0.5em] − mo`
			`type "5" + Enter: + mo[rspace=5] − mo`

Appendix H — Design tokens

Token	Value	Used for
frame	`#778e9a`	toolbar frame, tab top border, button hover/focus border (2 px, radius 3 px), handwriting strip
formula border	`#969696 (1 px)`	formula display
separator	`#bbbdbe (1 px)`	group separators
select border/text	`#b1b3b4`	font/size boxes
selected tab bg	`#f2f3f5`	selected tab (+ selected_tab_background image)
tab bg	`#556e7b`	non-selected tabs (+ tab_background image)
contextual tab bg	`#dc9e9e`	Contextual tab (+ context_tab_background image)
selection	`#c3c3c3 / #3c3c3c`	selection highlight light / dark
edit-context highlight	`#cddce3`	current row underline band
error	`#ffaaaa`	invalid region
caret	`#000 1 px × 20 px (white on dark; #888 inverse caret; underline caret black 50 %)`	caret
empty slot	`#006400`	empty box outline
touch handles	`#00aaff`	selection / caret precision handles (24 px square + 12 px arrow)
toolbar font	`11 px bold sans-serif`	panel text, pop-up titles
formula font size	`16 px (default)`	math
hand stroke	`#284078`	handwriting ink
hand grid	`#d4d4d4, 20 px pitch`	handwriting paper
hand toolbar	`#e5edf0 / keyboard button #4e5861`	handwriting UI
combo hover	`#a4cae7`	pop-up item hover
pop-up panel	`#fff, 1 px solid #000`	drop-down panels
help link	`rgb(24,33,47)`	? icon

Sizes: large button 33 × 33 px (2-row layout), small button 22 × 22 px (3-row layout), expand arrow 7 × 60 px, tab 44 × 22/25 px, separator height 60 px, multi-row panel height 66 px. Scales: toolbarSize 1 / 1.5 / 2 (font 11 / 17 / 22 px).

Appendix I — Behaviour differences caused by global modes

Details with MathML are in FR-DOC §26. Selection in RTL mode uses Shift+ArrowRight.

Right to left editing ON (tabs 1–9): 687 controls re-tested, 670 identical, 17 differ.

Control	Tooltip	States that differ
t4_b10	2 rows column with left curly brackets	A,A2,B,C
t4_b11	Piecewise function	A,A2,B,C
t4_b12	2 rows column with right curly brackets	A,A2,B,C
t4_b13	Aligned equations	A,A2,B,C
t8_b8	Limit to infinity	A,A2,B,C
t8_b18	Sine	A,A2,B,C
t8_b19	Cosine	A,A2,B,C
t8_b20	Tangent	A,A2,B,C
t8_b21	Logarithm	A,A2,B,C
t8_b22	Logarithm to base n	A,A2,B,C
t8_b23	Natural logarithm	A,A2,B,C
t8_x1_i0	Cosecant	A,A2,B,C
t8_x1_i1	Secant	A,A2,B,C
t8_x1_i2	Cotangent	A,A2,B,C
t8_x1_i3	Arcsine	A,A2,B,C
t8_x1_i4	Arccosine	A,A2,B,C
t8_x1_i5	Arctangent	A,A2,B,C

Automatic italic OFF (tabs 1 and 4): 379 controls re-tested, 0 identical, 379 differ.

Control	Tooltip	States that differ
t0_b0	Fraction (Ctrl+/)	B,C
t0_b1	Square root	B,C
t0_b2	Bevelled fraction	B,C
t0_b3	Root	B,C
t0_b4	Superscript (Ctrl+Up)	B,C
t0_b5	Subscript (Ctrl+Down)	B,C
t0_b6	Parentheses	B,C
t0_b7	Square brackets	B,C
t0_b8	Vertical bars	B,C
t0_b9	Curly brackets	B,C
t0_b10	Plus sign	B
t0_b11	Forward slash	B
t0_b12	Multiplication sign	B
t0_b13	Plus-minus sign	B
t0_b14	Minus sign	B
t0_b15	Division sign	B
t0_b16	Greater-than or equal to	B
t0_b17	Less-than or equal to	B
t0_b18	Element of	B
t0_b19	Subset of	B
t0_b20	Union	B
t0_b21	Intersection	B
t0_b22	Empty set	B
t0_b23	Infinity	B
t0_b24	Number pi	B
t3_b0	Alpha	A,A2,B,C

t3_b1	Beta	A,A2,B,C
t3_b2	Gamma	A,A2,B,C
t3_b3	Delta	A,A2,B,C
t3_b4	Epsilon	A,A2,B,C
t3_b5	Zeta	A,A2,B,C
t3_b6	Eta	A,A2,B,C
t3_b7	Theta	A,A2,B,C
t3_b8	Theta	A,A2,B,C
t3_b9	Iota	A,A2,B,C
t3_b10	Kappa	A,A2,B,C
t3_b11	Lambda	A,A2,B,C
t3_b12	Mu	A,A2,B,C
t3_b13	Nu	A,A2,B,C
t3_b14	Xi	A,A2,B,C
t3_b15	Omicron	A,A2,B,C
t3_b16	Pi	A,A2,B,C
t3_b17	Pi	A,A2,B,C
t3_b18	Rho	A,A2,B,C
t3_b19	Final sigma	A,A2,B,C
t3_b20	Sigma	A,A2,B,C
t3_b21	Tau	A,A2,B,C
t3_b22	Upsilon	A,A2,B,C
t3_b23	Phi	A,A2,B,C
t3_b24	Phi	A,A2,B,C
t3_b25	Chi	A,A2,B,C
t3_b26	Psi	A,A2,B,C
t3_b27	Omega	A,A2,B,C
t3_b28	Natural numbers (double-struck capital n)	B
t3_b29	Integer numbers (double-struck capital z)	B
t3_b30	Rational numbers (double-struck capital q)	B
t3_b31	Complex numbers (double-struck capital c)	B
t3_b32	Real numbers (double-struck capital r)	B
t3_b33	Prime numbers (double-struck capital p)	B
t3_b34	Fraktur capital a	B
t3_b35	Fraktur capital b	B
t3_b36	Fraktur capital c	B
t3_b37	Script capital a	B
t3_b38	Script capital b	B
t3_b39	Script capital c	B
t3_b40	Imaginary part (black-letter capital i)	B
t3_b41	Real part (black-letter capital r)	B
t3_b42	Ell	B
t3_b43	Arabic-Indic numerals	B,C
t3_b44	Eastern Arabic-Indic numerals	B,C
t3_b45	Hydrogen	B
t3_b46	Carbon	B
t3_b47	Nitrogen	B
t3_b48	Oxygen	B
t3_b49	Fluorine	B
t3_b50	Sulfur	B
t3_x0_i0	Capital alpha	A,A2,B,C
t3_x0_i1	Capital beta	A,A2,B,C
t3_x0_i2	Capital gamma	A,A2,B,C

t3_x0_i3	Capital delta	A,A2,B,C
t3_x0_i4	Capital epsilon	A,A2,B,C
t3_x0_i5	Capital zeta	A,A2,B,C
t3_x0_i6	Capital eta	A,A2,B,C
t3_x0_i7	Capital theta	A,A2,B,C
t3_x0_i8	Capital iota	A,A2,B,C
t3_x0_i9	Capital kappa	A,A2,B,C
t3_x0_i10	Capital lambda	A,A2,B,C
t3_x0_i11	Capital mu	A,A2,B,C
t3_x0_i12	Capital nu	A,A2,B,C
t3_x0_i13	Capital xi	A,A2,B,C
t3_x0_i14	Capital omicron	A,A2,B,C
t3_x0_i15	Capital pi	A,A2,B,C
t3_x0_i16	Capital rho	A,A2,B,C
t3_x0_i17	Capital sigma	A,A2,B,C
t3_x0_i18	Capital tau	A,A2,B,C
t3_x0_i19	Capital upsilon	A,A2,B,C
t3_x0_i20	Capital phi	A,A2,B,C
t3_x0_i21	Capital chi	A,A2,B,C
t3_x0_i22	Capital psi	A,A2,B,C
t3_x0_i23	Capital omega	A,A2,B,C
t3_x1_i0	Double struck capital a	B
t3_x1_i1	Double struck capital d	B
t3_x1_i2	Double struck capital g	B
t3_x1_i3	Double struck capital j	B
t3_x1_i4	Double struck capital m	B
t3_x1_i5	Prime numbers (double-struck capital p)	B
t3_x1_i6	Double struck capital s	B
t3_x1_i7	Double struck capital v	B
t3_x1_i8	Double struck capital y	B
t3_x1_i9	Double struck a	B
t3_x1_i10	Double struck d	B
t3_x1_i11	Double struck g	B
t3_x1_i12	Double struck j	B
t3_x1_i13	Double struck m	B
t3_x1_i14	Double struck p	B
t3_x1_i15	Double struck s	B
t3_x1_i16	Double struck v	B
t3_x1_i17	Double struck y	B
t3_x1_i18	Double struck capital b	B
t3_x1_i19	Double struck capital e	B
t3_x1_i20	Double struck capital h	B
t3_x1_i21	Double struck capital k	B
t3_x1_i22	Natural numbers (double-struck capital n)	B
t3_x1_i23	Rational numbers (double-struck capital q)	B
t3_x1_i24	Double struck capital t	B
t3_x1_i25	Double struck capital w	B
t3_x1_i26	Integer numbers (double-struck capital z)	B
t3_x1_i27	Double struck b	B
t3_x1_i28	Double struck e	B
t3_x1_i29	Double struck h	B
t3_x1_i30	Double struck k	B
t3_x1_i31	Double struck n	B

t3_x1_i32	Double struck q	B
t3_x1_i33	Double struck t	B
t3_x1_i34	Double struck w	B
t3_x1_i35	Double struck z	B
t3_x1_i36	Complex numbers (double-struck capital c)	B
t3_x1_i37	Double struck capital f	B
t3_x1_i38	Imaginary numbers (double-struck capital i)	B
t3_x1_i39	Double struck capital l	B
t3_x1_i40	Double struck capital o	B
t3_x1_i41	Real numbers (double-struck capital r)	B
t3_x1_i42	Double struck capital u	B
t3_x1_i43	Double struck capital x	B
t3_x1_i44	Double struck c	B
t3_x1_i45	Double struck f	B
t3_x1_i46	Double struck i	B
t3_x1_i47	Double struck l	B
t3_x1_i48	Double struck o	B
t3_x1_i49	Double struck r	B

… 229 more (see FR-DOC §26).

Appendix J — Layout measurements of the reference

Measured in the formula display at device scale 1 (coordinates in page px; only differences matter). Visible glyph boxes: text, font size, width × height, top-left.

Formula	Glyph boxes (text · font-size · w×h · x,y)	Rules (w×h)
x	`x · 16px · 8×18 · 604,772`	`9×1`
x^2	`x · 16px · 8×18 · 604,776` `2 · 12px · 6.7×14 · 613,772`	`16×1`

x^(2^3)	`x · 16px · 8×18 · 604,779`
	`2 · 12px · 6.7×14 · 613,775`
	`3 · 10px · 5.6×11 · 620,771`
		`22×1`

x_1	`x · 16px · 8×18 · 604,772` `1 · 12px · 6.7×14 · 613,780`	`16×1`
a/b	`a · 16px · 8.9×18 · 608,772` `b · 16px · 8.9×18 · 608,796`	`14×1, 18×1`

(a/b)/c	`a · 12px · 6.7×14 · 612,772`
	`b · 12px · 6.7×14 · 612,791`
	`c · 16px · 8×18 · 612,810`
		`20×1, 12×1, 24×1`

√x	`x · 16px · 8×18 · 619,776`	`13×1, 26×1`

∑ with limits	`n · 12px · 6.7×14 · 611,774`
	`∑ · 24px · 22×27 · 604,785`
	`i · 12px · 2.7×14 · 613,811`
		`22×1`

(x)	`( · 16px · 4.3×17 · 605,772`
	`) · 16px · 4.3×17 · 618,772`
	`x · 16px · 8×18 · 609,772`
		`19×1`